台大獸醫學系
獸醫臨床及影像診斷學

Dental system



普羅動物醫院
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- 國立嘉義大學 兼任講師
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- 桃園市獸醫師公會課程 講師
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- 歐洲獸醫高級學程ESAVS牙科課程 Ⅰ Ⅱ Ⅲ Ⅳ Ⅴ 結訓

犬貓完整的口腔評估與治COHAT

C= Comprehesive
O= Oral
H= Health
A= Assessment and
T= Treatment

COHAT

- 安全而完整的評估牙齒牙齦上及牙齦下的狀況
- 需要麻醉及止痛
- 麻醉中一定要氣管插管
- 牙齦下檢查需要X光
- 專業的技術

COHAT 八步

- 1.門診麻醉前口腔檢查、理學檢查、血液檢查
- 2.麻醉計劃、IV catheter(for IV fluid and anesthetic drugs)
- 3.麻醉監控: 至少血氧、心跳、血壓、呼吸速率、體溫
- 4.檢查每顆牙齒、用牙周探針(periodontal probe)檢查囊袋、檢查牙齦是否發炎
- 5.移除牙齦上及牙齦下的牙菌斑和牙結石
- 6. 牙科X光檢查牙齦下的牙根與骨頭
- 7. 治療牙周病的牙齒或拔除
- 8. 術後照顧與居家口腔保健

牙周疾病



牙周疾病發生率

- 牙周疾病在小動物病患是最常見的健康問題之一（Lund et al. 1999）。
- 2歲時，70%的貓和80% 的犬患有不同形式的牙周疾病（Marshall 2014）。
- 更近期的研究描述所有病患具有近90% 的發病率（Fernandes 2012, Queck et al. 2018, Stella et al. 2018）
- 小型犬和玩具犬特別容易發病（Homann & Gaengler 1996），其發病率隨年齡增長而增加（Stella et al. 2018）。

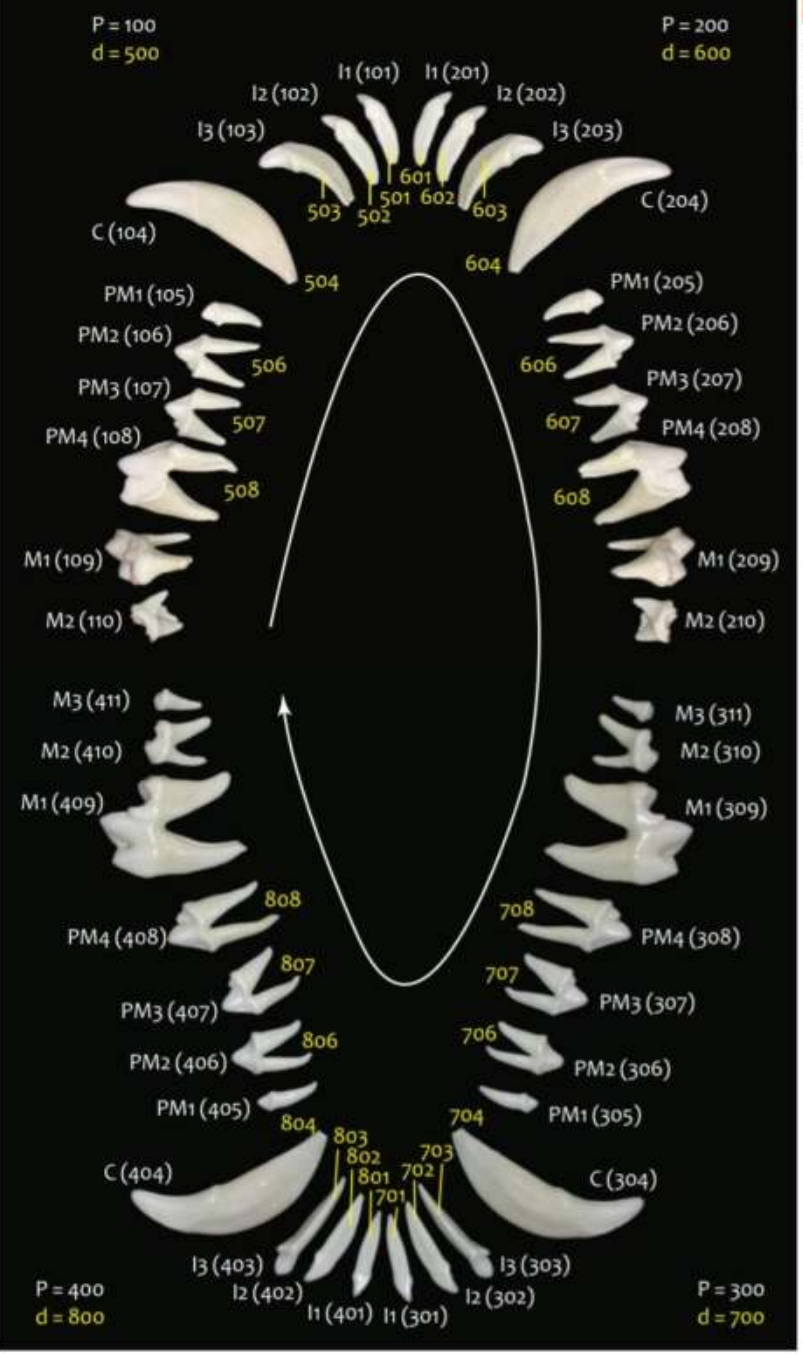
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牙周疾病的臨床症狀

表 1. 牙痛可能引起之明顯可見變化。

- 寵物與飼主之間的互動模式改變
- 過度流涎
- 攻擊性
- 躲藏
- 睡眠型態受影響
- 理毛次數減少
- 進食行為改變
- 食物偏好改變：從硬食變成軟食
- 狼吞虎嚥：吞下整塊食物而沒有咀嚼
- 單側咀嚼
- 舔嘴唇
- 牙齒打顫
- 磨牙：特別是在貓
- 舌頭垂露在嘴巴外
- 玩耍行為改變
- 食物或水盆中有血
- 鼻子流出血液分泌物
- 摩擦臉部或用手掌搔抓臉部
- 口鼻部周圍脫毛
- 患貓不再對人磨蹭臉部表達情感
- 食物掉到食盆外：不願意咀嚼

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2.7 canine dentiti
mandibular teeth and th
The numbering for the d
yellow. C = canine tooth;
tooth; M = molar tooth; I
tooth.
(© Dr Margherita Gracis)

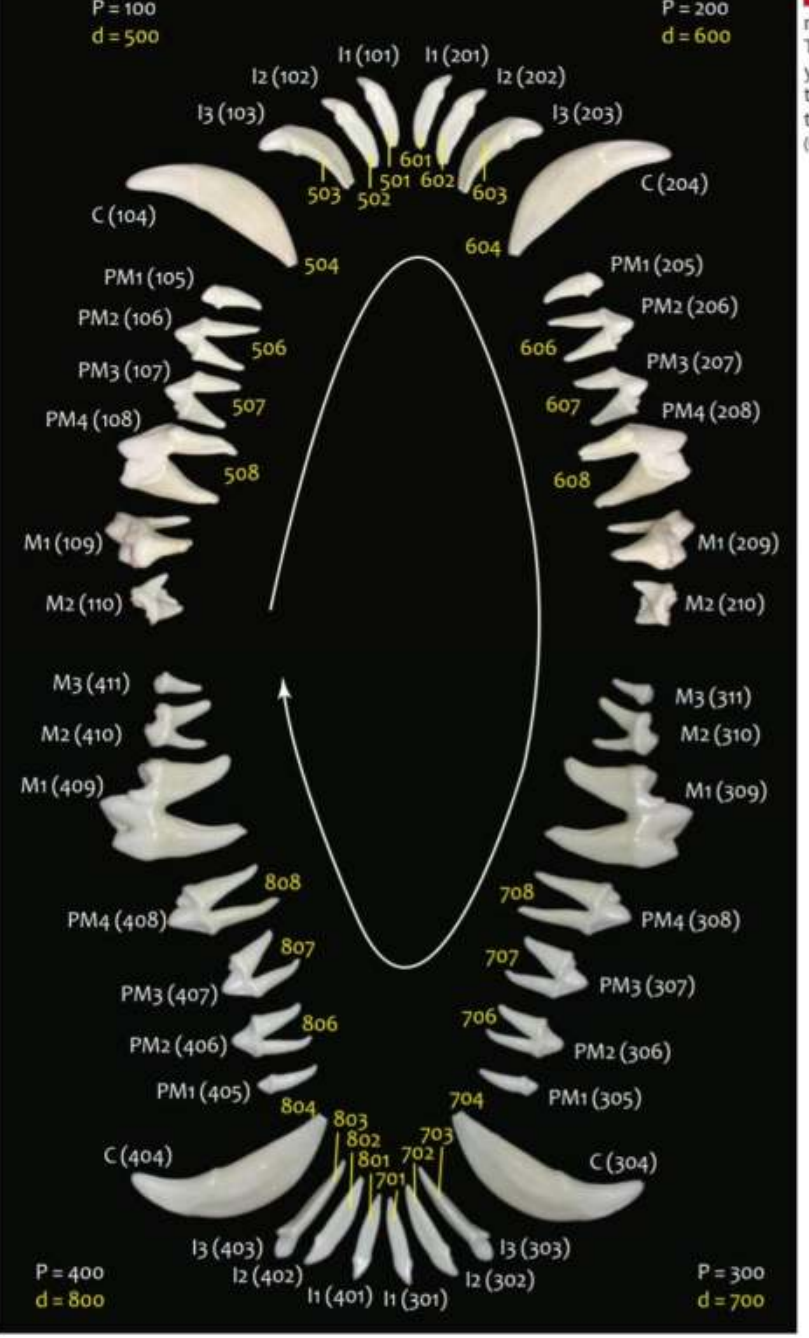
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門牙incisor: 單牙根

犬齒canine:單牙根

前臼齒premolar: 單牙根&多牙根

臼齒molar:多牙根&單牙根

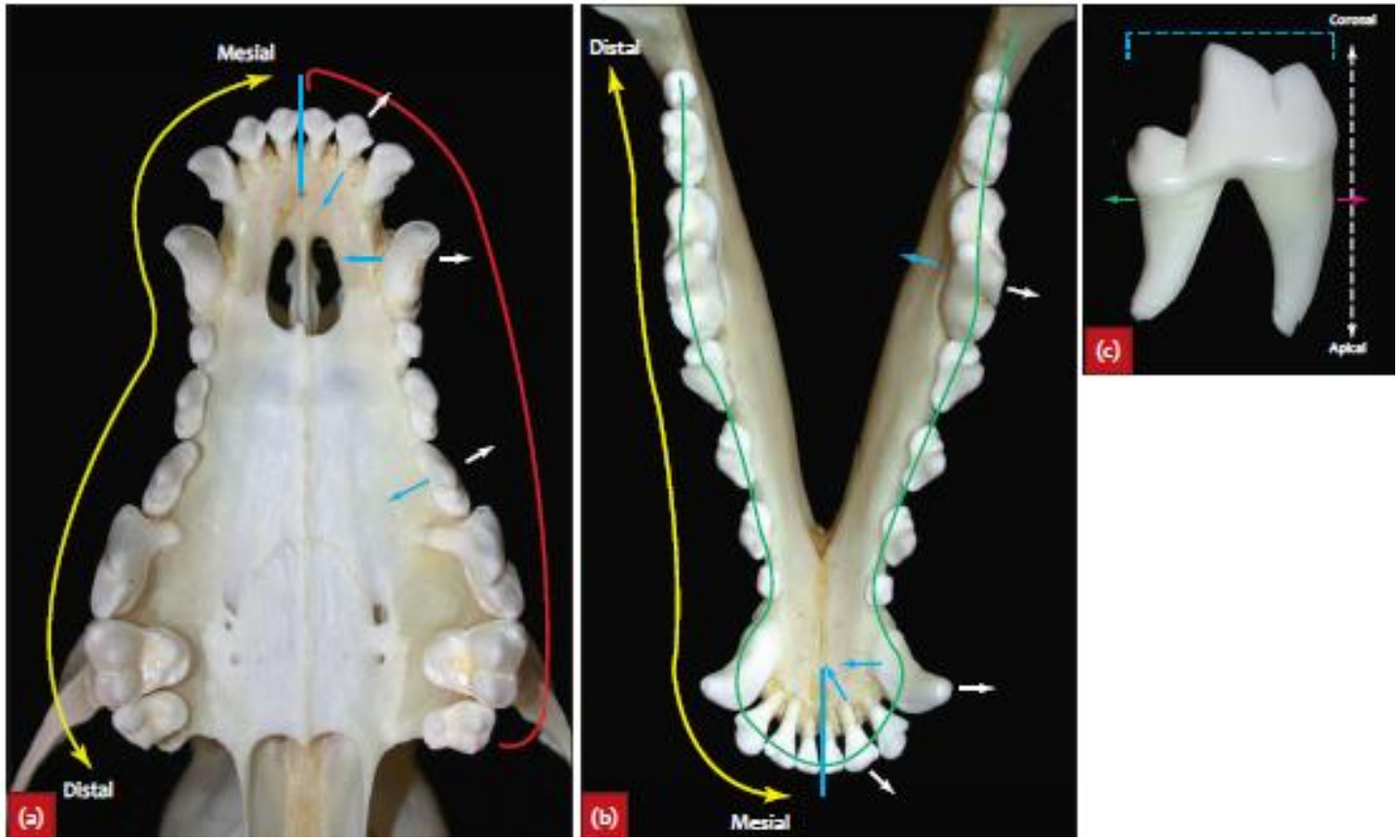


mandibular teeth and th
The numbering for the d
yellow. C = canine tooth;
tooth; M = molar tooth; I
tooth.
(© Dr Margherita Gracis)

Dental formula

		Incisor	Canine	Premolar	Molar	total
Adult Dog	Maxillary	3	1	4	2	42
	Mandubular	3	1	4	3	
Puppy	Maxillary	3	1	3		28
	Mandubular	3	1	3		
Adult Cat	Maxillary	3	1	3	1	30
	Mandubular	3	1	2	1	
Kitten	Maxillary	3	1	3		26
	Mandubular	3	1	2		

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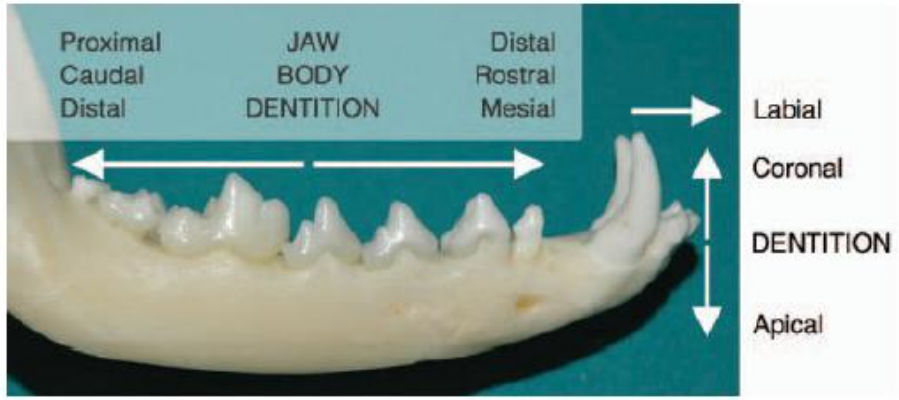


Figure 1.40 Directional anatomic terms for the mandible and its dentition (lateral view).

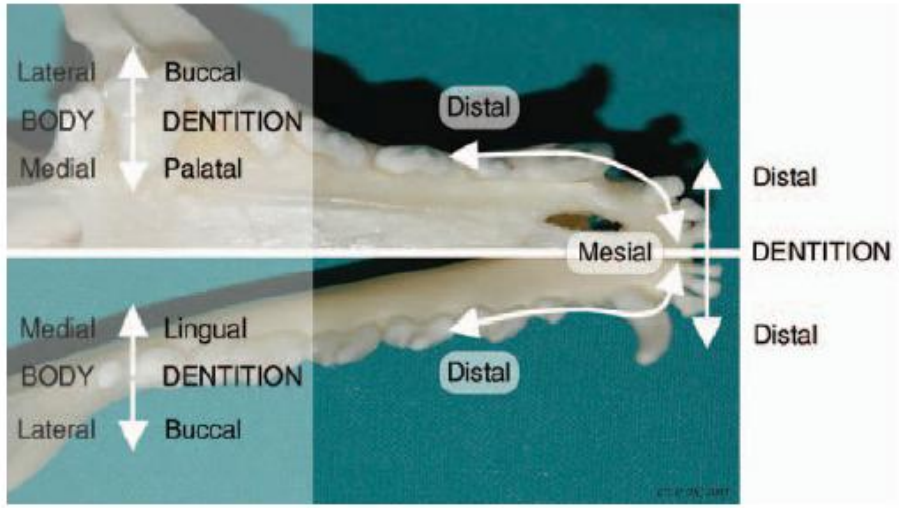
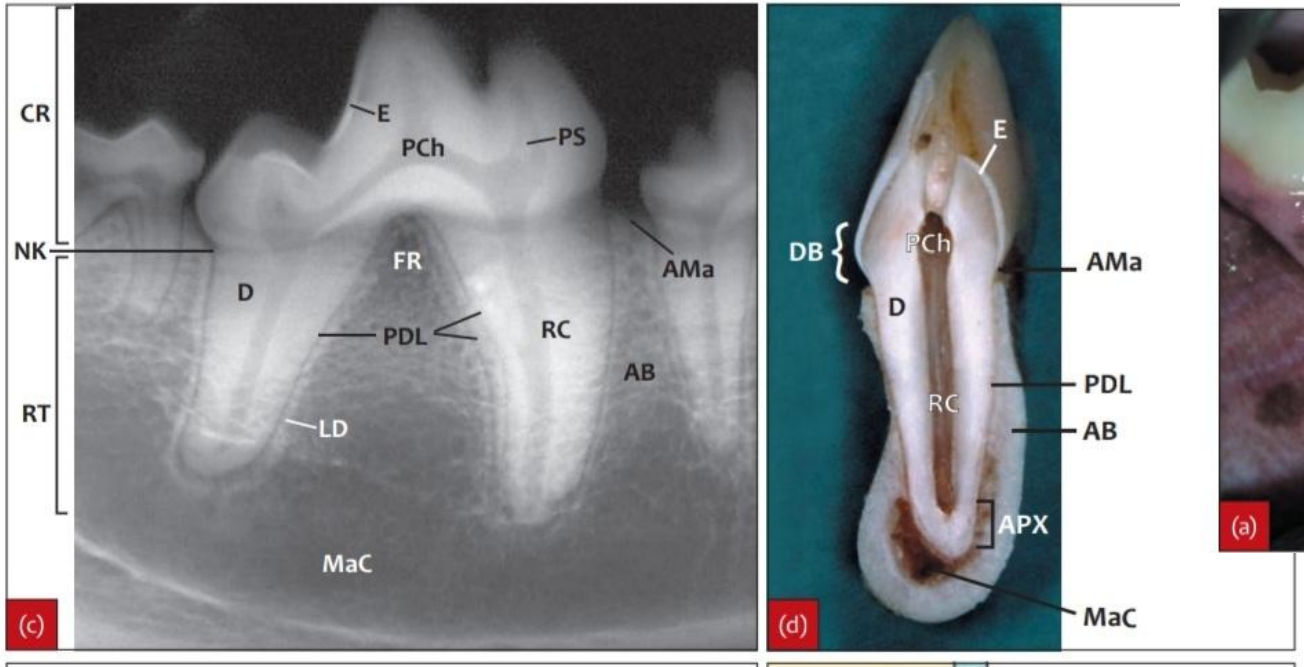
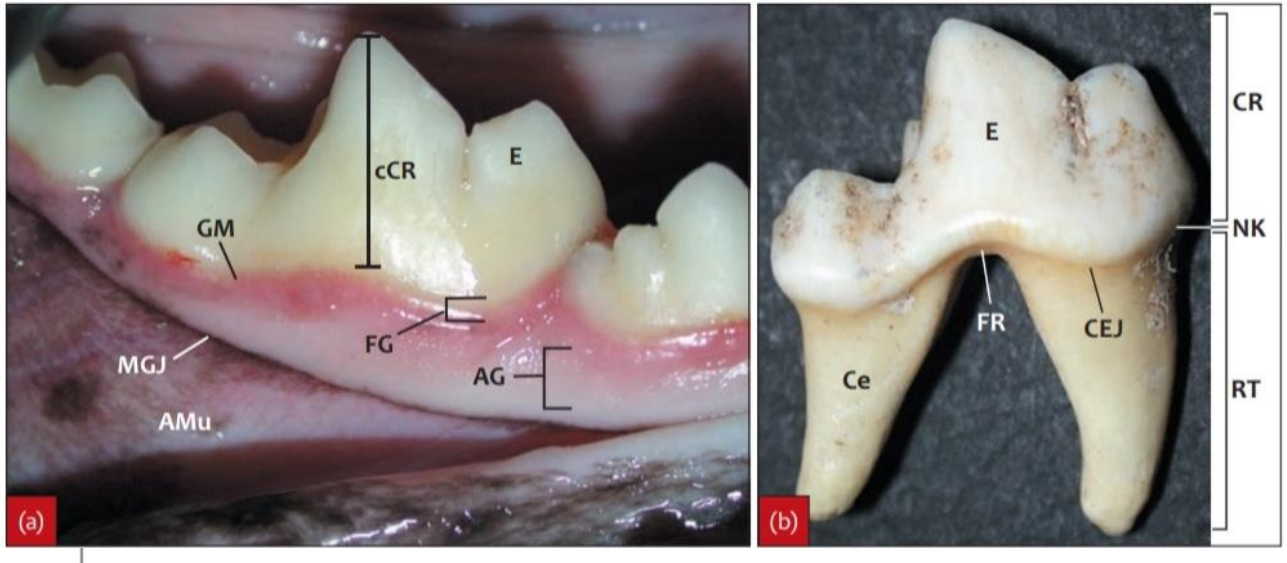


Figure 1.41 Directional anatomic terms for the mandible and maxilla and their dentition (occlusal view).

牙齒組織

- 牙冠crown&牙根root
- 牙釉質enamel、牙本質dentin、牙骨質cementum
- 牙髓腔pulp



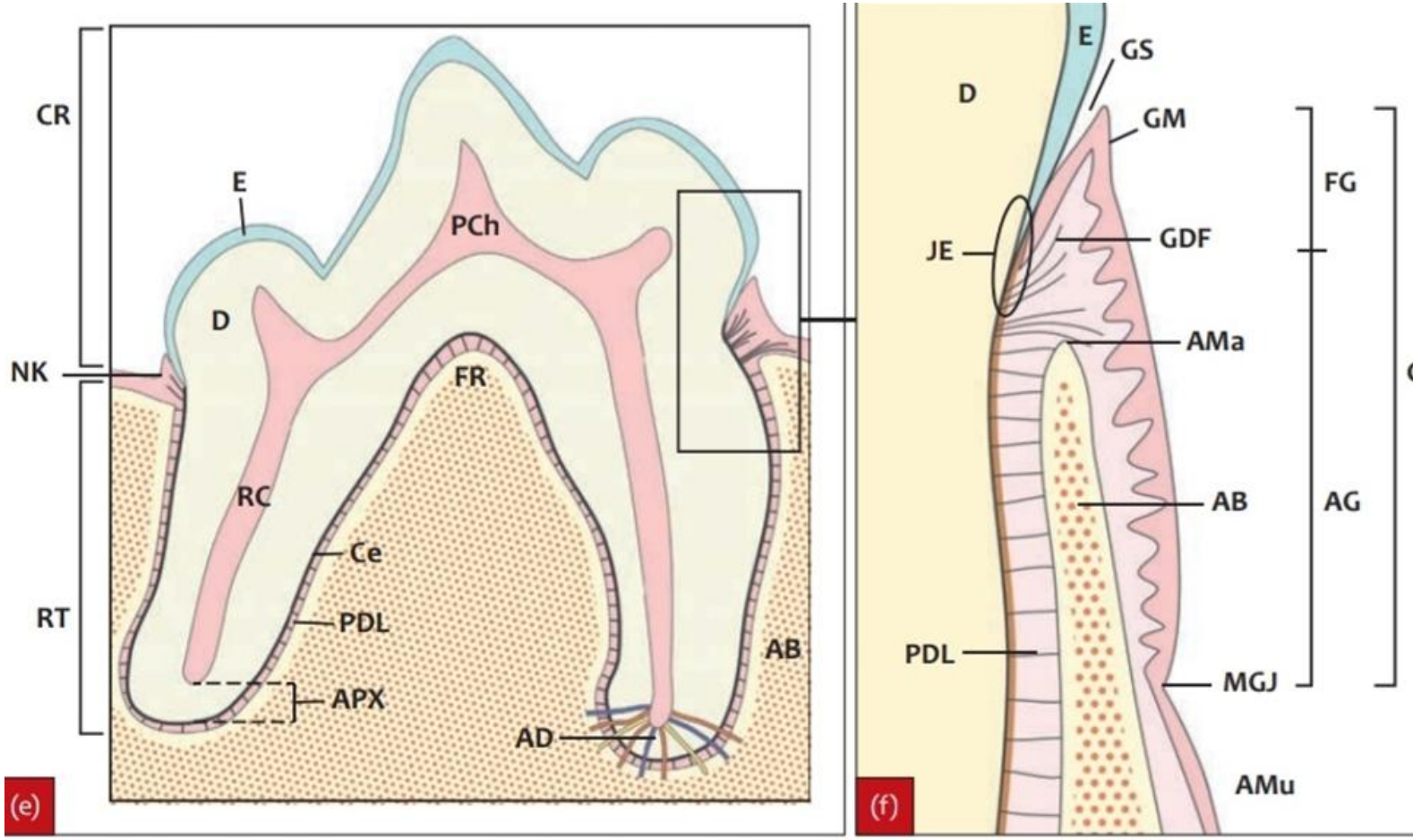


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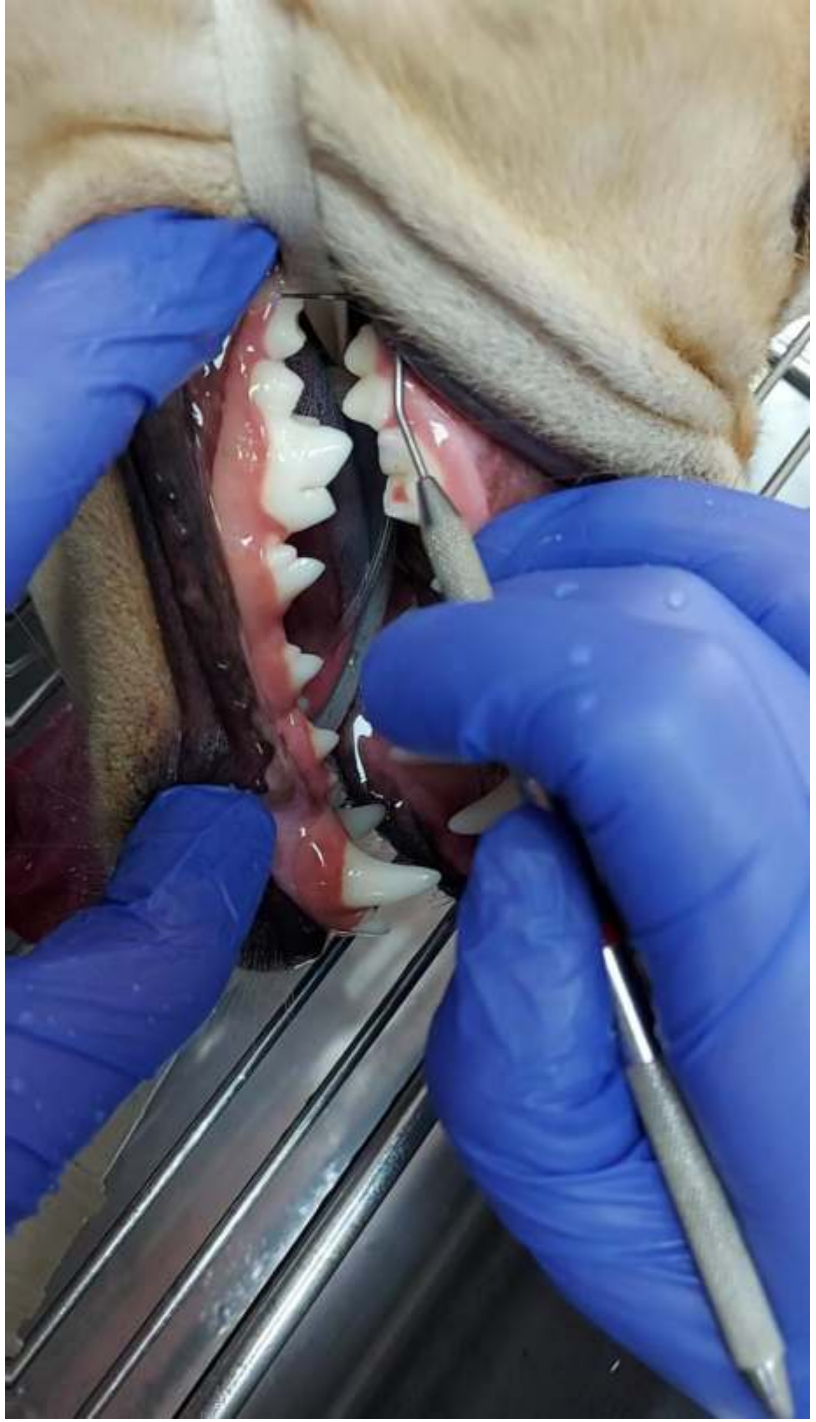
牙周組織

- 牙齦 :游離牙齦(FG)&
  附著牙齦(AG)
- 牙齦溝(GS): 貓小於0.5mm，
  狗小於3mm
- 齒槽骨(AB):容納牙根的骨頭
- 牙周韌帶(PDL):拉住牙根和齒槽骨



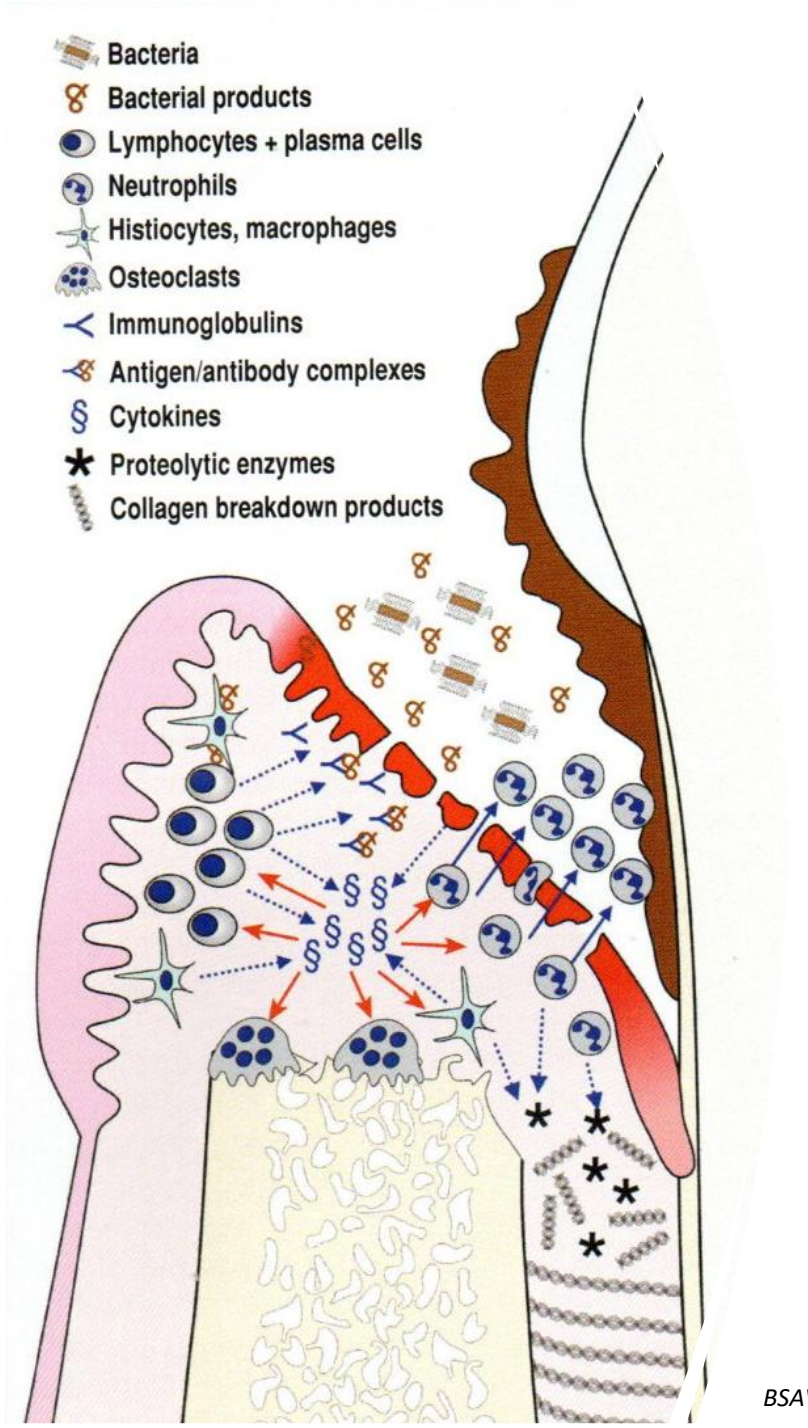
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犬牙齦溝





- 貓牙齦溝



牙周病的致病機制

- 牙菌斑累積

  牙齦邊緣造成牙齦炎

  牙齦下(牙齦溝或囊袋)造成牙周炎，骨頭流失，牙齒鬆動
- 菌血症

  影響其他器官，如心、腎、肝...等

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- 大魔王是細菌，不是結石

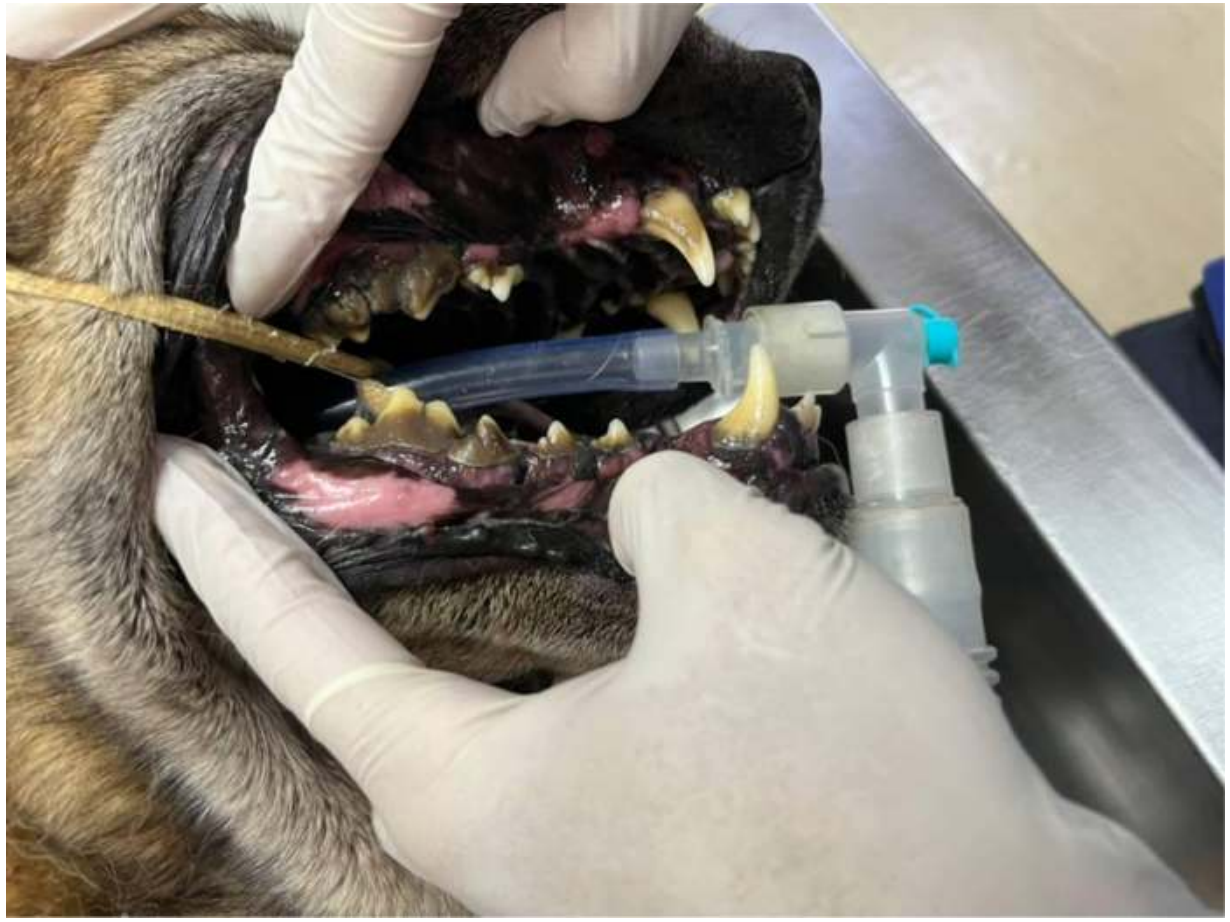
- 牙齦炎gingivitis
  牙齦紅腫流血
  可逆的
- 牙周炎periodontitis
  牙周組織被破壞
  進一步造成骨頭流失
  不可逆的

牙結石

礦物質加牙菌斑

略鹼性

提供粗糙表面讓更多牙菌斑附著



COHAT 第一步 門診麻醉前口腔檢查

- 1.門診麻醉前口腔檢查、理學檢查、血液檢查
- 2.麻醉計劃、IV catheter(for IV fluid and anesthetic drugs)
- 3.麻醉監控: 至少血氧、心跳、血壓、呼吸速率、體溫
- 4.檢查每顆牙齒、用牙周探針(periodontal probe)檢查囊袋、檢查牙齦是否發炎
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- 7.治療牙周病的牙齒或拔除
- 8.術後照顧與居家口腔保健

COHAT 第一步
門診&麻醉前檢查

口腔檢查
理學檢查
血液檢查



COHAT 第一步 門診&麻醉前檢查

- 清醒病患的口腔檢查

  安全第一，手不可放動物咬合面

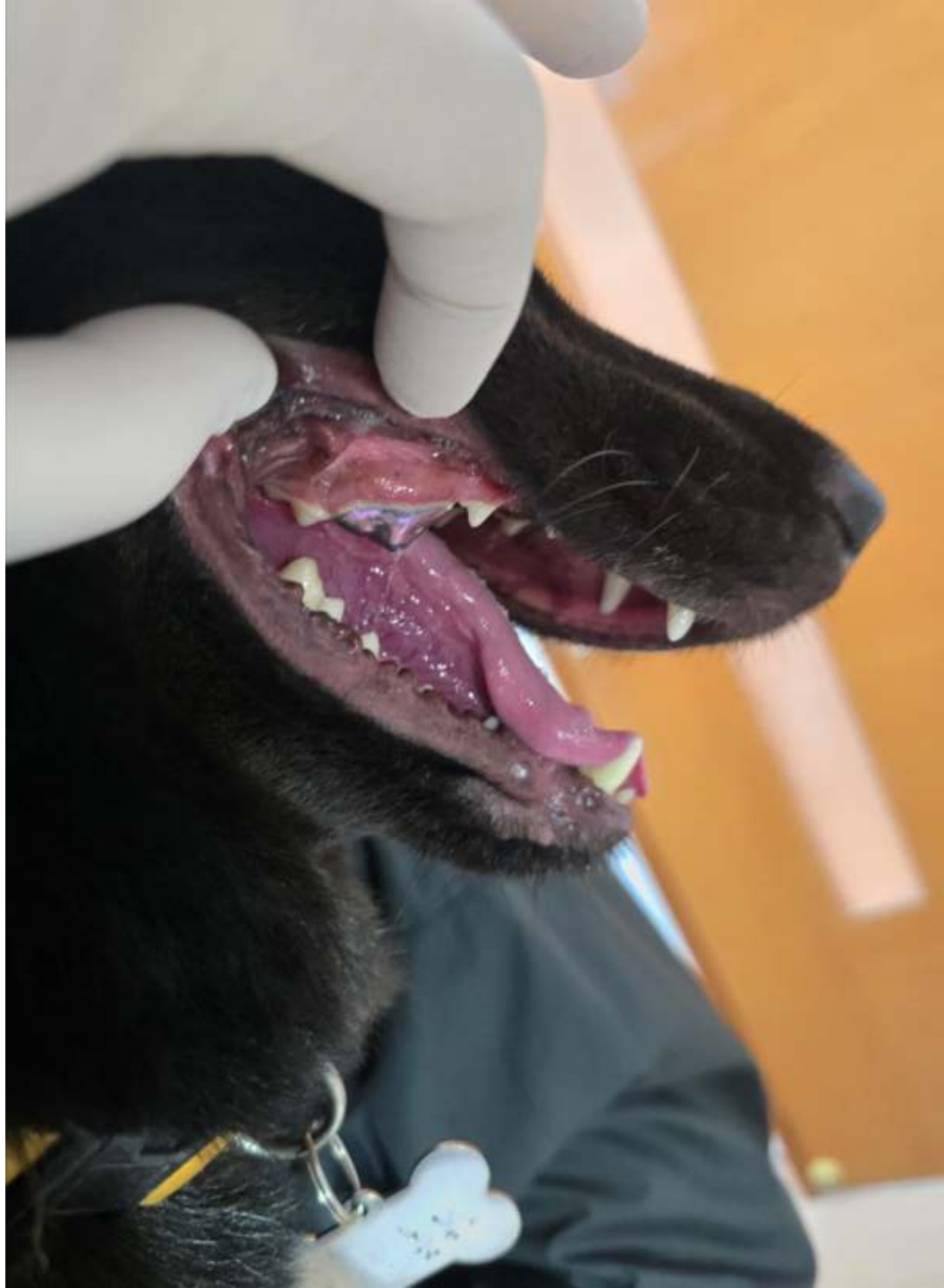
  良好的光源

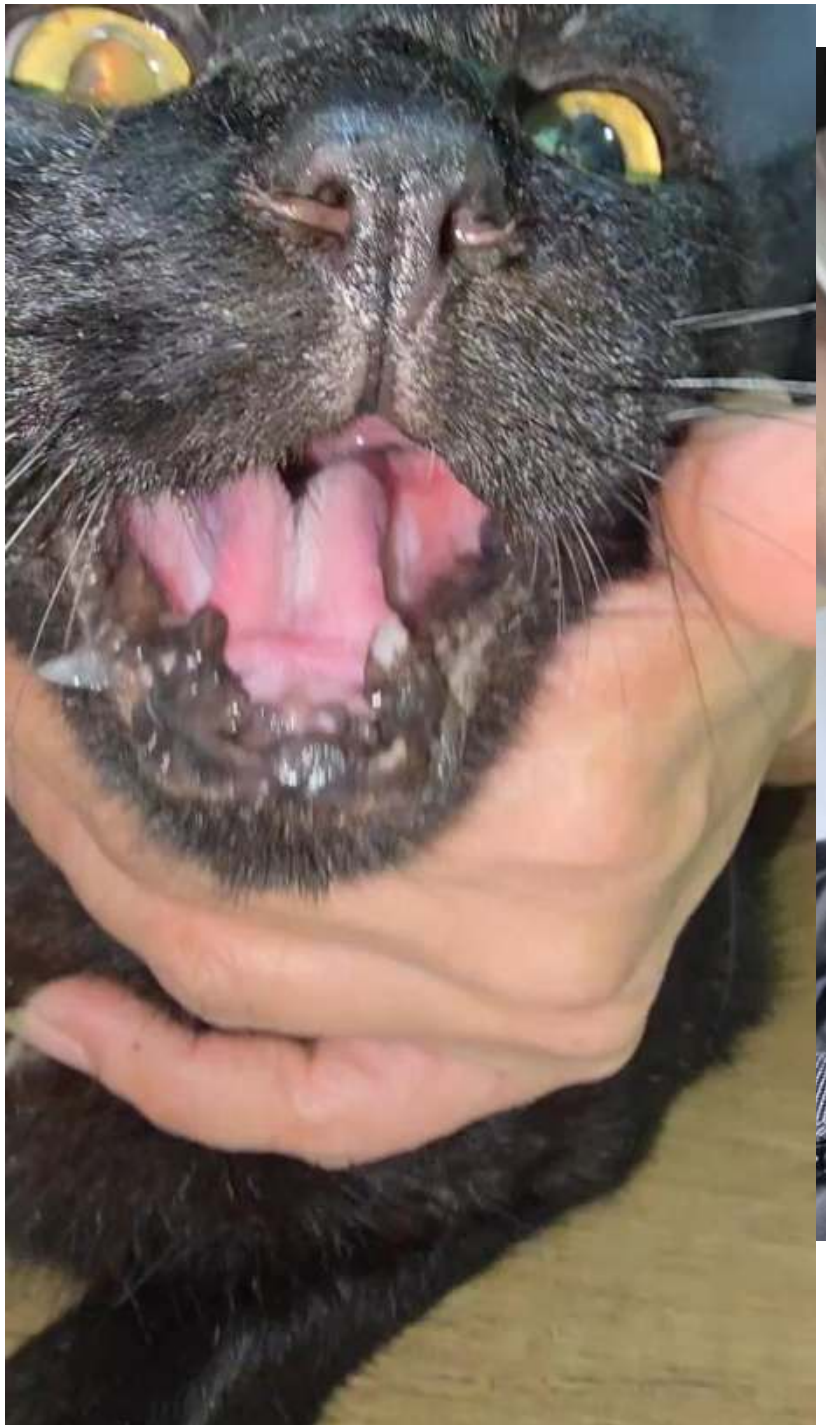
  直接接觸，記得戴手套

  間接可用棉棒、壓舌板、冰棒棍

  打開口腔

- 咬合、牙齒缺損、斷裂、牙結石覆蓋程度、牙齦退縮、牙齦炎、牙根暴露狀況、口腔黏膜狀況、有無口腔腫瘤.....











3.11 Demonstration of the lifting of the upper lip of a small crossbreed dog with the aid of a wooden tongue depressor.

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COHAT 第一步 門診&麻醉前檢查

麻醉前評估

理學檢查:必須

血液檢查:必須

X光檢查:視狀況而定

超音波檢查:視狀況而定

麻醉風險分級ASA

門診
理學檢查

- 外觀—左右是否對稱、步態、有無外傷
- 眼
- 耳
- 黏膜顏色
- 是否脫水
- 皮膚

- 淋巴結
- 呼吸系統
- 循環系統
- 消化系統
- 泌尿生殖系統
- 骨骼肌肉系統
- 神經系統

ASA 麻醉分級

- ASA Ⅰ: 健康正常病患
- ASAⅡ: 輕度系統性疾病,但無功能障礙
- ASAⅢ:中度至重度全身性疾病，且部分功能障礙
- ASAⅣ:重度全身性疾病，有功能障礙，且危及生命
- ASAⅤ:瀕死且如果不立即進行手術，預期無法存活

COHAT 第二步 麻醉前準備及計畫

- 1.**門診麻醉前口腔檢查、理學檢查、血液檢查**
- 2.**麻醉計畫、IV catheter(for IV fluid and anesthetic drugs)**
- 3.麻醉監控: 至少血氧、心跳、血壓、呼吸速率、體溫
- 4.檢查每顆牙齒、用牙周探針(periodontal probe)檢查囊袋、檢查牙齦是否發炎
- 5.移除牙齦上及牙齦下的牙菌斑和牙結石
- 6. 牙科X光檢查牙齦下的牙根與骨頭
- 7.治療牙周病的牙齒或拔除
- 8.術後照顧與居家口腔保健

COHAT 第二步 麻醉前準備及計畫

再次理學檢查，確認體重、體溫、心跳...等等

詢問空腹狀況

簽手術同意書

IV catheter

全身麻醉下呼吸道管理

- 氣管內插管是必要的
- 提供氧氣及氣體麻醉劑給病患，以達到良好而穩定的全身麻醉

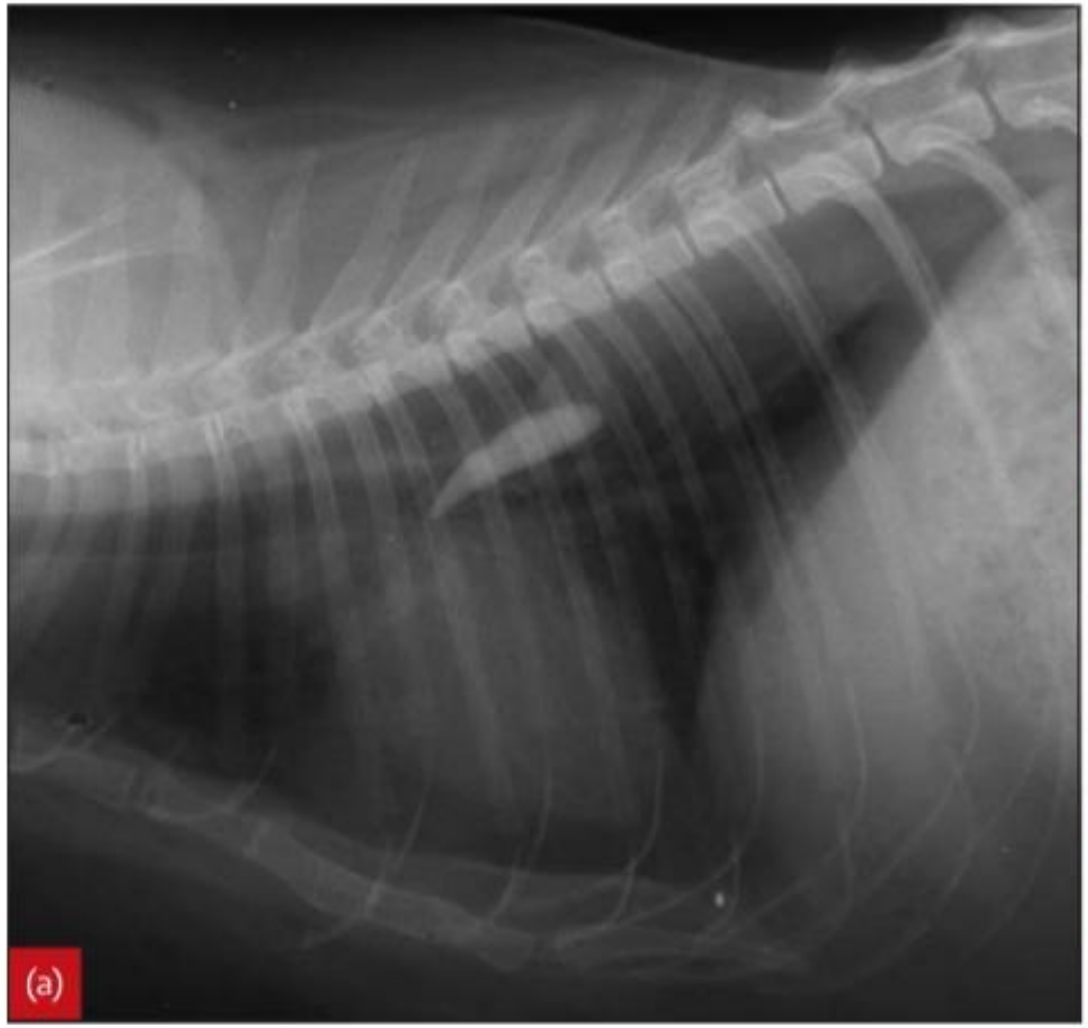
- 防止病患吸入

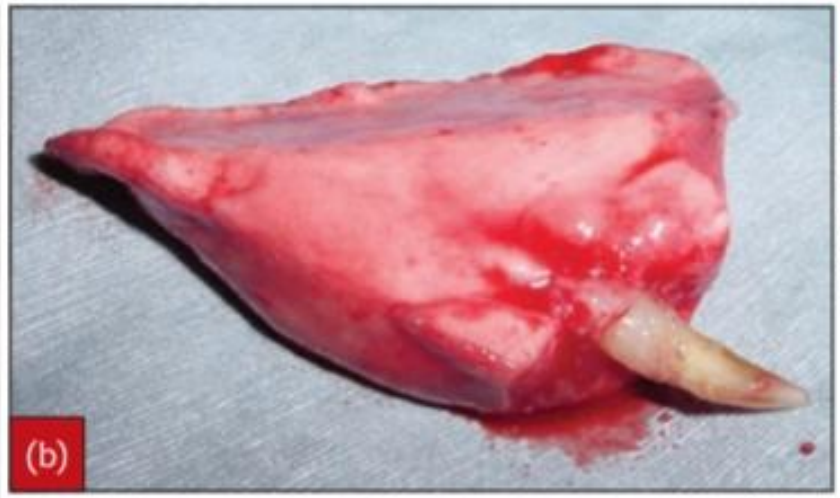
液體 如:冷卻水、血液、唾液、藥水(尤其根管治療)

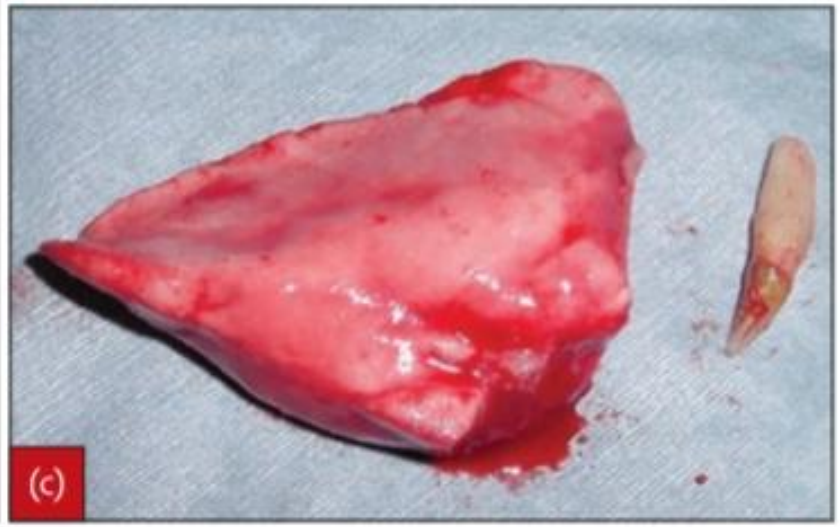
固體 如:牙結石、部分牙齒、拋光膏、器械...



氣管內插管很重要







13.30 (a–c) Accidental aspiration of a maxillary canine tooth in a cat, requiring lobectomy for its removal.
(© Dr Alexander M. Reiter)

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水很重要

- 冷卻:動力機器會產熱
- 清潔口腔:食物渣、毛髮、拋光膏、藥水、唾液
- 移除洗牙後的細菌、牙結石碎屑
- 冬天溫水減少體溫散失



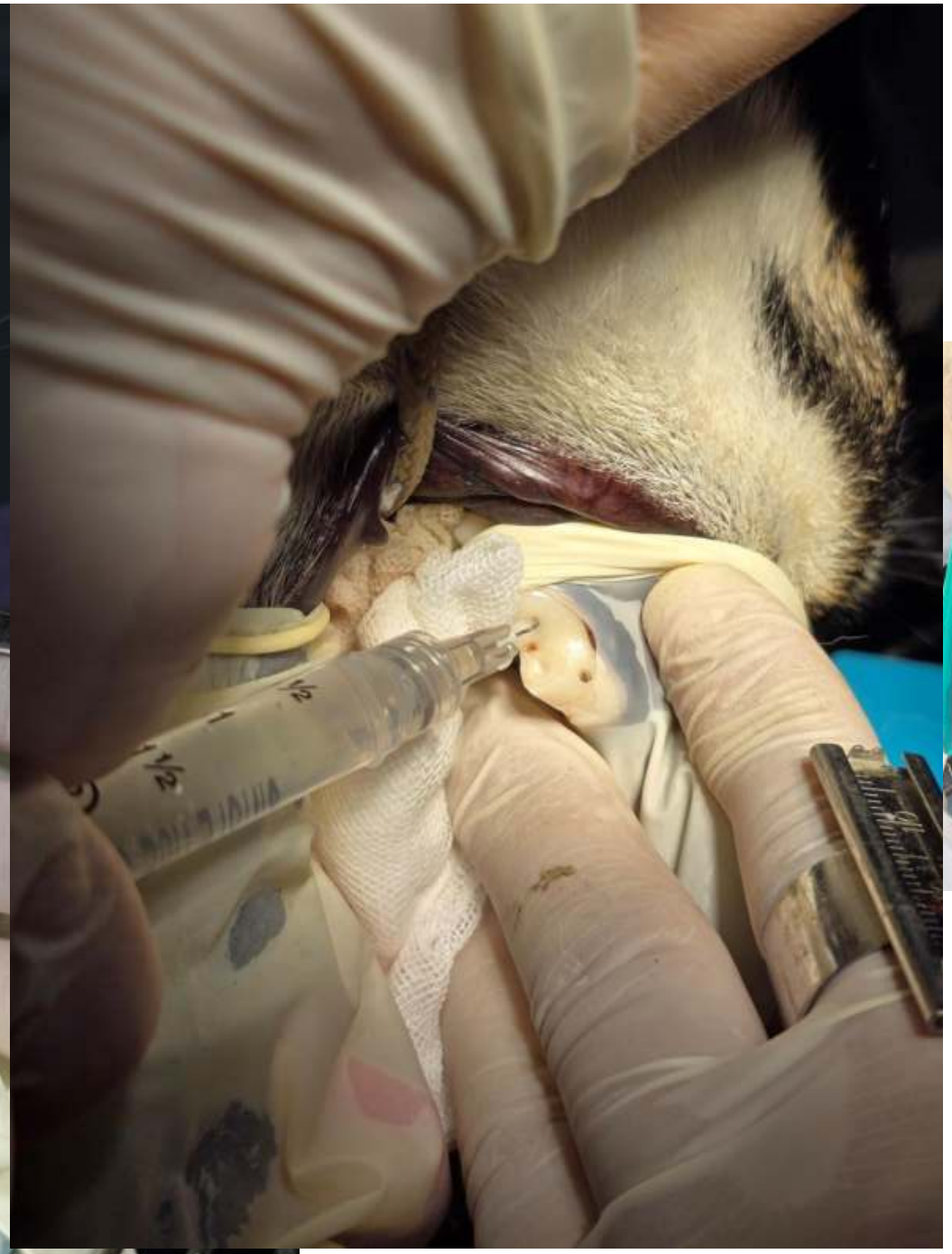
避免水進入呼吸道

- 氣管插管!!!
- 脖子墊高，吻部要低
- 操作時勿抬頭
- 口腔放棉球或紗布塊(術後記得一定要取出)
- 牙科治療時，唾液可能會不斷分泌，拔管時，清潔口腔的動作要重覆進行，確定喉頭乾淨，至麻醉後恢復吞嚥反射











COHAT 第三步 麻醉監控

- 1.**門診麻醉前口腔檢查、理學檢查、血液檢查**
- 2.麻醉計劃、IV catheter(for IV fluid and anesthetic drugs)
- 3.**麻醉監控: 至少血氧、心跳、血壓、呼吸速率、體溫**
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- 7.治療牙周病的牙齒或拔除
- 8.術後照顧與居家口腔保健

麻醉後注意事項

- 回家後神經反射恢復才可吃喝
  先從水開始喝,如果不會嗆到或吐，才可以吃東西，且當天都要少量多餐
- 口腔若有傷口或縫線，泥狀或太硬的食物皆不適合，兩周內避免玩具、潔牙骨、乾燥肉乾

疼痛管理

- 術前:

  止痛藥只能暫時舒緩，要解決問題需要牙周治療
- 術中疼痛控制:

  針劑: NSAIDs, 嗎啡類止痛

  神經阻斷

  輕柔的操作技術
- 術後:

  口服 NSAIDs, 嗎啡類止痛, gabapentin,冰敷

神經阻斷

表 5. 經常用於動物麻醉和疼痛管理的局部麻醉劑

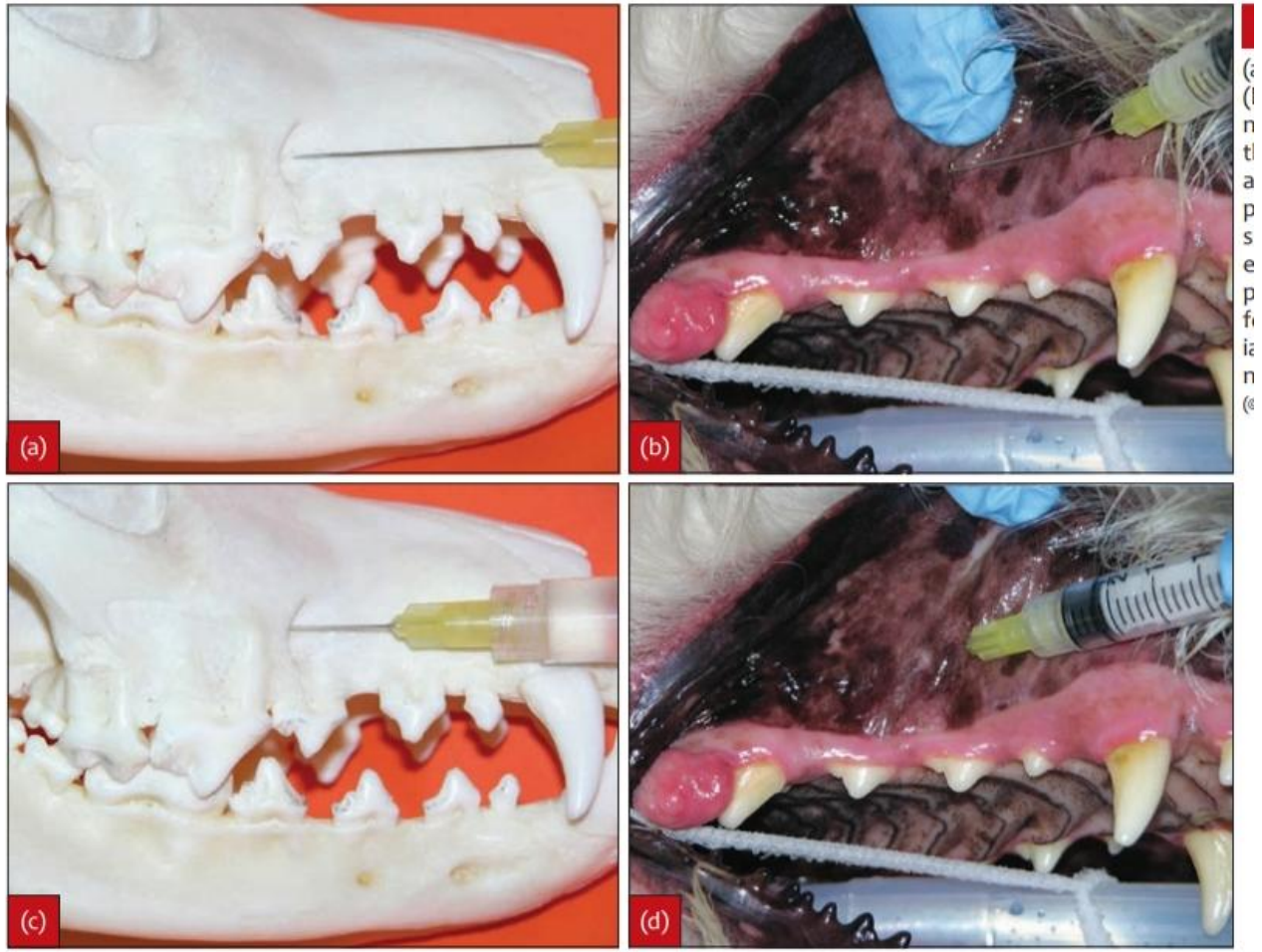
局部麻醉劑 *	起效時間（分鐘）	常用濃度（%）	阻斷時間（小時）	相對效力（lidocaine = 1）	建議最高劑量（mg/kg）
Lidocaine	5–15	1, 2	1–2	1	10（犬） 5（貓）
Mepivacaine	5–15	1, 2	1.5–2.5	1	4（犬） 2（貓）
Bupivacaine	10–20	0.25, 0.5, 0.75	4–6	4	4（犬） 2（貓）
Ropivacaine	10–20	0.5, 0.75	3–5	3	3（犬） 1.5（貓）
Levobupivacaine	10–20	0.5, 0.75	4–6	4	2（犬） 1（貓）

* 注射量（0.25 – 1 mL）根據病患的體型大小、解剖結構和體重而有所差異。可以根據手術時間長短，術後止痛必要性，重複注射麻醉阻斷劑，並使用低於最高建議劑量的注射量（請參見正文）

較常使用lidocaine 和bupivacaine ，每個位置的用量通常犬:0.2-1 mL； 貓: 0.1- 0.2 mL

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神經阻斷Infraorbital foramen



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神經阻斷mandibular foramen inferior alveolar nerve block





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神經阻斷Middle mental foramen





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COHAT 第四步 麻醉下 軟&硬組織檢查

- 1.門診麻醉前口腔檢查、理學檢查、血液檢查
- 2.麻醉計劃、IV catheter(for IV fluid and anesthetic drugs)
- 3.麻醉監控: 至少血氧、心跳、血壓、呼吸速率、體溫
- 4.**檢查每顆牙齒、用牙周探針(periodontal probe)檢查囊袋、檢查牙齦是否發炎**
- 5.移除牙齦上及牙齦下的牙菌斑和牙結石
- 6. 牙科X光檢查牙齦下的牙根與骨頭
- 7.治療牙周病的牙齒或拔除
- 8.術後照顧與居家口腔保健

COHAT 第四步 麻醉下 軟&硬組織檢查

- 麻醉誘導後，應仔細評估軟組織，包括舌頭、牙齦、黏膜、口咽與扁桃腺區域。
- 接下來，應該整體 和個別評估硬組織（包括上顎與下顎）和齒列，包括任何牙齒缺失、歪斜或斷裂的情況。
- 接續評估牙周的健康狀況，包括探測深度、牙齦萎縮與增生、活動性、根叉是否受影響，以及其他口腔病變。

全身麻醉下口腔的檢查

- 1. 檢查口咽：建議在進行氣管插管並填塞喉嚨之前
- 2. 拍攝術前照片：
- 3. 減少細菌數量：使用0.12% 的 chlorhexidine 漱口液沖洗口腔。
- 4. 評估並確定齒列：乳牙、恆牙、或混合存在，有無缺牙。
- 5. 評估軟組織：應檢查整個口腔，包含口腔黏膜（顏色、濕潤度、腫脹）、嘴唇與臉頰、腭、舌頭與舌下組織是否出現異常以及口腔團塊。
- 6. 專業洗牙與拋光
- 7. 使用探針進行牙科與牙周檢查
- 8. 拍攝牙科放射學影像：
- 9. 牙周疾病分級並記錄：合併臨床表現和牙科 X 光影像進行分級
- 10. 牙周治療、拔牙或手術
- 11. 術後 X 光影像及照片

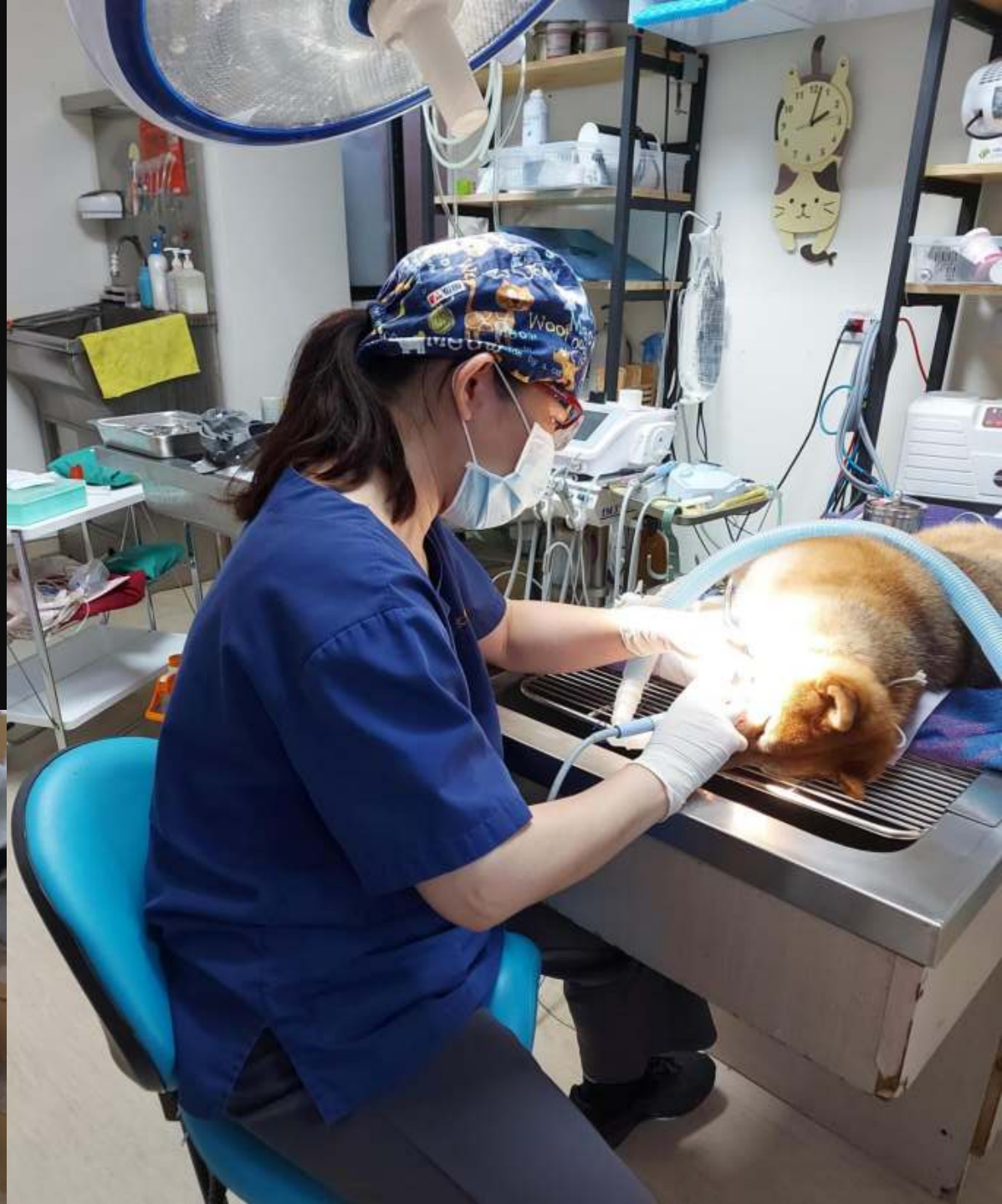
牙科治療獨立的空間



人的準備

- 1. 良好的光源
- 2. 放大設備（放大眼鏡或放大鏡）
- 3. 個人防護裝備（眼鏡、面罩與檢診手套）。
- 4.舒適座椅

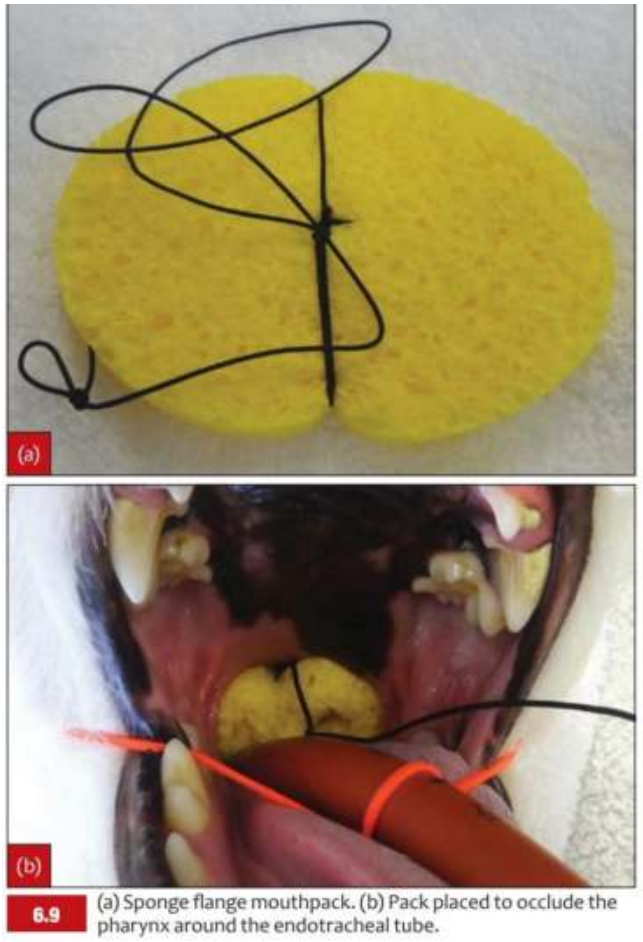




動物防護的準備

- 動物眼睛的防護
- 減少身體沾濕
- 避免水或物體進入呼吸道
- 舌頭與軟組織的防護
- 避免氣管內插管移動或破壞

動物防護的準備



6.9 (a) Sponge flange mouthpack. (b) Pack placed to occlude the pharynx around the endotracheal tube.

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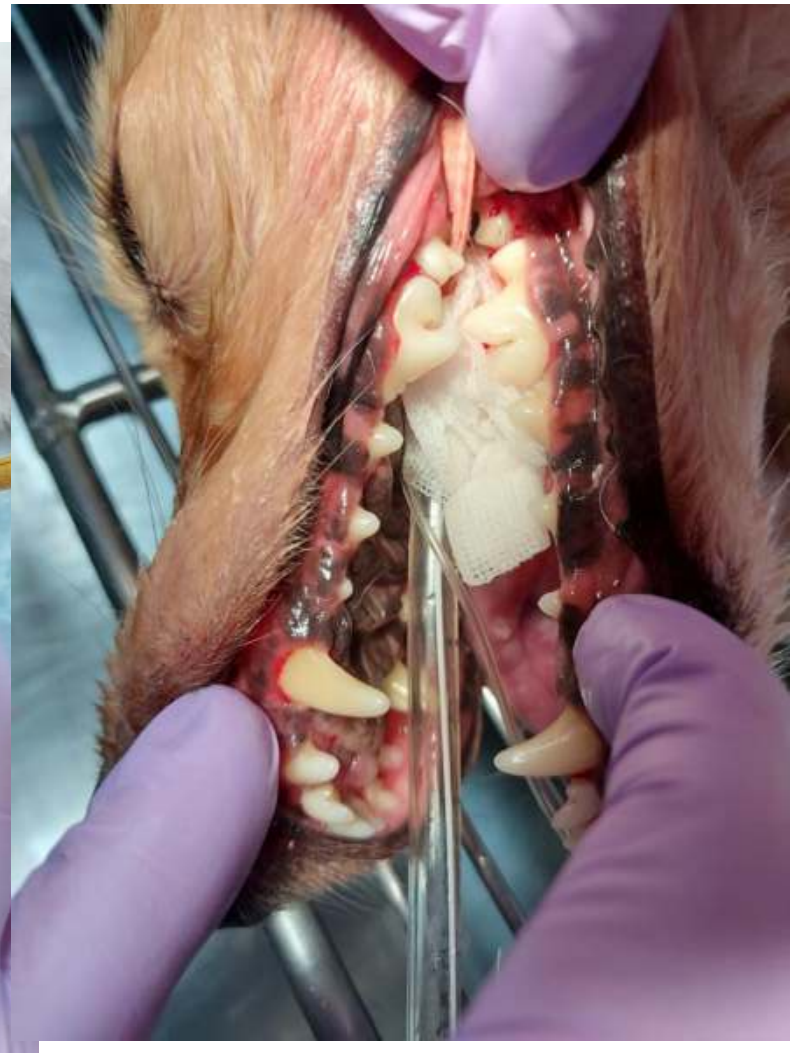




舌頭與軟組織的防護







舌頭與軟組織的防護



可利用的開嘴器





禁用的開嘴器



牙周檢查

- 牙結石
- 牙菌斑
- 牙齦炎與牙齦出血
- 牙齦退縮
- 牙齦腫大或增生
- 牙周囊袋
- 牙根分叉侵犯程度
- 牙齒鬆動程度

探查硬組織 (explorer)







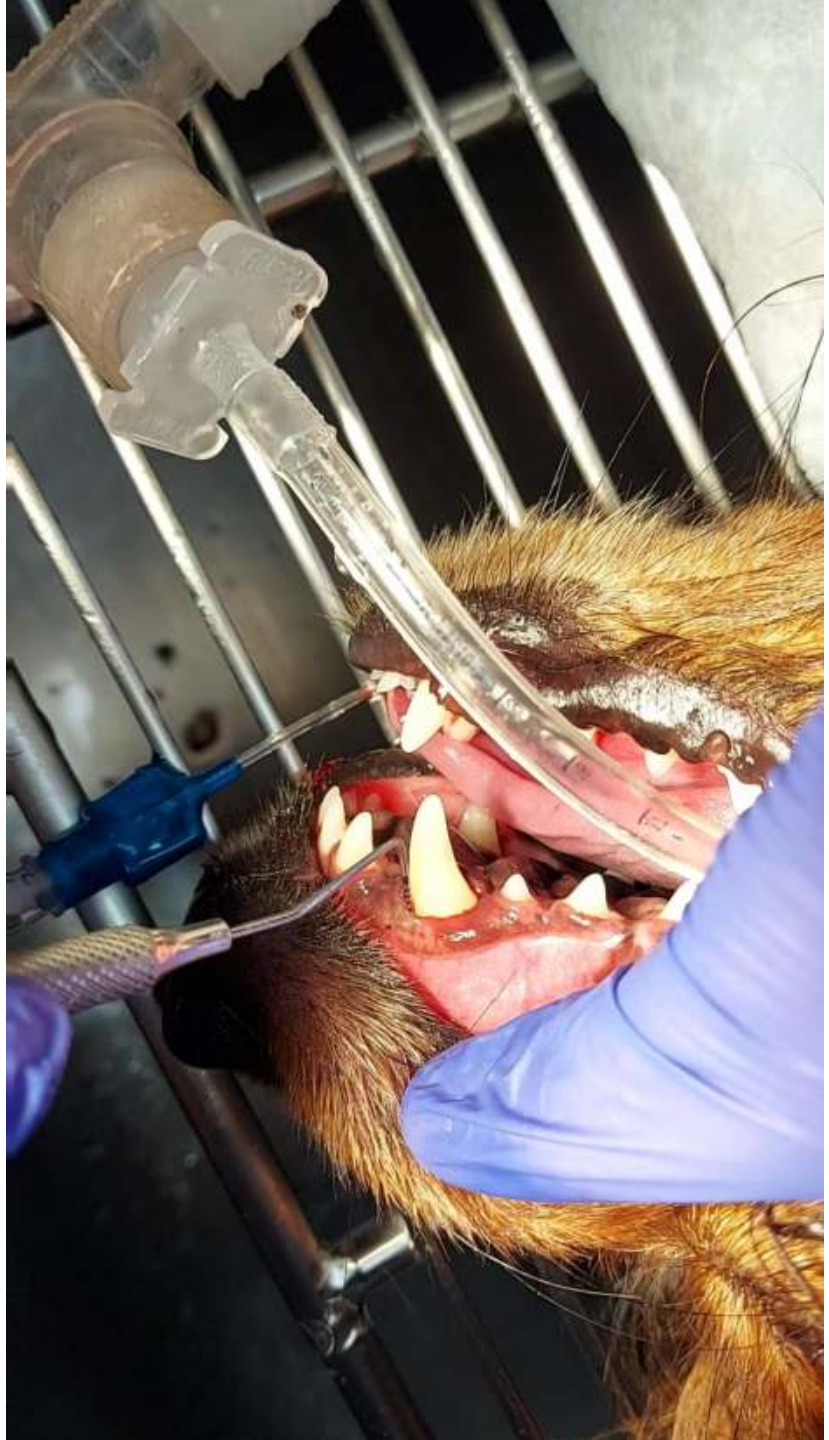
牙周探査
(probe；深度計)







牙周囊袋的檢測



口鼻廔管



牙結石程度 dental calculi

- 0 級 C0 : 無牙結石
- 1 級 C1 : 牙齦上的結石 佔表面1/3 以下
- 2 級 C2 : 牙齦上的結石 佔表面1/3 以上，不超過2/3，或牙齦下有部分結石
- 3 級 C3 : 牙齦上的結石 佔表面2/3 以上，或牙齦下有連續厚實結石



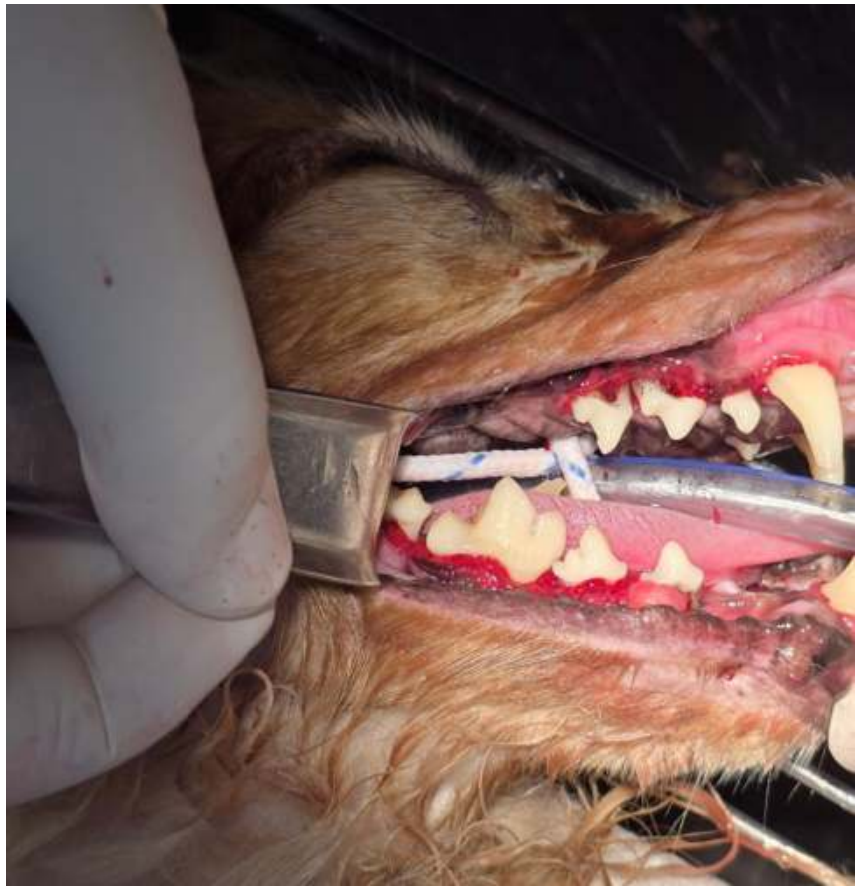
牙齦炎程度gingivitis

- 0 級 G0 : 正常牙齦
- 1 級 G1 : 輕微發炎: 顏色稍有變化與腫脹，探查時”未”出血
- 2 級 G2 : 中度發炎: 發紅與腫脹，探查時”有”出血
- 3 級 G3 : 嚴重發炎 :明顯紅腫與潰瘍，有自發性出血的傾向



牙根分叉侵犯程度 Furcation exposure

- 0 級 : 正常
- 1級 F1 : 牙周探針伸入未超過一半
- 2級 F2 : 牙周探針伸入超過一半，但未貫穿
- 3級 F3 : 牙周探針伸入會貫穿



牙根分岔暴露程度 Furcation exposure







牙鬆動程度Mobility

- 0級 M0 : 生理性移動少於 0.2mm
- 1級 M1 : 水平移動超過0.2mm，未超過0.5mm
- 2級 M2 :水平移動超過0.5mm，未超過1mm
- 3級 M3 :水平移動超過1 mm，或可上下移動



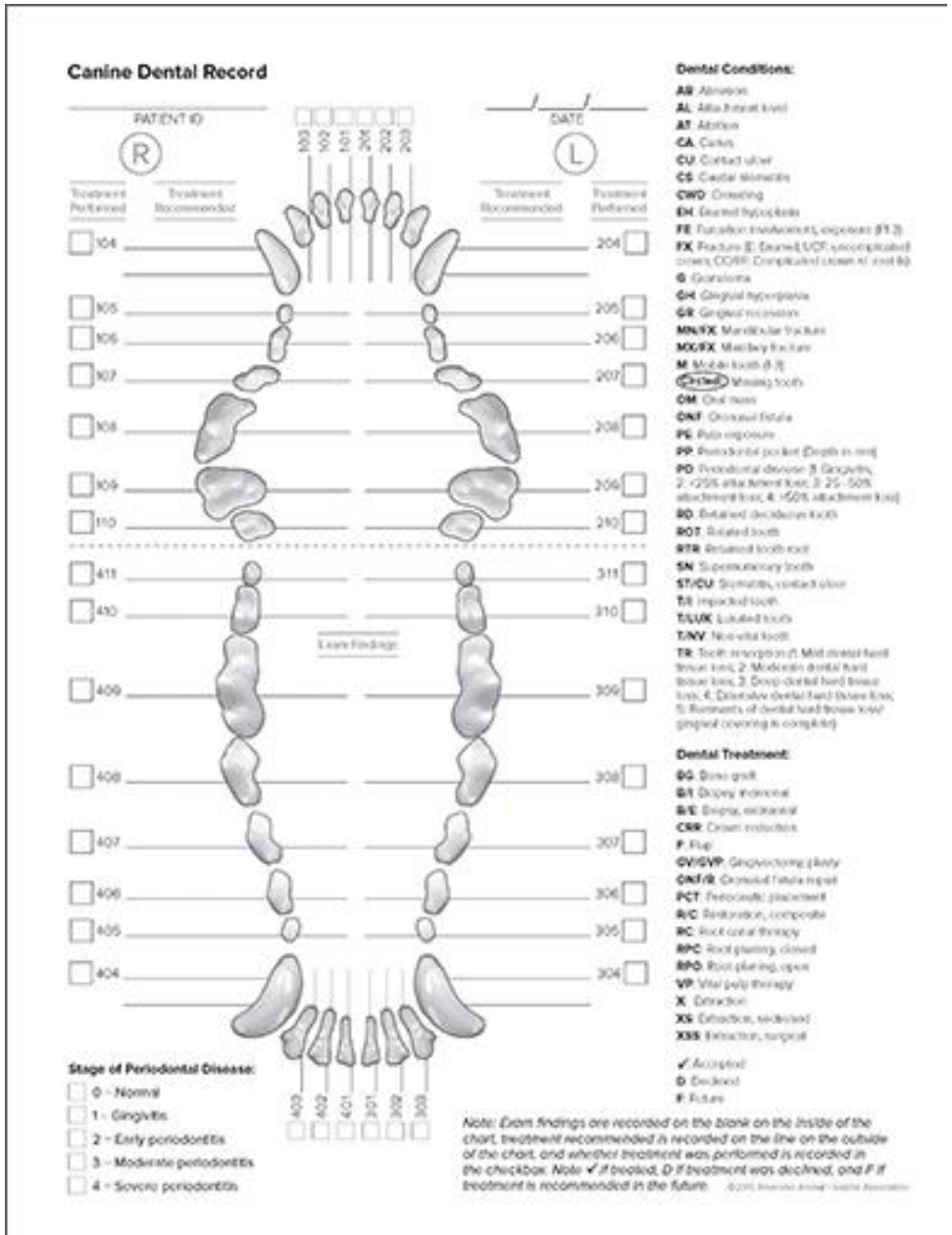
牙周病分級

- 0級 (PD0) ： 正常
- 1級 (PD1) ：只有牙齦炎
- 2級 (PD2) ： 早期牙周炎,F1,
  <25 % 附著流失
- 3級 (PD3) : 中度牙周炎,F2,
  25~50 % 附著流失
- 4級 (PD4) : 末期牙周炎,F3,
  >50 % 附著流失



FIG 158. Determine the stage of periodontal disease with clinical findings and dental radiographs for each tooth.

Dental Chart (Dog)
From American Animal Hospital Association (AAHA)



Canine Dental Record

PATIENT ID

DATE

R

L

Treatment Performed

Treatment Recommended

Treatment Recommended

Treatment Performed

104 105 106 107 108 109 110 411 410 409 408 407 406 405 404

204 205 206 207 208 209 210 311 310 309 308 307 306 305 304

103 102 101 201 202 203

403 402 401 301 302 303

Exam Findings

Stage of Periodontal Disease:

- 0 - Normal
- 1 - Gingivitis
- 2 - Early periodontitis
- 3 - Moderate periodontitis
- 4 - Severe periodontitis

Dental Conditions:

AB: Abrasion
AL: Attachment level
AT: Attrition
CA: Caries
CU: Contact ulcer
CS: Crestal resorption
CWD: Crowding
EH: Enamel hypoplasia
FE: Furcation involvement, exposure (F1-3)
FX: Fracture (E: Enamel; UCF: uncomplicated crown; CCRF: Complicated crown +/- root fx)
G: Granuloma
GH: Gingival hyperplasia
GR: Gingival recession
MN/FX: Mandibular fracture
MX/FX: Maxillary fracture
M: Missing tooth (# 3)
OM: Oral mass
ONF: Oronasal fistula
PE: Pulp exposure
PP: Periodontal pocket (Depth in mm)
PD: Periodontal disease (1: Gingivitis; 2: <25% attachment loss; 3: 25 - 50% attachment loss; 4: >50% attachment loss)
RD: Retained deciduous tooth
ROT: Rotated tooth
RTR: Retained tooth root
SN: Supernumerary tooth
ST/CU: Stomatitis, contact ulcer
TI: Impacted tooth
T/LUX: Luxated tooth
T/NV: Non-vital tooth
TR: Tooth resorption (1: Mild dental hard tissue loss; 2: Moderate dental hard tissue loss; 3: Deep dental hard tissue loss; 4: Extensive dental hard tissue loss; 5: Remnants of dental hard tissue lost/gingival covering is complete)

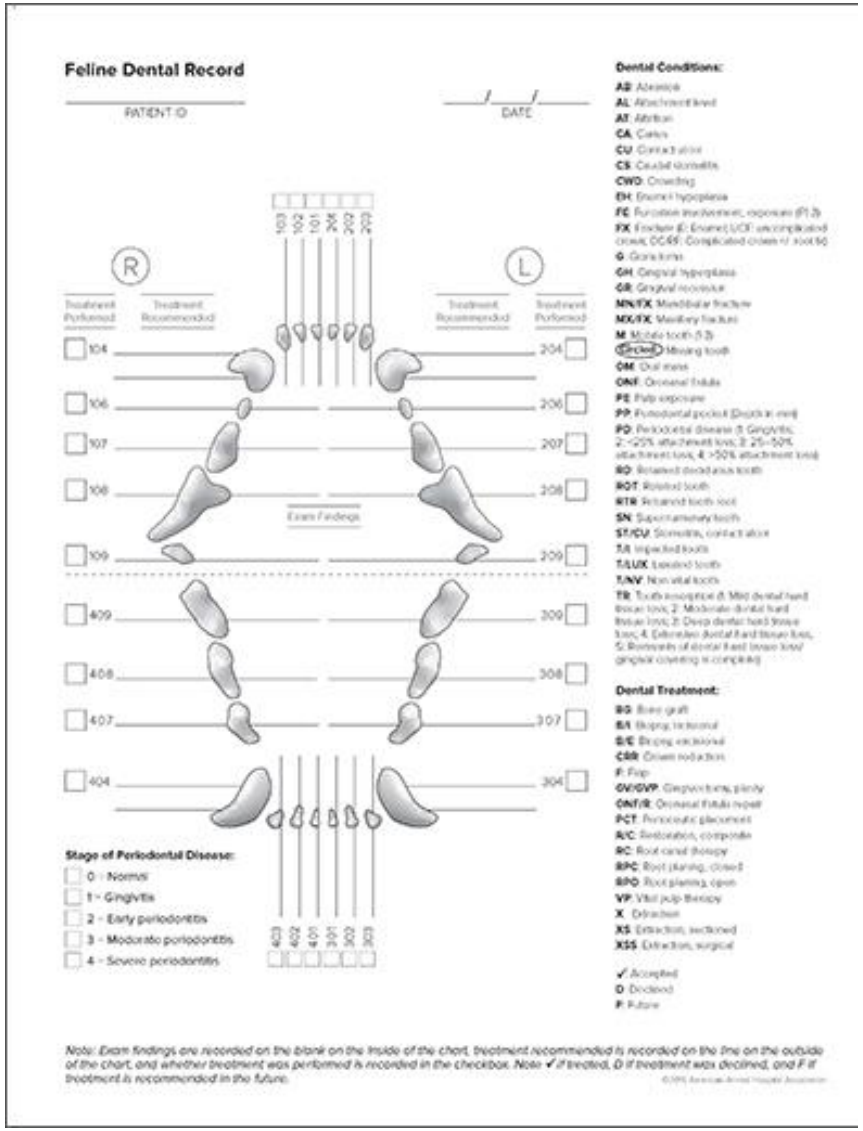
Dental Treatment:

BG: Bone graft
BI: Biopsy, incisional
BE: Biopsy, excisional
CRR: Crown reduction
F: Flap
GV/GVP: Gingivectomy (Ablate)
ONF/R: Oronasal fistula repair
PCT: Periosteal placement
R/C: Restoration, composite
RC: Root canal therapy
RPC: Root planing, closed
RPO: Root planing, open
VP: Vital pulp therapy
X: Extraction
XS: Extraction, sectioned
XSS: Extraction, surgical

✓ Accepted
D Declined
F Future

Note: Exam findings are recorded on the blank on the inside of the chart, treatment recommended is recorded on the line on the outside of the chart, and whether treatment was performed is recorded in the check box. Note ✓ if treated, D if treatment was declined, and F if treatment is recommended in the future.

Dental Chart (Cat)
From American Animal Hospital Association (AAHA)



Feline Dental Record

PATIENT ID

DATE

Dental Conditions:

AB: Abscess
AL: Attachment loss
AT: Attrition
CA: Caries
CU: Contact ulcer
CS: Caudal stomatitis
CWD: Crowding
EH: Enamel hypoplasia
FC: Furcation involvement, exposure (F1,2)
FX: Fracture (E: Enamel; UCE: uncomplicated crown; CCE: Complicated crown +/- root fx)
G: Gingivitis
GH: Gingival hyperplasia
GR: Gingival recession
MN/FX: Mandibular fracture
MX/FX: Maxillary fracture
M: Missing tooth (1,2)
OM: Oral mass
ONF: Oronasal fistula
PE: Pulp exposure
PP: Periodontal pocket (Depth in mm)
PD: Periodontal disease (1: Gingivitis; 2: <25% attachment loss; 3: 25–50% attachment loss; 4: >50% attachment loss)
RD: Retained deciduous tooth
ROT: Rotated tooth
RTR: Retained tooth root
SN: Supernumerary tooth
ST/CU: Stomatitis, contact ulcer
TA: Impacted tooth
TJ/LUX: Luxated tooth
TJ/NV: Non vital tooth
TR: Tooth resorption (1: Mild dental hard tissue loss; 2: Moderate dental hard tissue loss; 3: Deep dentin hard tissue loss; 4: Extensive dental hard tissue loss, 5: Remnants of dental hard tissue loss/gingival covering is complete)

Dental Treatment:

BG: Bone graft
BI/I: Biopsy, incisional
BI/E: Biopsy, excisional
CBR: Crown reduction
F: Flap
GV/GVP: Gingivectomy, gingivoplasty
ONF/R: Oronasal fistula repair
PCT: Periodontal placement
R/C: Restoration, composite
RC: Root canal therapy
RPC: Root planing, closed
RPO: Root planing, open
VP: Vital pulp therapy
X: Extraction
XS: Extraction, sectioned
XSS: Extraction, surgical

✓ Accepted
D: Declined
F: Future

Exam Findings

Stage of Periodontal Disease:

0 - Normal
1 - Gingivitis
2 - Early periodontitis
3 - Moderate periodontitis
4 - Severe periodontitis

Note: Exam findings are recorded on the blank on the inside of the chart, treatment recommended is recorded on the line on the outside of the chart, and whether treatment was performed is recorded in the checkbox. Note ✓ if treated, D if treatment was declined, and F if treatment is recommended in the future.

COHAT 第五步 移除牙菌斑和牙結石

- 1.門診麻醉前口腔檢查、理學檢查、血液檢查
- 2.麻醉計劃、IV catheter(for IV fluid and anesthetic drugs)
- 3.麻醉監控: 至少血氧、心跳、血壓、呼吸速率、體溫
- 4.檢查每顆牙齒、用牙周探針(periodontal probe)檢查囊袋、檢查牙齦是否發炎
- 5.移除牙齦上及牙齦下的牙菌斑和牙結石
- 6. 牙科X光檢查牙齦下的牙根與骨頭
- 7.治療牙周病的牙齒或拔除
- 8.術後照顧與居家口腔保健

COHAT 第五步 移除的牙菌斑和牙結石

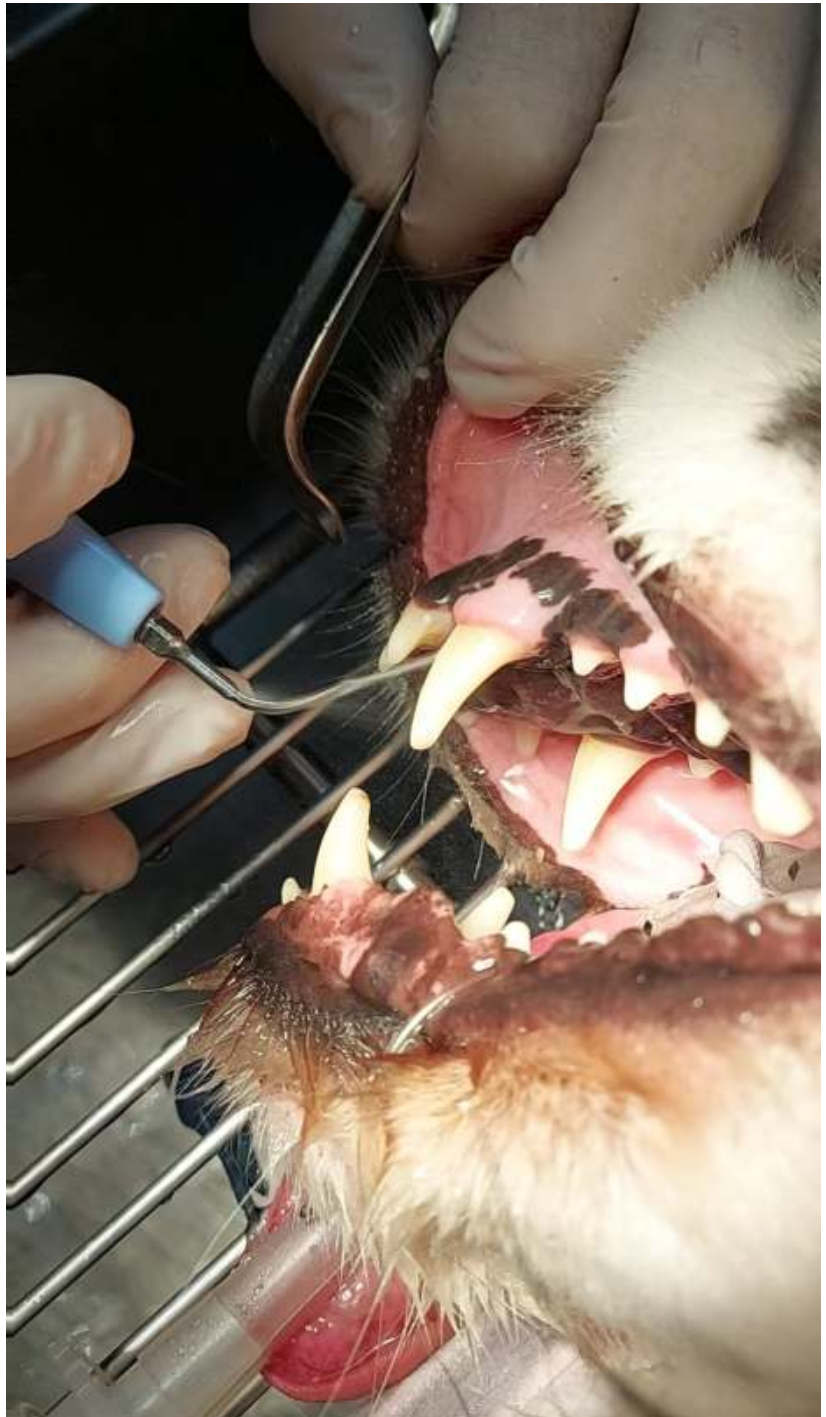
- 包括牙齦上及牙齦下
- 動力洗牙機
- 拋光
- 手持式器械

磁伸式 VS 壓電式



7.1 (a) Magnetostrictive scaler. (b) Metal stack inserts with





水噴出的位置很重要

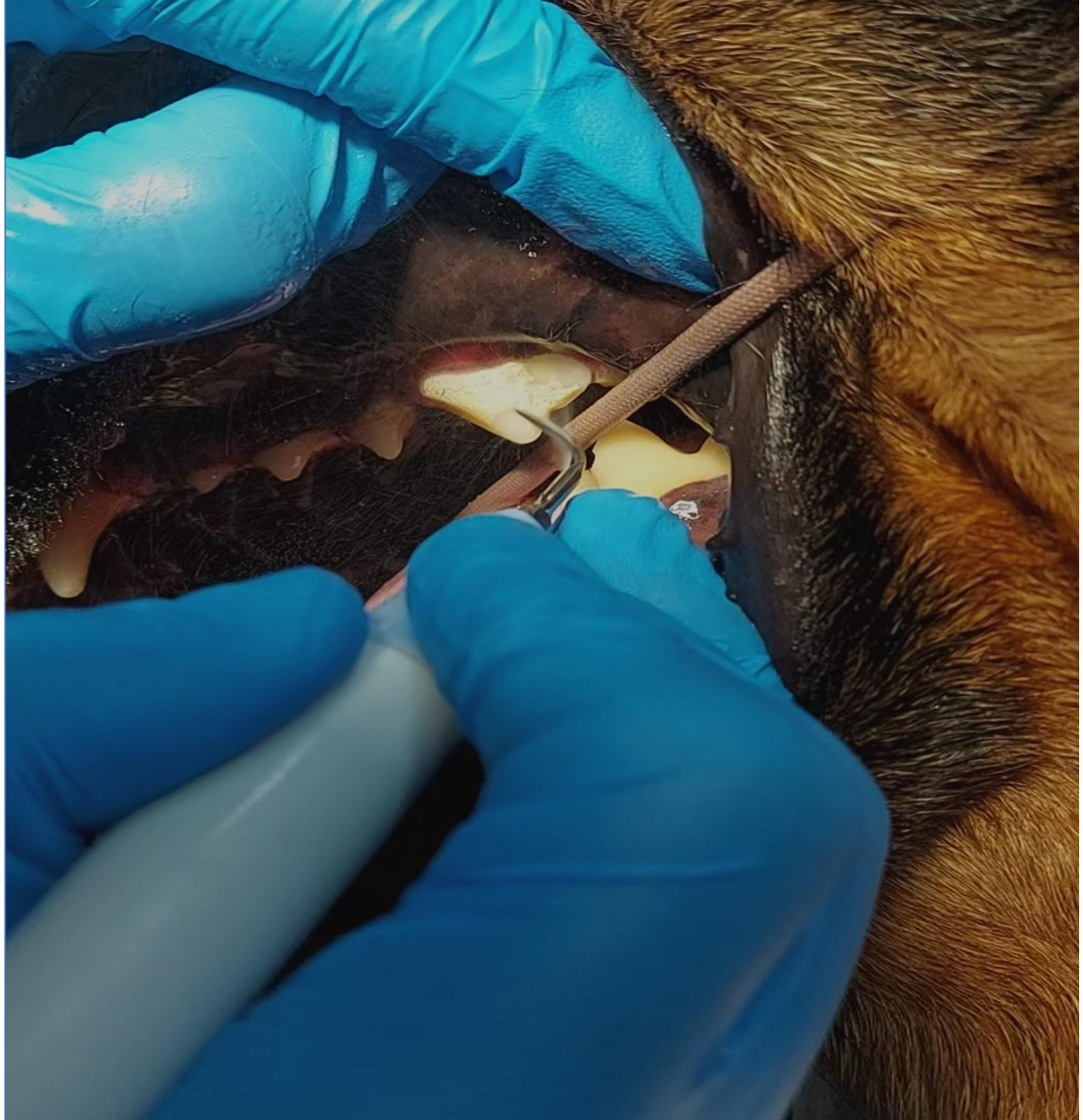
移除大塊牙結石



洗牙

牙菌斑、牙結石移除

超音波洗牙機



手持式器械

- Scaler: 銳利，使用在牙齦上
- Curettes: 圓鈍，使用在牙齦下



拋光

移除牙菌斑、污漬

讓牙齒表面平整

要移動位置，避免過熱



無麻醉洗牙的醫學與倫理問題

1. 動物在沒有麻醉下無法接受完整口腔牙周探查和放射線檢查

2.沒有麻醉的狀況下，根本無法移除牙齦下的牙菌斑和牙結石(僅移除牙齦上牙菌斑和牙結石是無效的，並無法改善寵物的健康)

3. 無麻醉是騙人(deceptive)的，會讓飼主產生虛假的成就感(accomplishment)

4.牙周相關器械是尖銳的，可能對清醒和移動的動物造成傷害

5.在沒有進行氣管插管下刮除結石，可能導致病患吸入牙結石

(BSAVA Manual of canine and feline dentistry and oral surgery, 4th ed 2018)

無麻醉洗牙

無麻醉洗牙，無法控制疼痛，未兼顧動物福利

若造成口腔出血，加上動物因疼痛而激動，出血不易控制

利器會傷害牙釉質，一旦牙釉質刮受傷不平整，細菌和結石就更容易附著...

操作人員也存在風險

COHAT 第六步 牙科X光檢查

- 1.門診麻醉前口腔檢查、理學檢查、血液檢查
- 2.麻醉計劃、IV catheter(for IV fluid and anesthetic drugs)
- 3.麻醉監控: 至少血氧、心跳、血壓、呼吸速率、體溫
- 4.檢查每顆牙齒、用牙周探針(periodontal probe)檢查囊袋、檢查牙齦是否發炎
- 5.移除牙齦上及牙齦下的牙菌斑和牙結石
- 6.**牙科X光檢查牙齦下的牙根與骨頭**
- 7.治療牙周病的牙齒或拔除
- 8.術後照顧與居家口腔保健

COHAT 第六步 牙X光檢查

DR

COHAT 第六步 牙X光檢查





- 牙科X光檢查牙齦下的牙根、牙周韌帶與骨頭
- X光機: 輻射發射，分落地式和攜帶式

  70KVP
- 影像接收: 傳統膠片 、CR、DR







FIG 217. Digital sensors.

影像接收：數位放射攝影（digital radiography, DR）





- 全口牙科 X 光影像: 全口牙科 X 光影像能夠顯現出比臨床檢查多 40% 的病理學發現（Verstraete et al. 1998a, b）。
- 由於費用考量，臨床獸醫師常常被迫在完整全口牙科 X 光影像與客戶費用限制之間取得平衡。然而，至少 應針對臨床異常的牙齒進行牙科 X 光檢查!



WSAVA-Global-Dental-Guidelines-JSAPMandarin.







No X ray
No density

No anesthesia
No periodontal treatment

X 光要在洗牙後進行



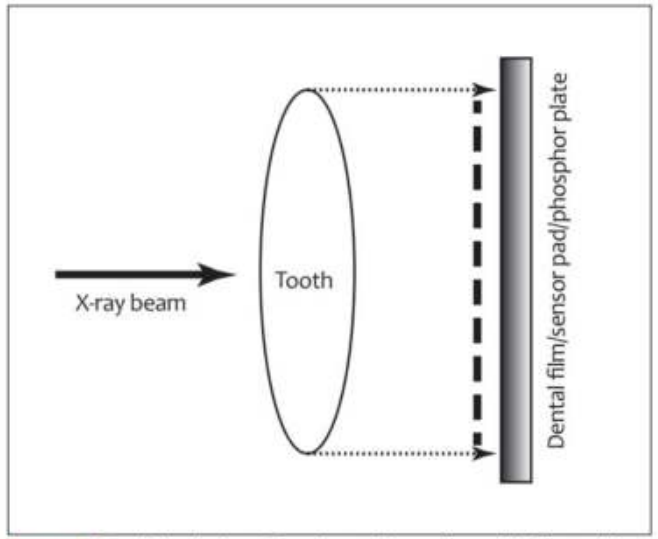




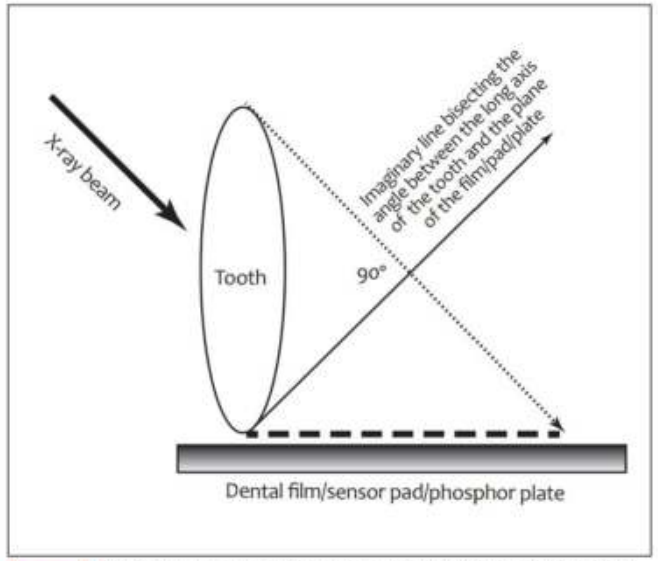
拍攝方法

- 牙科 X 光影像的標準技術：
  平行法: 適用位於中間至後
  側下顎骨的牙齒
  角平分線法: 大部分牙齒皆
  適用此方法

所有放射學影像都是在病患
全身麻醉下拍攝



4.9 Parallel technique. The plane of the radiographic film and the long axis of the teeth of interest are parallel to each other, and the X-ray beam is aimed perpendicular to them.
(© Dr Alexander M. Reiter)

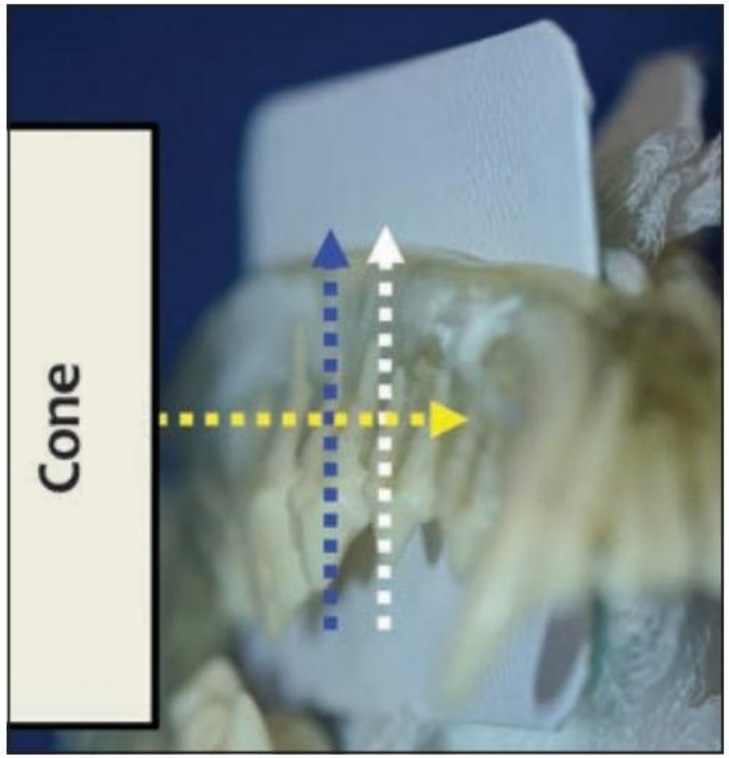


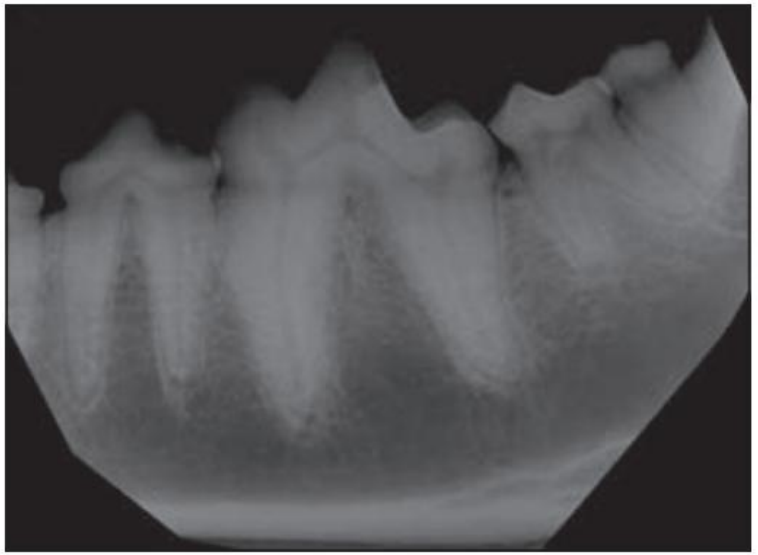
4.10 Bisecting angle technique. The radiographic film is placed at an angle to the long axis of the teeth of interest, and the X-ray beam is aimed perpendicular to an imaginary line bisecting that angle.
(© Dr Alexander M. Reiter)

BSAVA Manual of Canine and Feline Dentistry and Oral Surgery, 4th ed., 2018

平行法

n) 14: Lateral views of caudal mandibular cheek teeth

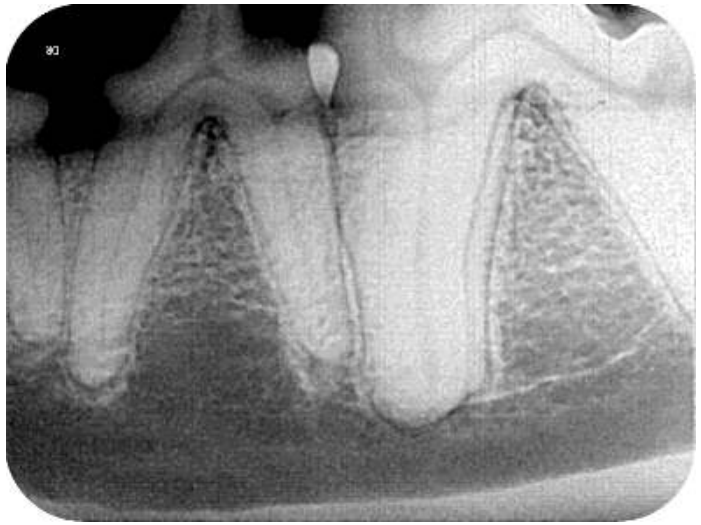


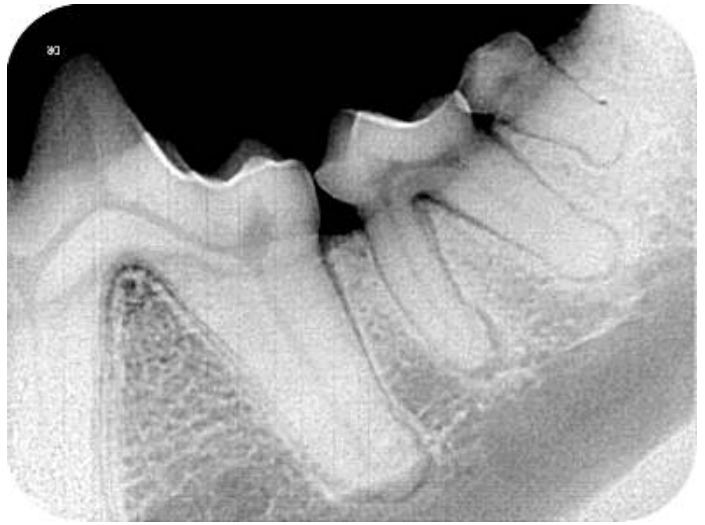


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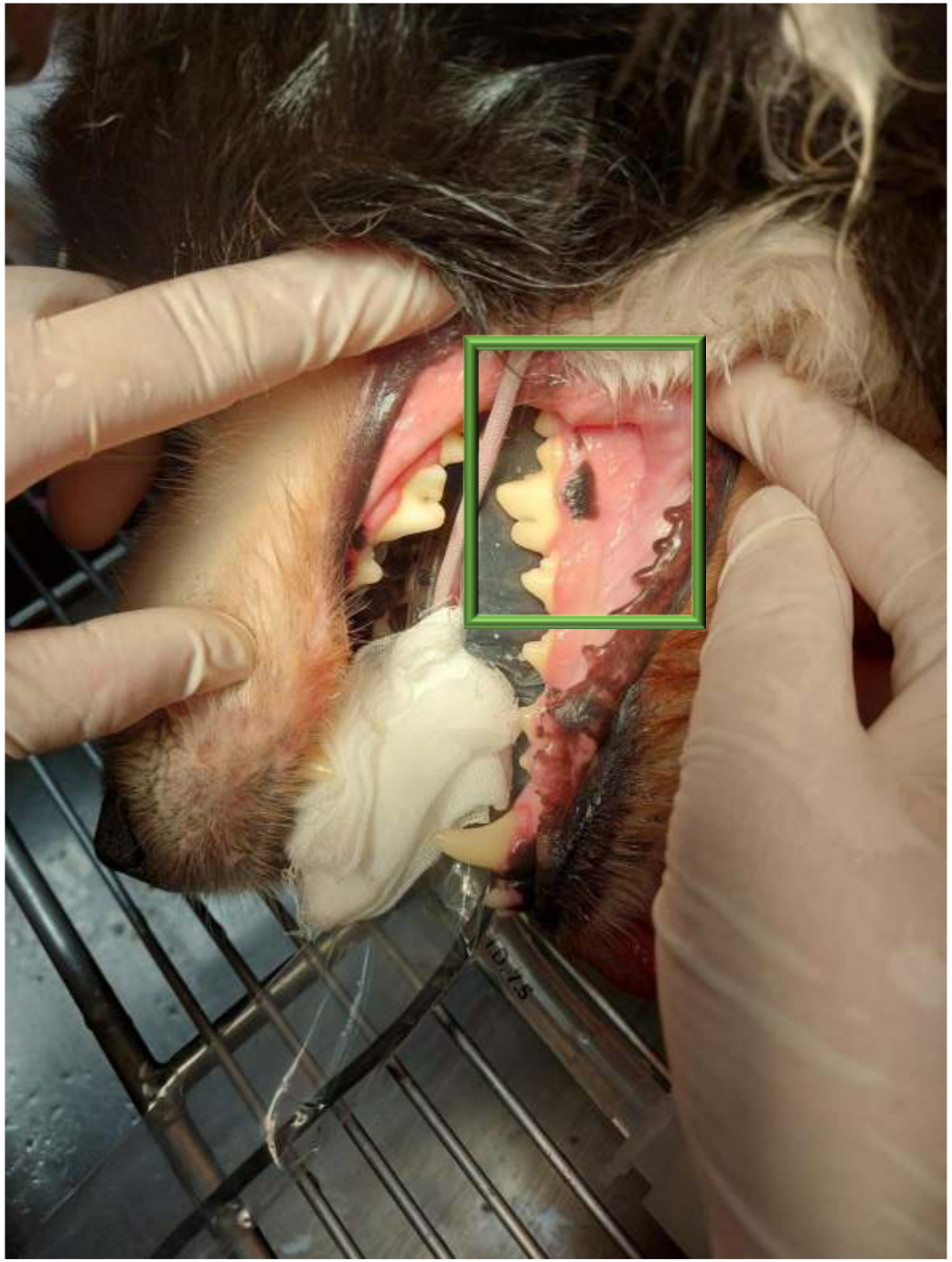
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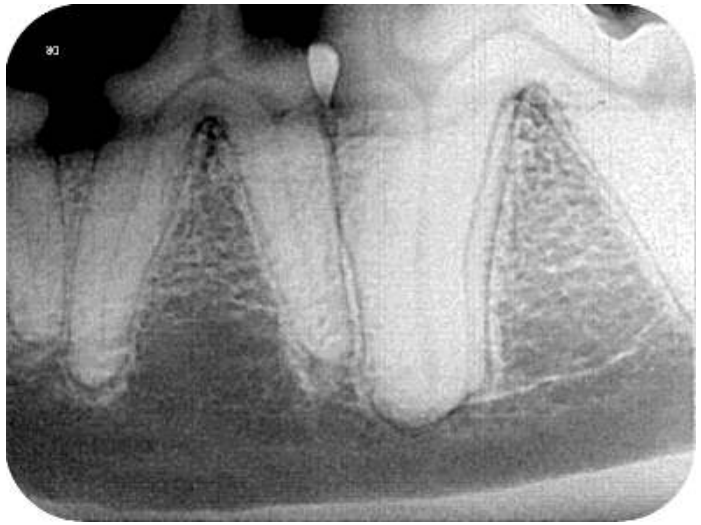


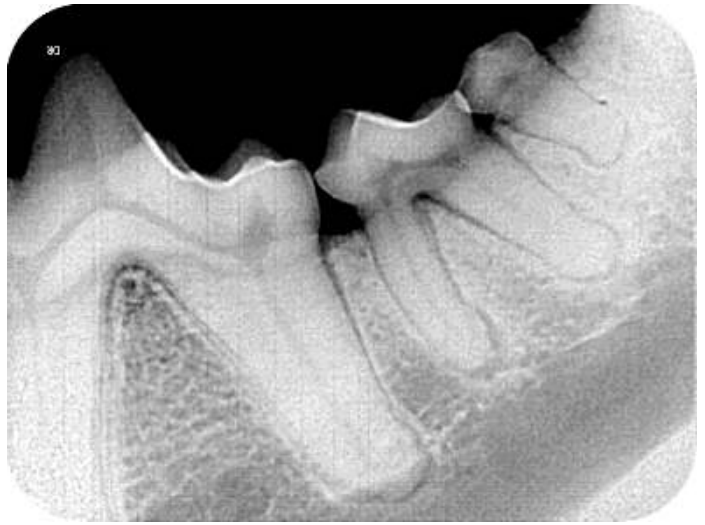




平行法



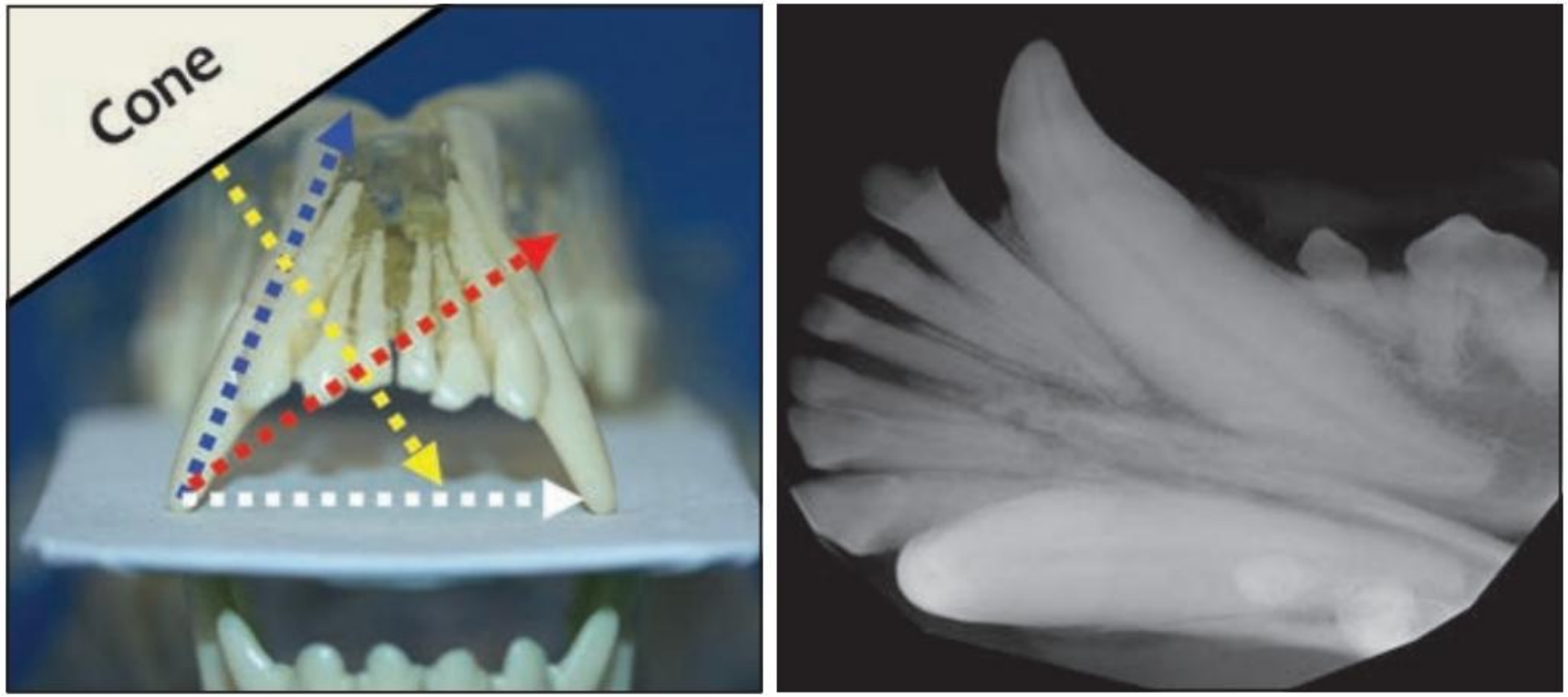




平行法

角平分線法

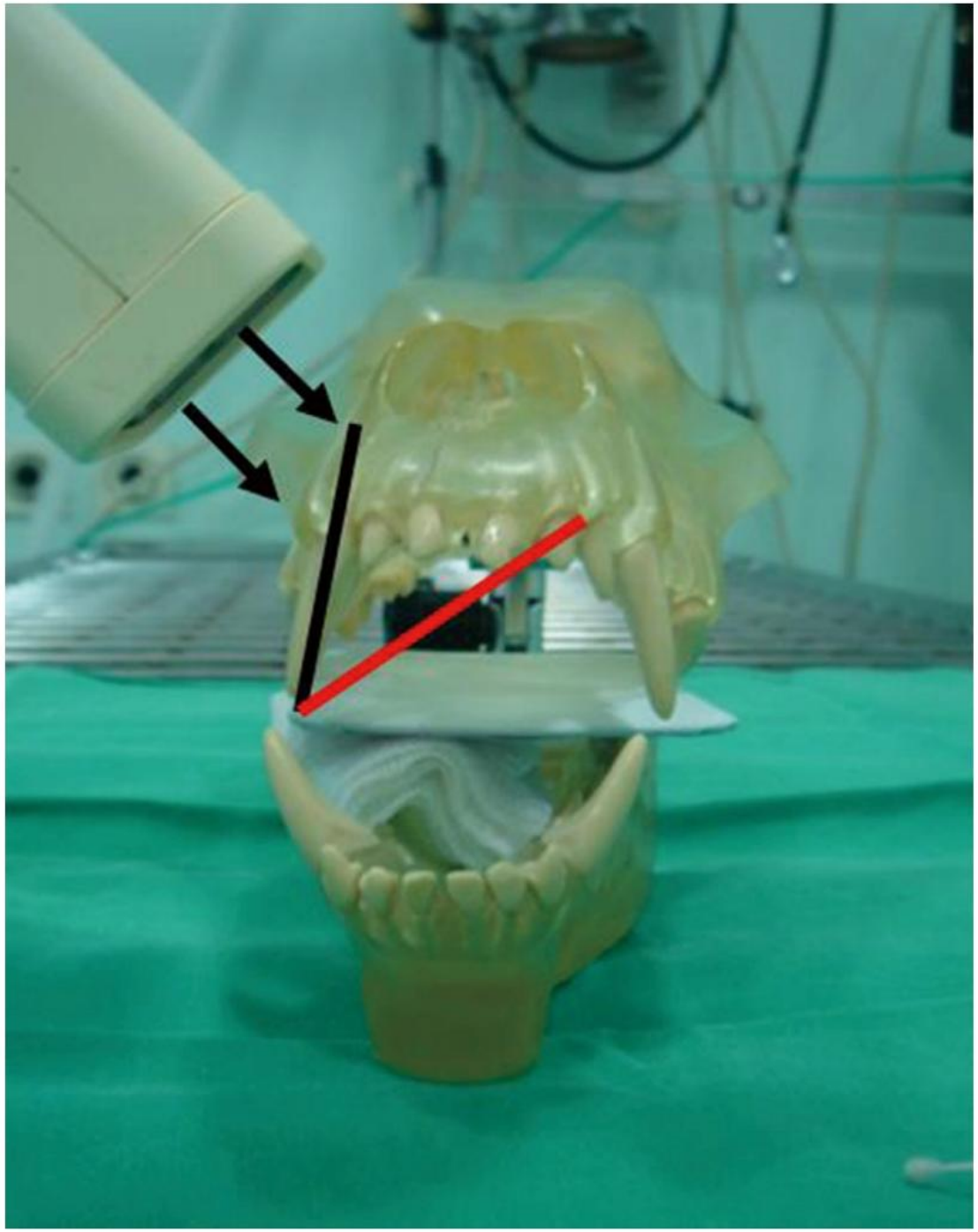
) 10: Lateral views of mandibular canine and incisor teeth



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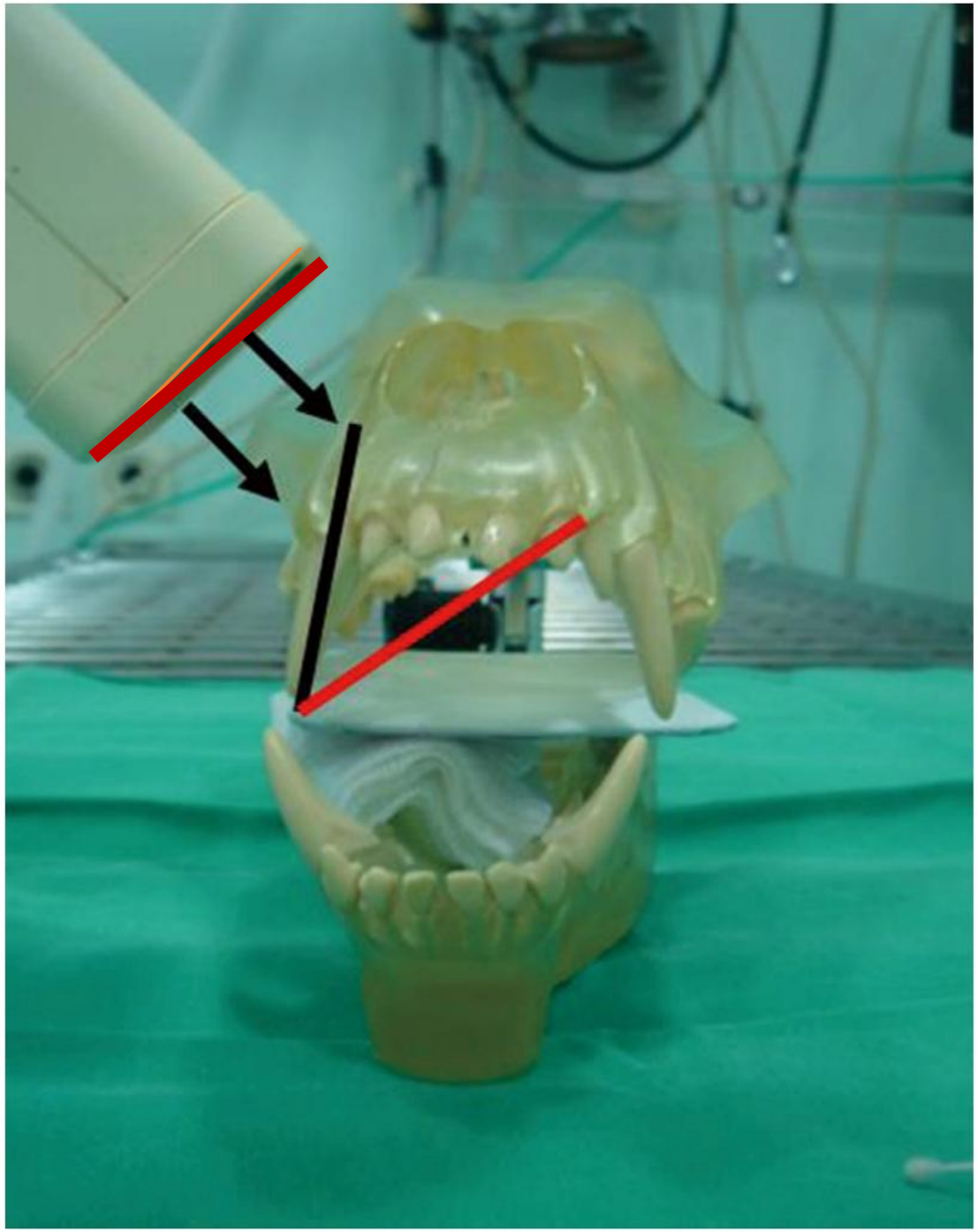
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角平分線法



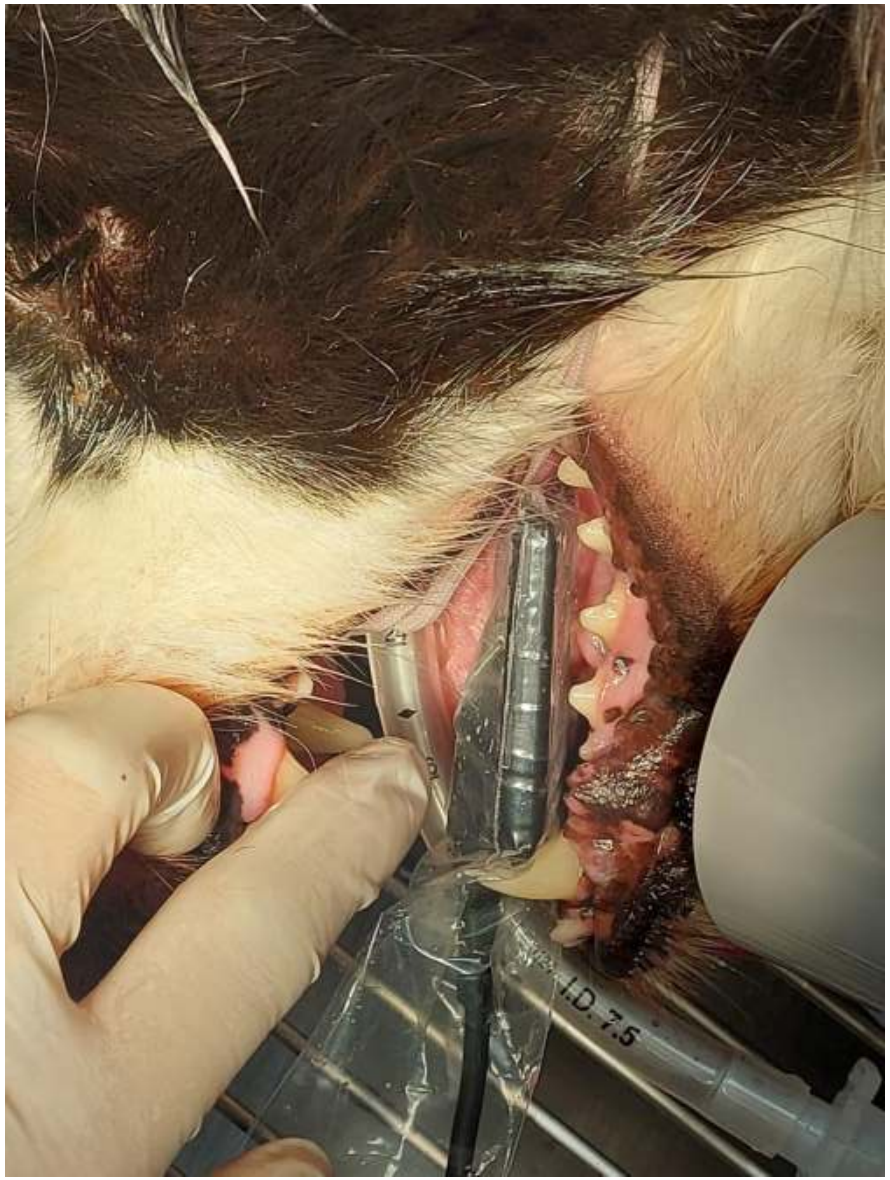
BSAVA Manual of Canine and Feline Dentistry and Oral Surgery, 4th ed., 2018

角平分線法



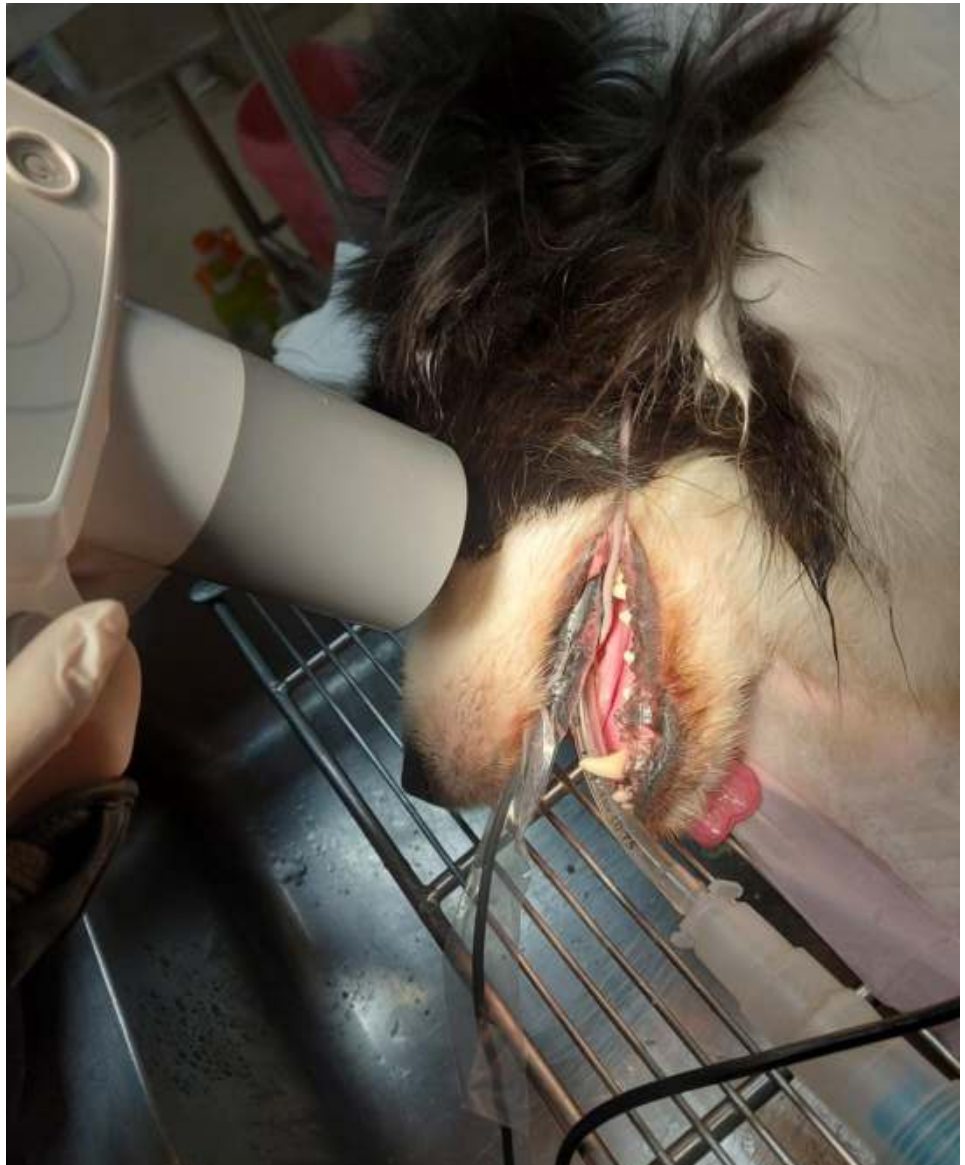
BSAVA Manual of Canine and Feline Dentistry and Oral Surgery, 4th ed., 2018

下顎前臼歯:角平分線法



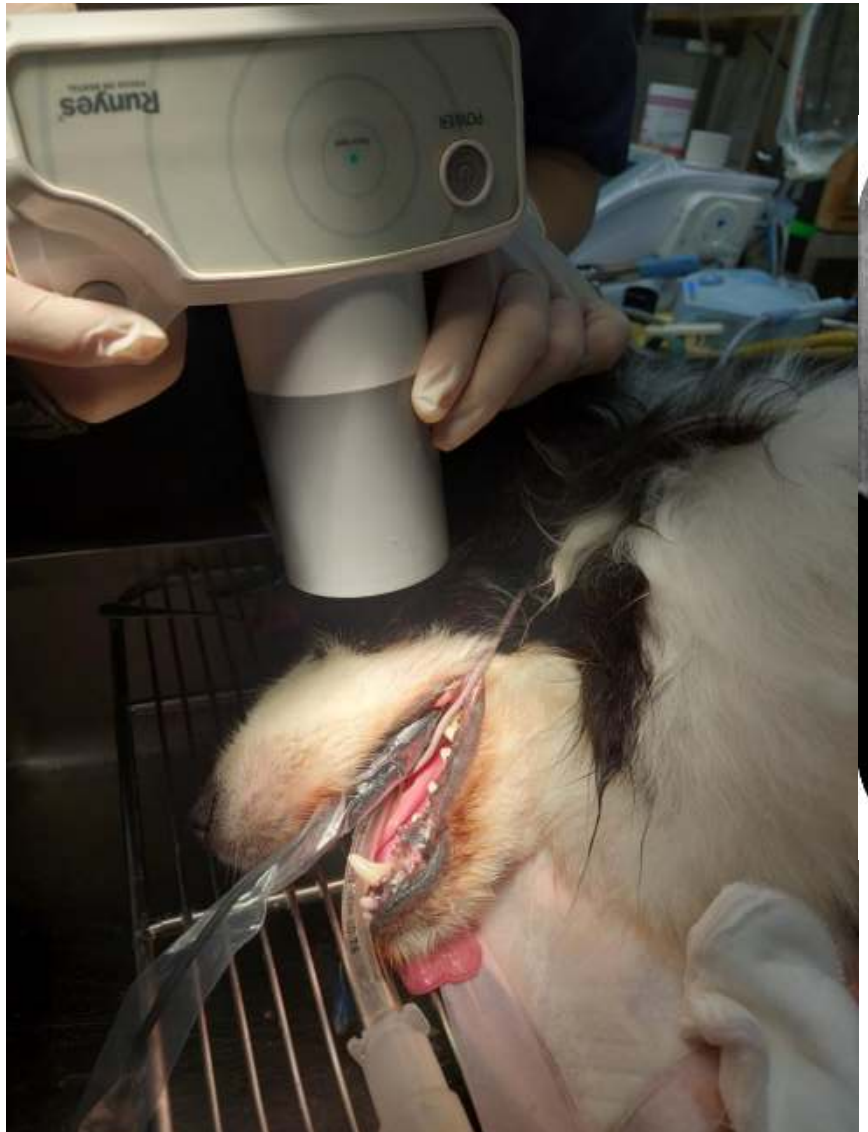


上顎前臼歯



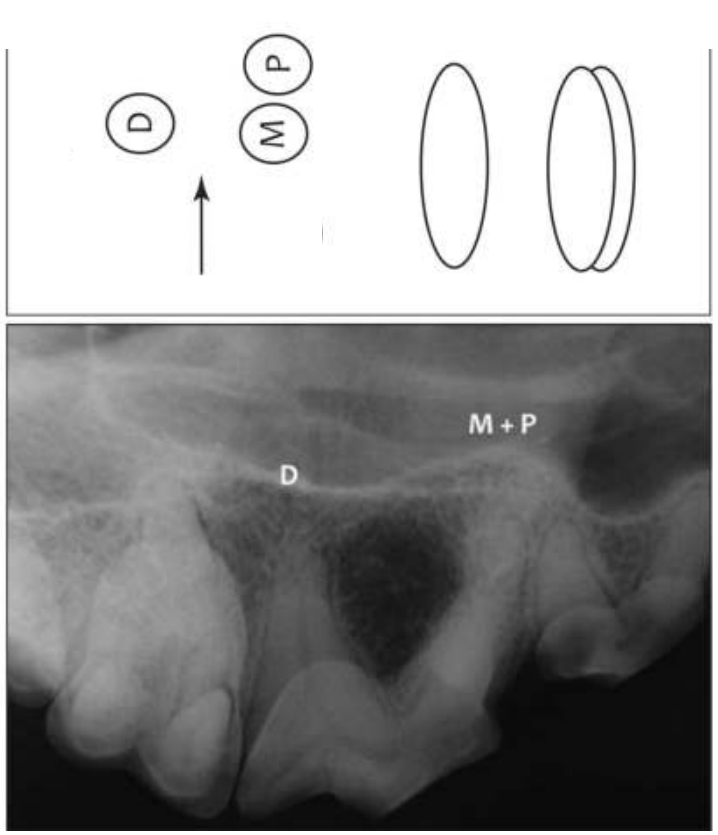


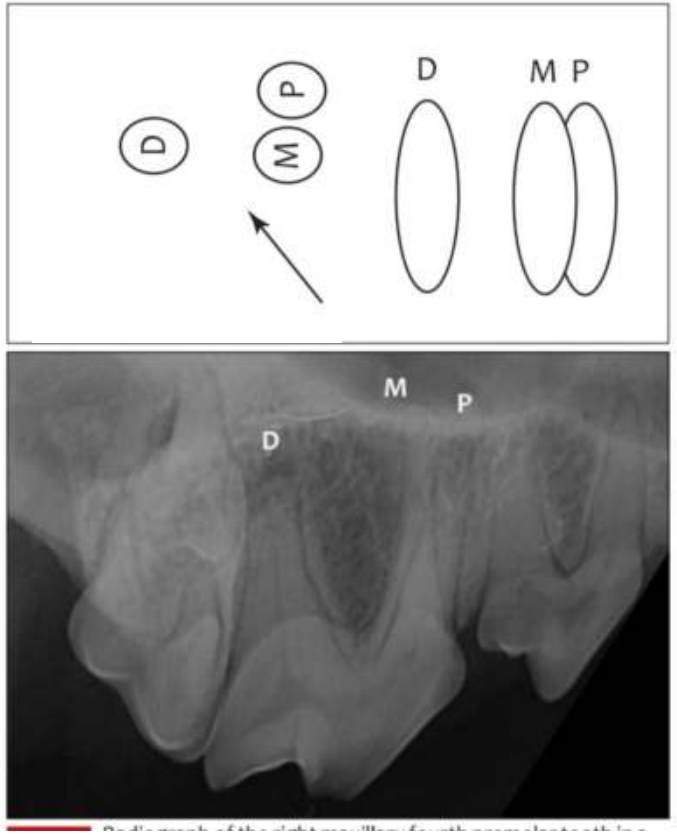
上顎臼齒與第四前臼齒

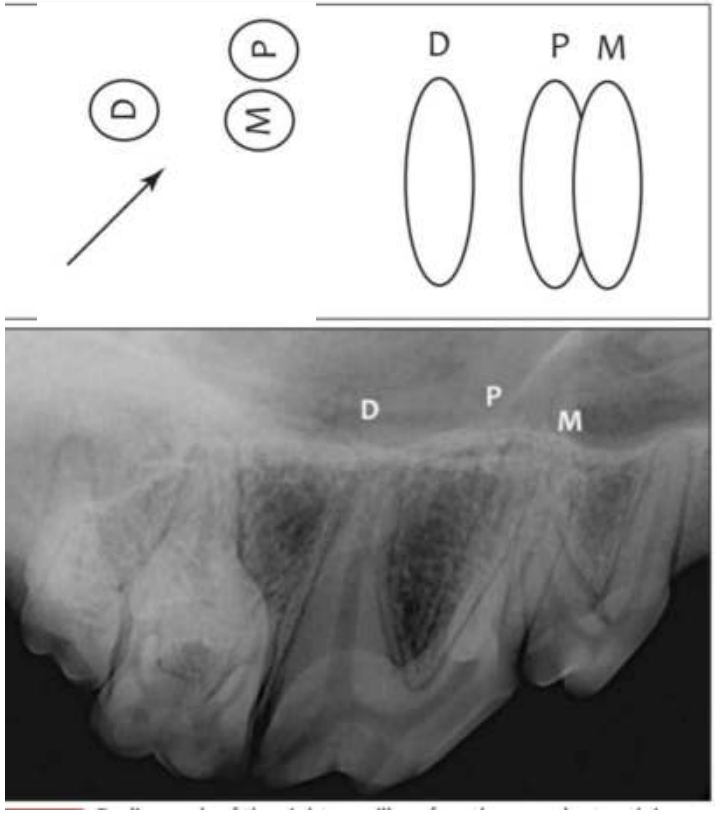




多牙根牙的判讀







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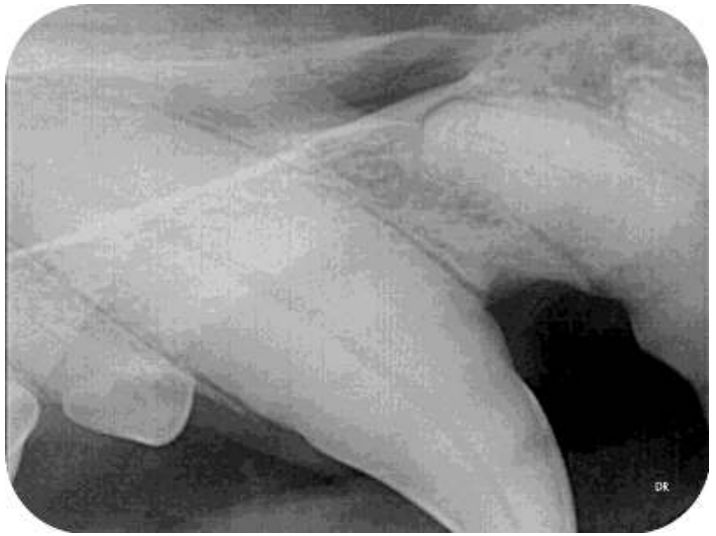


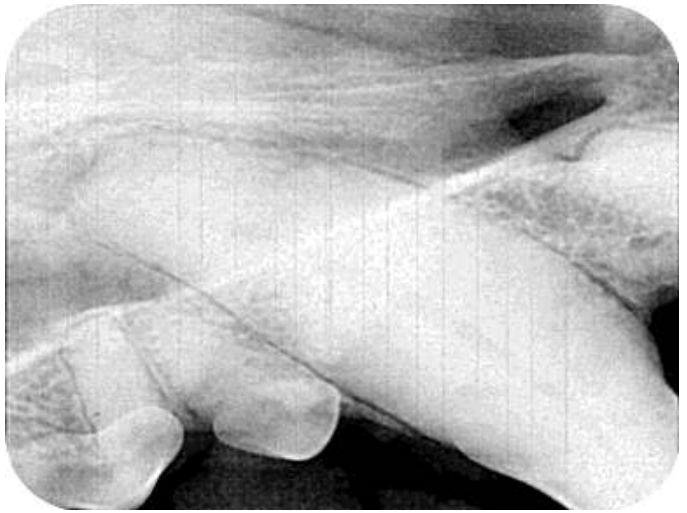
上顎犬歯



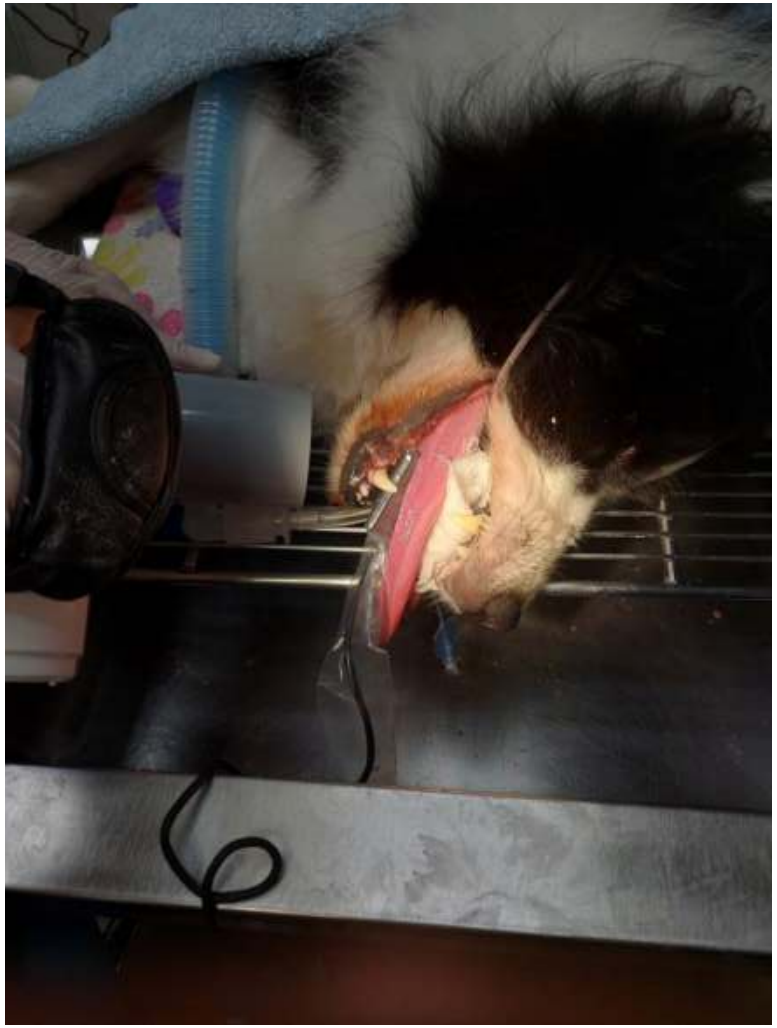


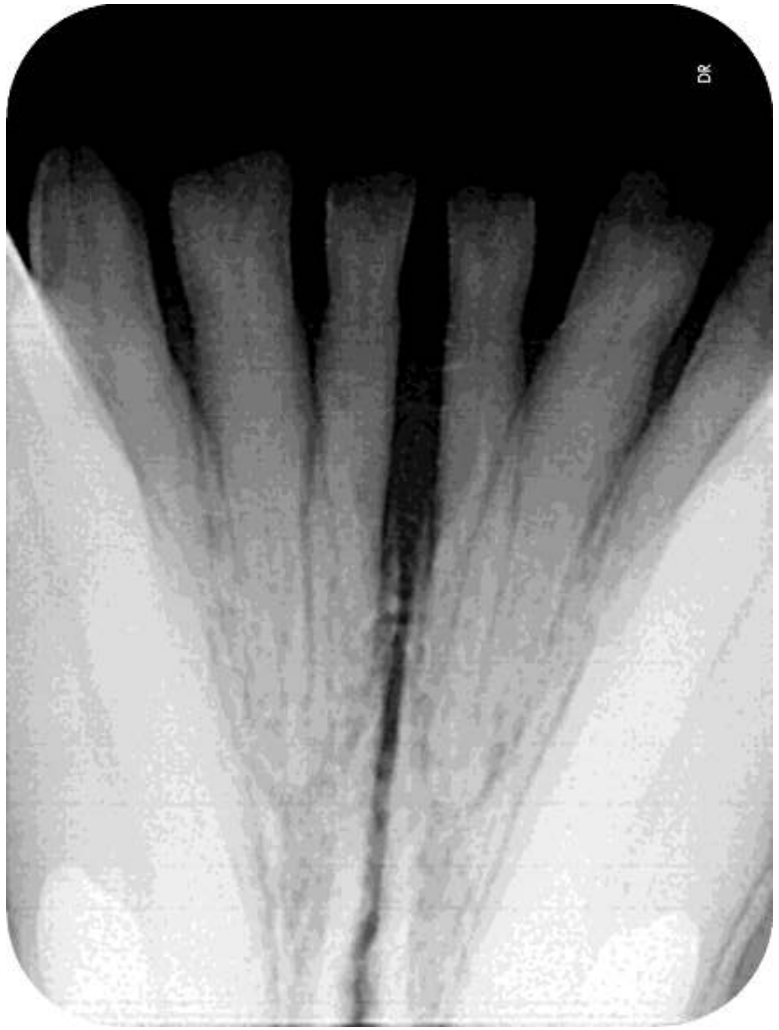




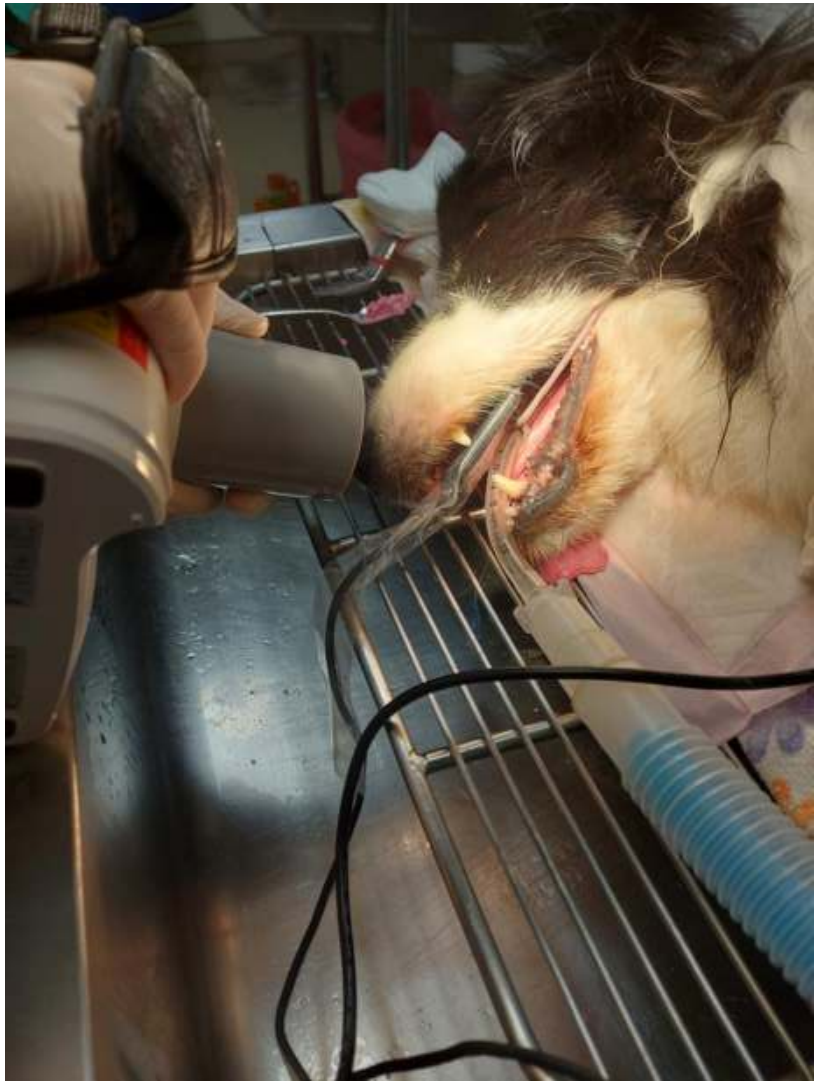


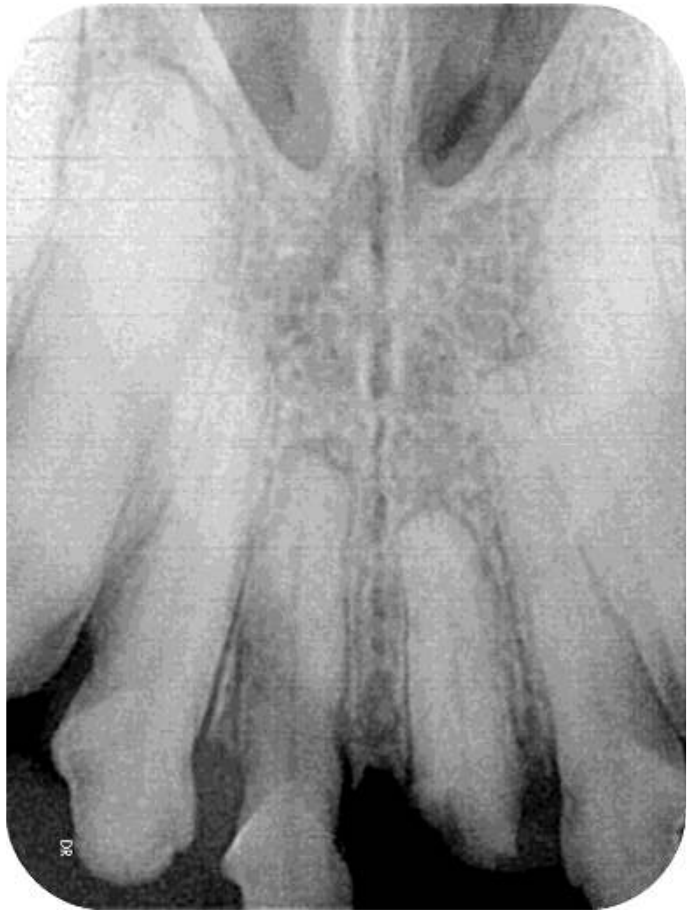
下顎門牙



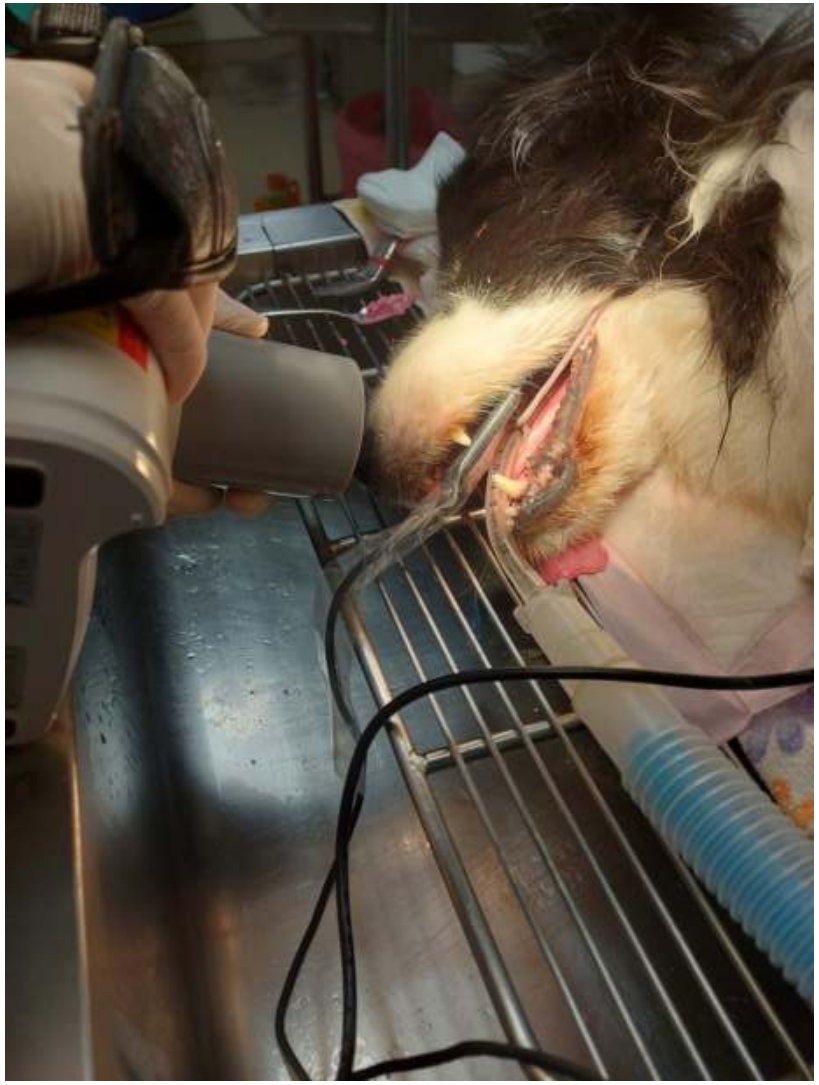


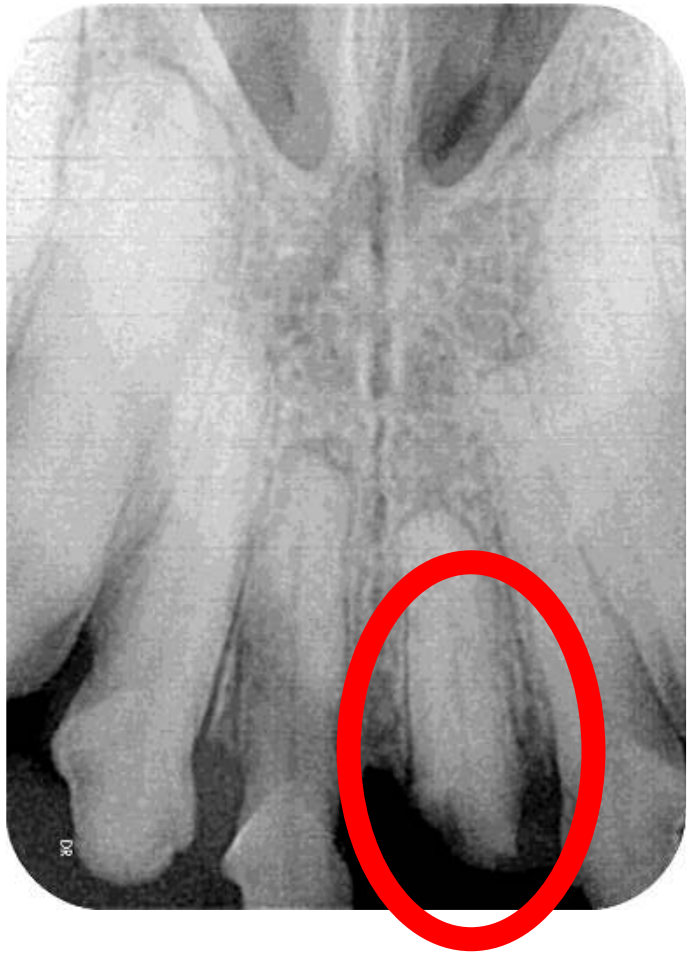
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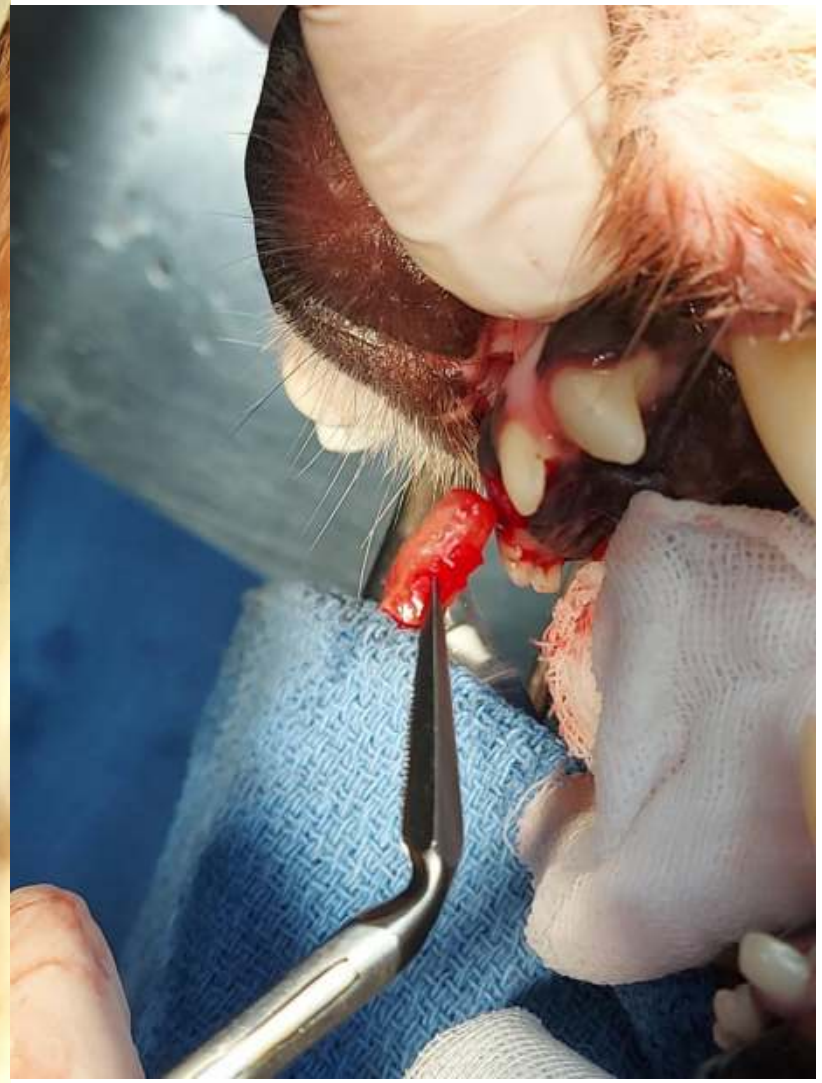


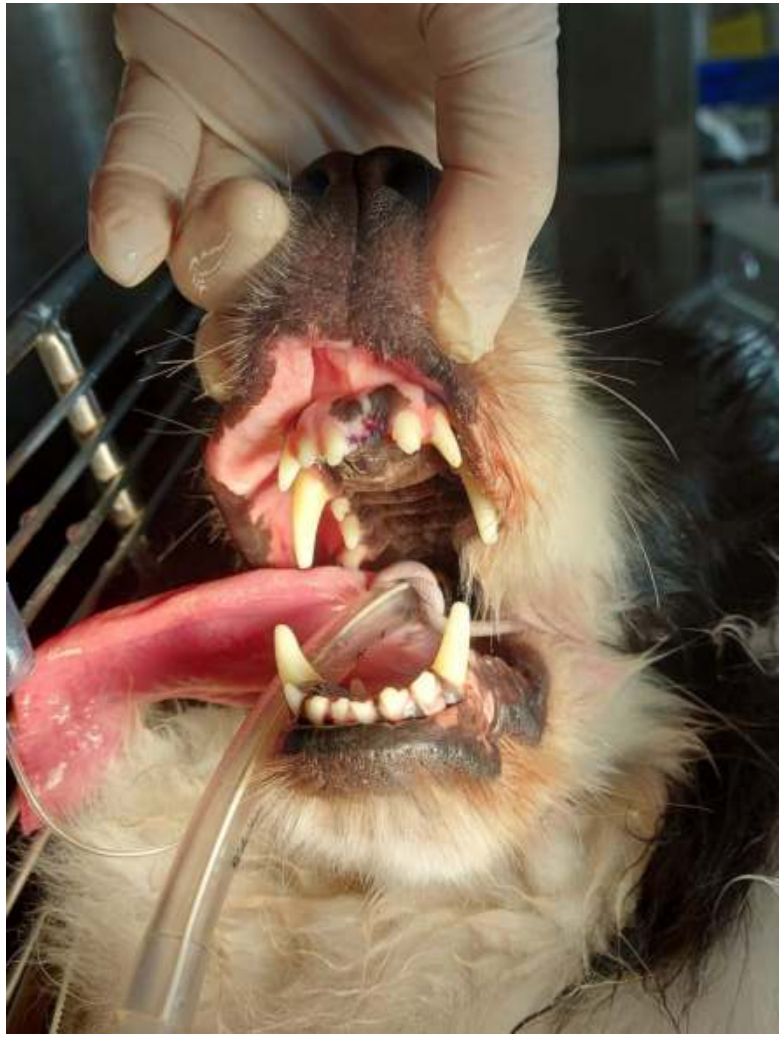
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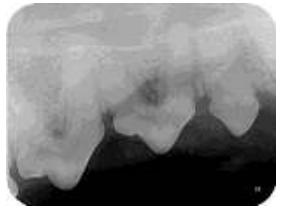


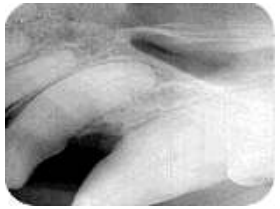
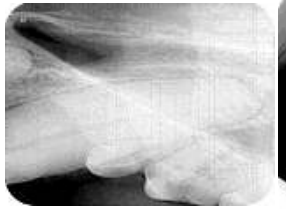
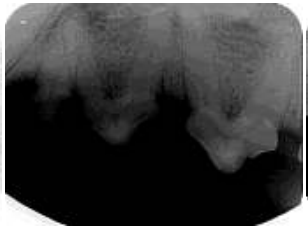






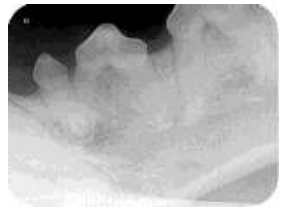











技術品質

延長
縮短
曝光條件

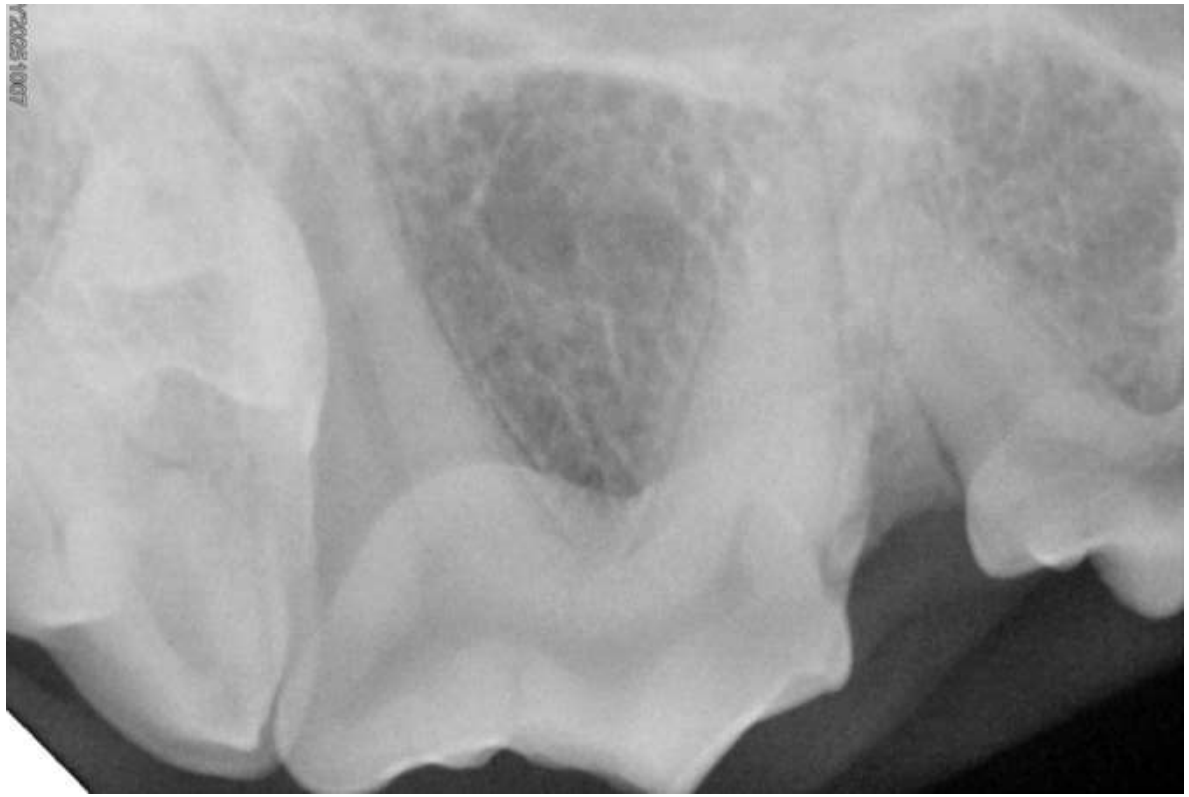
延長

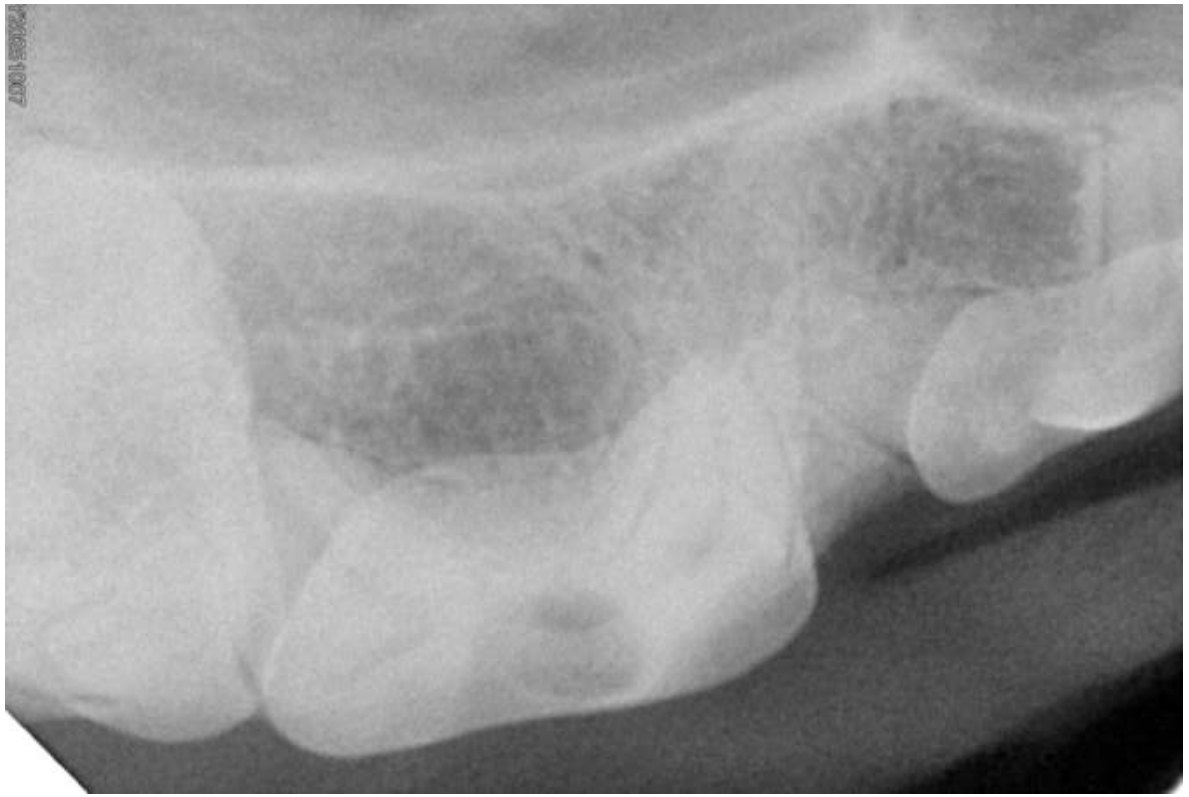




改變X光位置

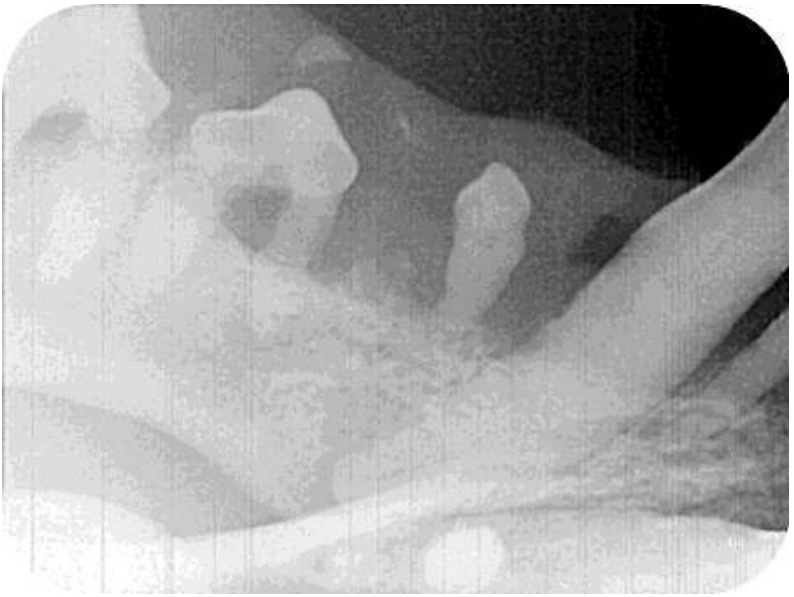
縮短





改變X光位置

曝光條件







- 改變X光秒數

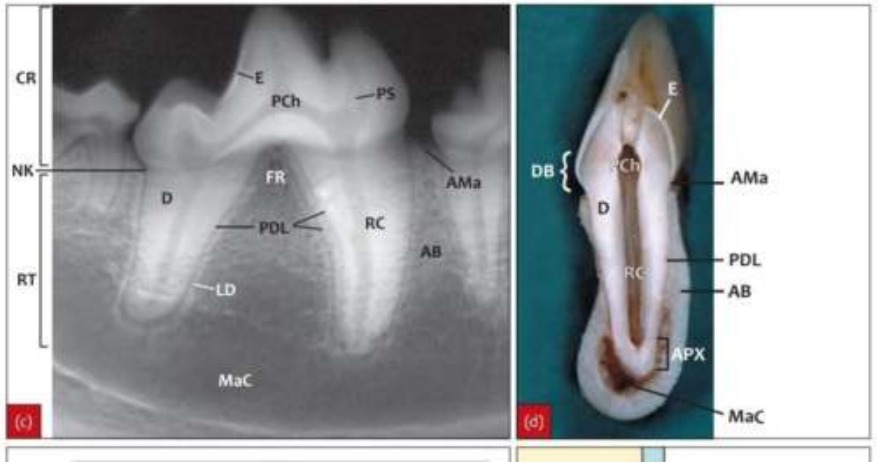
Dental radiograph interpretation

- 1.determining which teeth were imaged
- 2.Normal anatomy

✓ Alveolar bone, periodontal ligament, lamina dura

✓ Mandibular : ventral cortex of mandible, mandibular canal , mental foramen , mandibular symphysis

✓Maxillary: palatine fissure, three-root teeth



mandibular

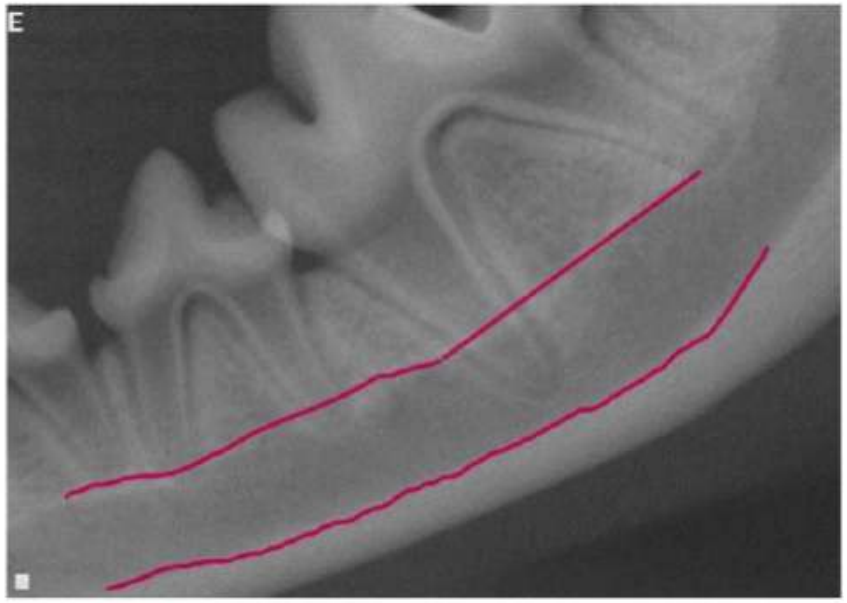


FIG 177. Mandibular canal radiographically.

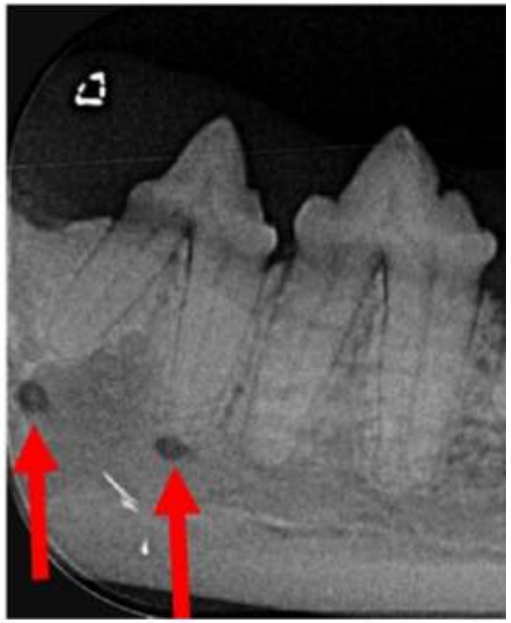


FIG 178. Mental foramina radiographically.

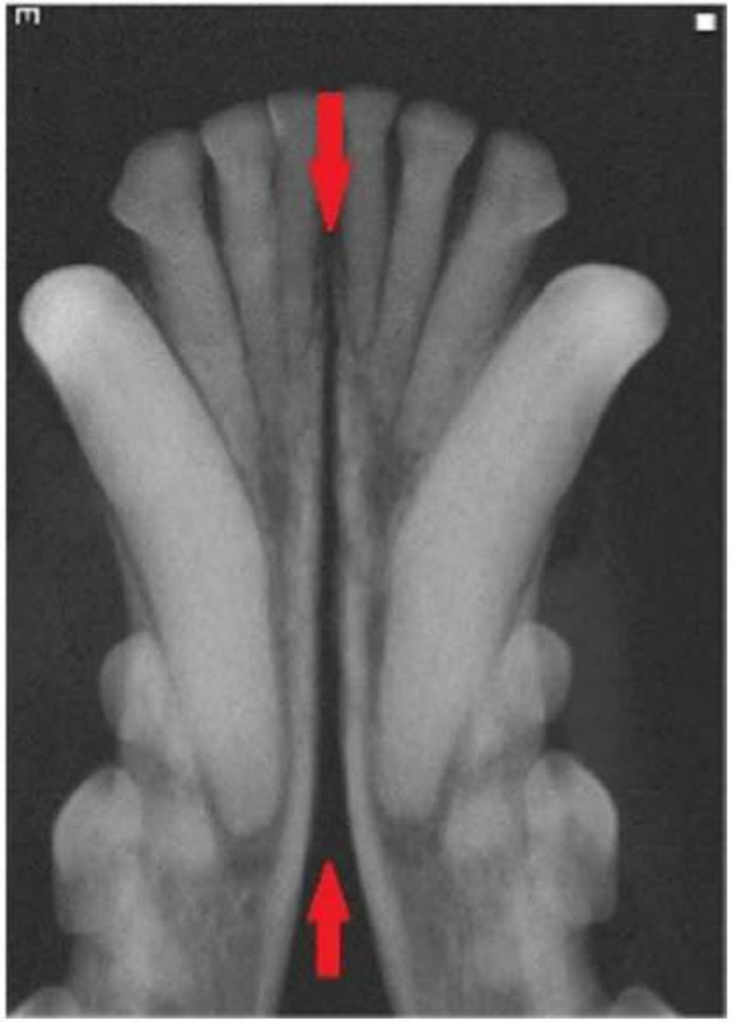


FIG 179. Mandibular symphysis radiographically.

maxillary

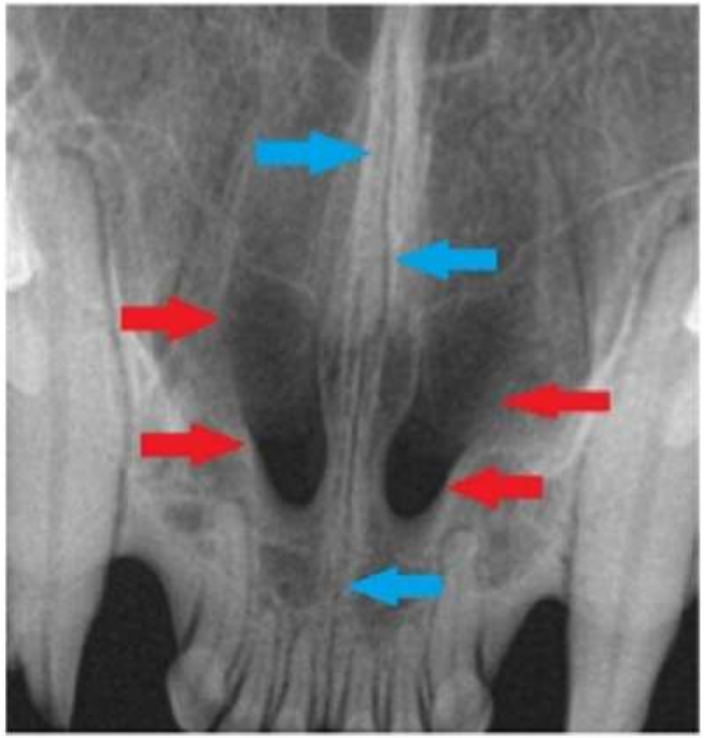


FIG 180. Maxillary palatine fissures (red arrows) and palatine suture (blue arrows) radiographically.

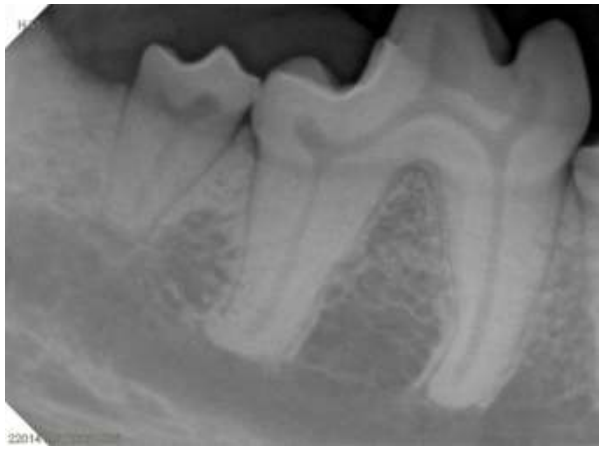


Dental radiograph interpretation

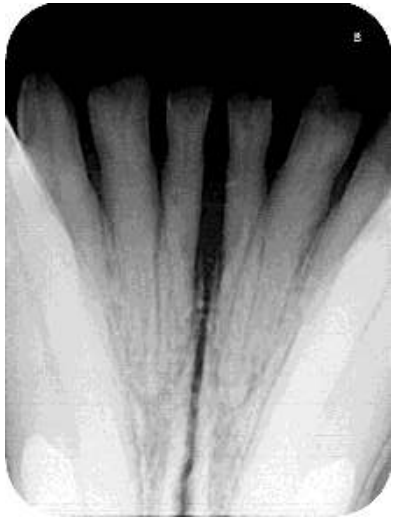


- Orientation:
- Premolar and molar:　if the mesial side is on the left side of the film=>the left side of the patient
- Canine and Incision: like VD abdomen X-ray ，right side of the mouth is on the left

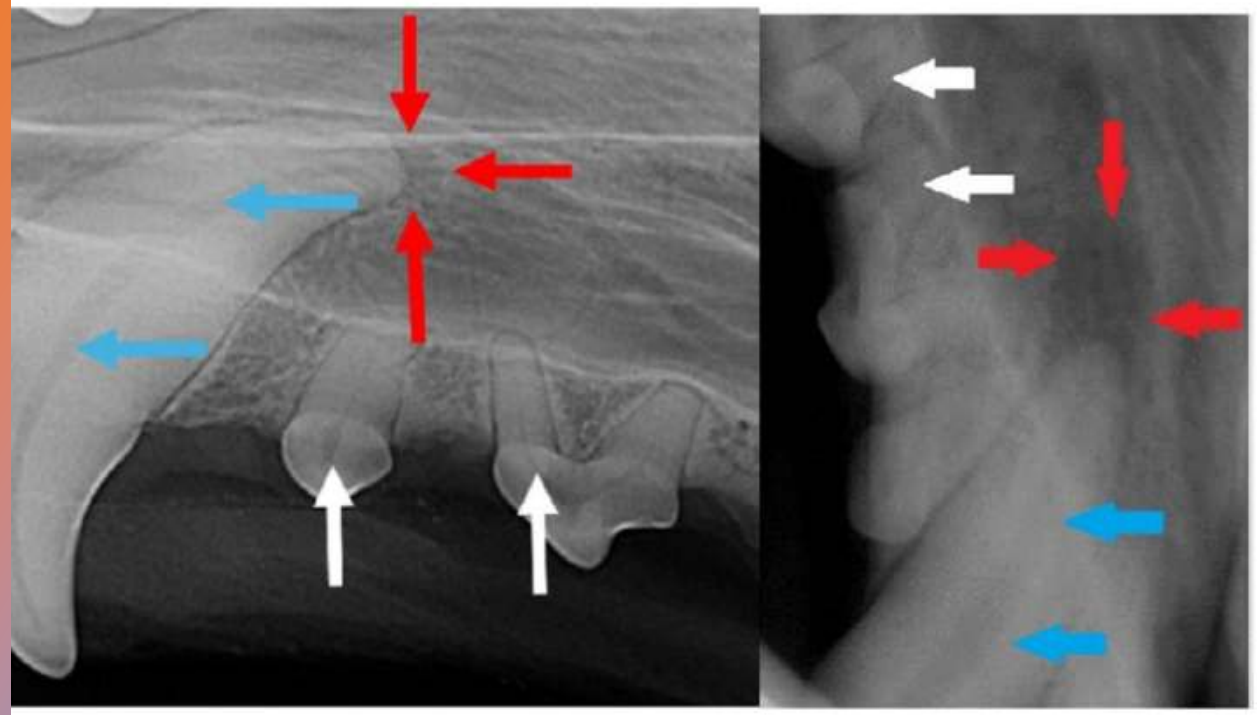




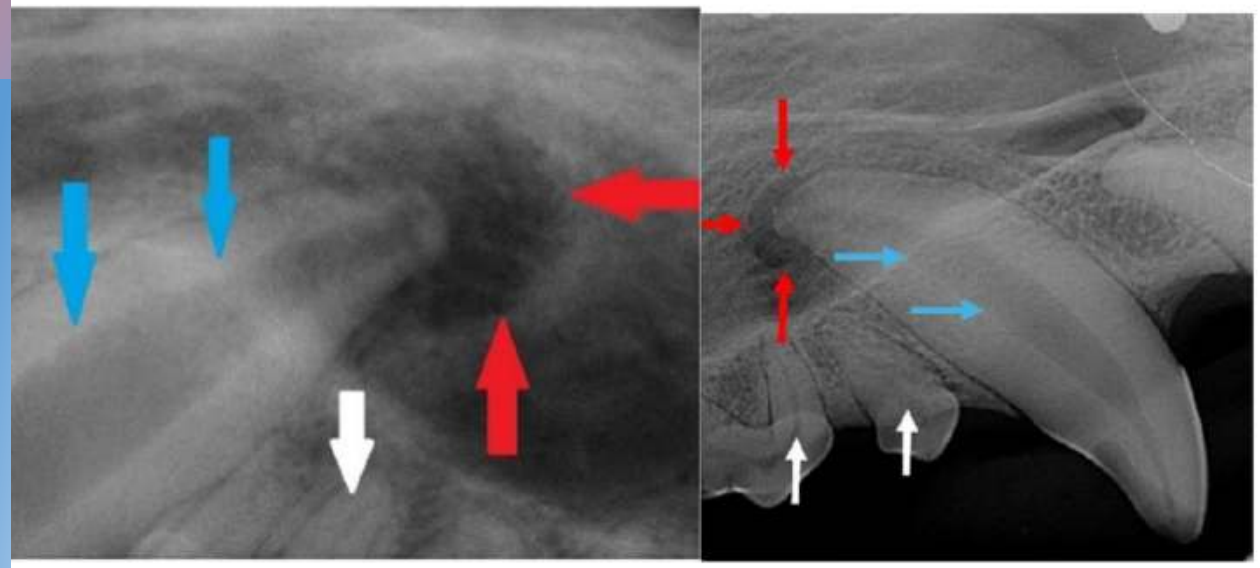




Chevron effect VS apical bone loss



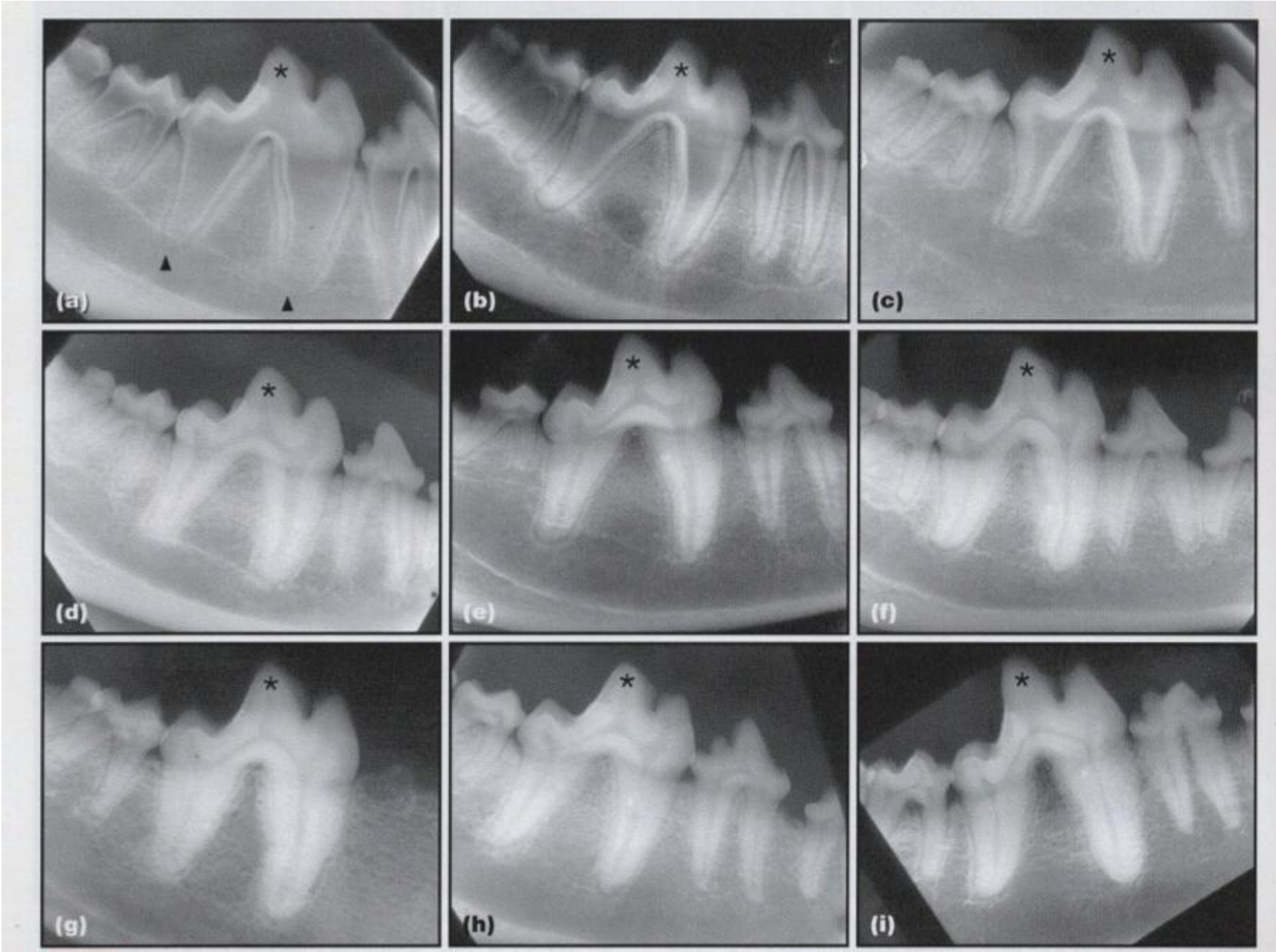
Chevron effect: regular and V-shaped



apical bone loss :irregular and round

sus (B) apical bone loss radiographically.

Ageing process



1.21 Radiographic series showing the ageing process (apexogenesis and dentine production) of the right mandibular first molar tooth (*) of the dog. **(a)** 6 months, open apex (arrowheads). **(b)** 9 months, apical closure. Following progressive thickening of the dentinal walls and narrowing of the pulp system. **(c)** 16 months. **(d)** 2 years. **(e)** 3 years. **(f)** 4 years. **(g)** 6 years. **(h)** 8 years. **(i)** 12 years.

COHAT 第七步 治療牙周病的牙齒或拔除

- 1.門診麻醉前口腔檢查、理學檢查、血液檢查
- 2.麻醉計劃、IV catheter(for IV fluid and anesthetic drugs)
- 3.麻醉監控: 至少血氧、心跳、血壓、呼吸速率、體溫
- 4.檢查每顆牙齒、用牙周探針(periodontal probe)檢查囊袋、檢查牙齦是否發炎
- 5.移除牙齦上及牙齦下的牙菌斑和牙結石
- 6. 牙科X光檢查牙齦下的牙根與骨頭
- **7.治療牙周病的牙齒或拔除**
- 8.術後照顧與居家口腔保健

牙周治療過程

麻醉下

洗牙

拋光

探查

照X光

拔牙

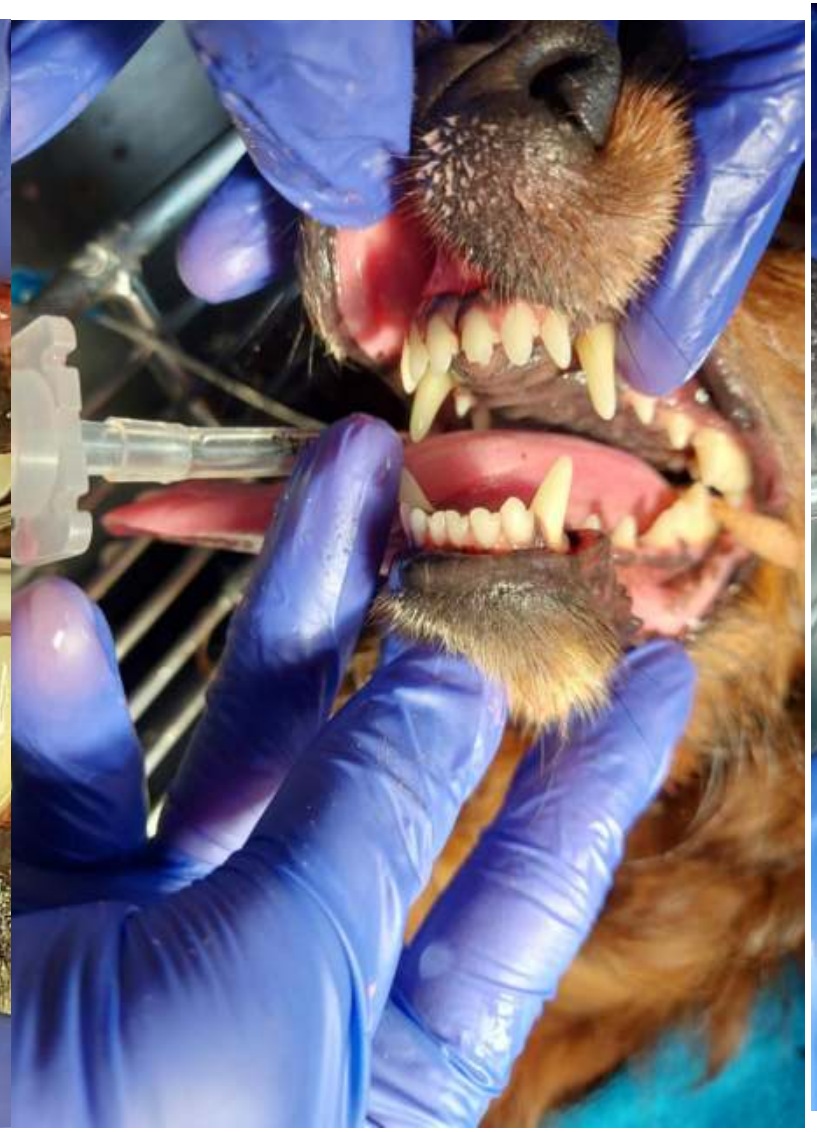
縫合

洗牙前





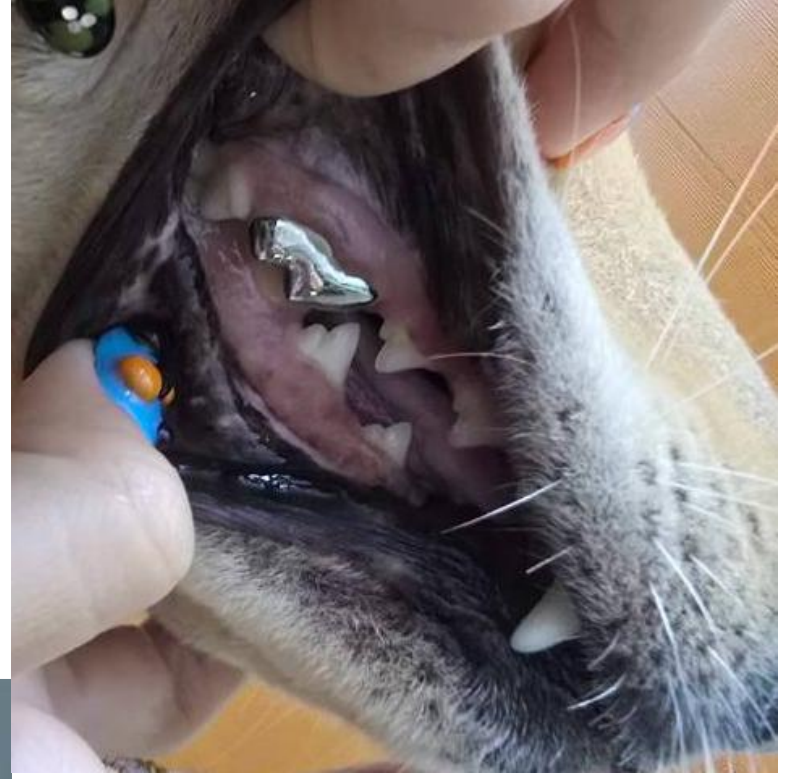



洗牙後













COHAT 第八步 術後照顧與居家口腔保健

- 1.門診麻醉前口腔檢查、理學檢查、血液檢查
- 2.麻醉計劃、IV catheter(for IV fluid and anesthetic drugs)
- 3.麻醉監控: 至少血氧、心跳、血壓、呼吸速率、體溫
- 4.檢查每顆牙齒、用牙周探針(periodontal probe)檢查囊袋、檢查牙齦是否發炎
- 5.移除牙齦上及牙齦下的牙菌斑和牙結石
- 6. 牙科X光檢查牙齦下的牙根與骨頭
- 7.治療牙周病的牙齒或拔除
- 8. 術後照顧與居家口腔保健



刷牙+潔牙產品>>**刷牙**>>擦拭>>潔牙產品>>沖水>>喝水>>什麼都沒做



黃金準則: 每天刷牙，刷毛進牙齦溝



每顆牙5~10秒以上

口腔保健







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潔牙產品認證


VETERINARY
ORAL HEALTH COUNCIL
VOHC
Accepted ®
Finding the most effective route to
good oral health for your dog or cat.

牙X光病例分享

Case 1

- 奶露
- DOG
- Old English sheepdog
- Spayed female
- 13Y/O
- Never tooth brush
- Never periodontal treatment





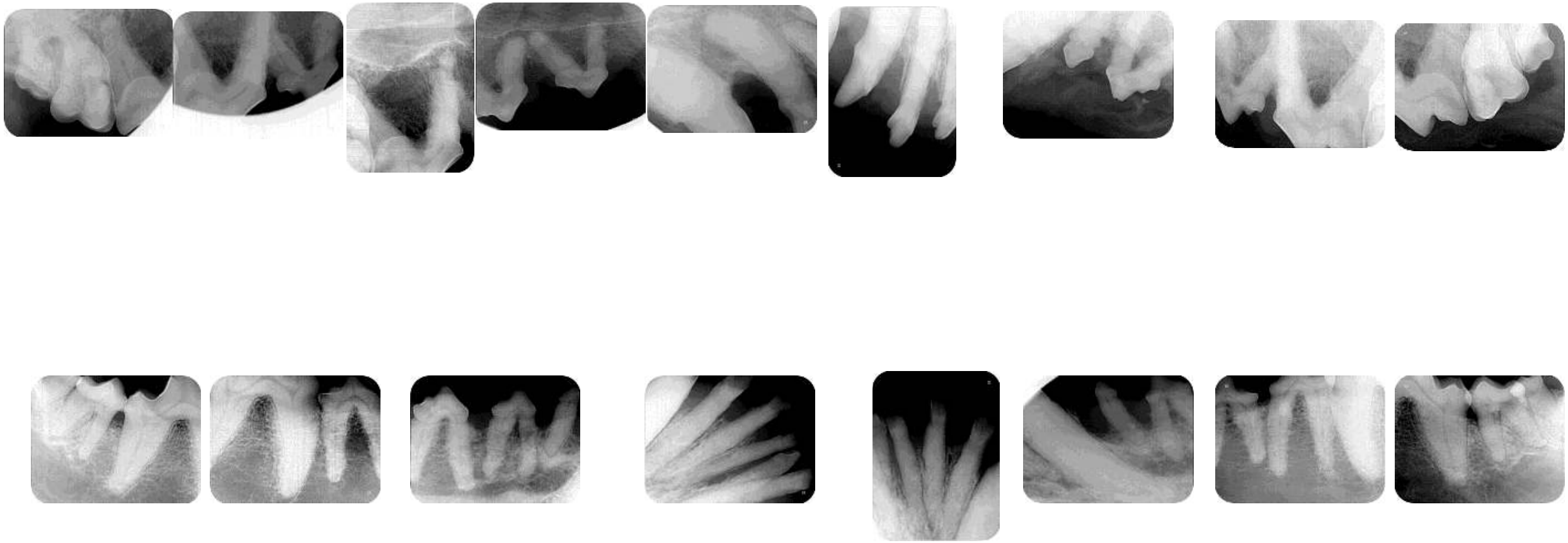


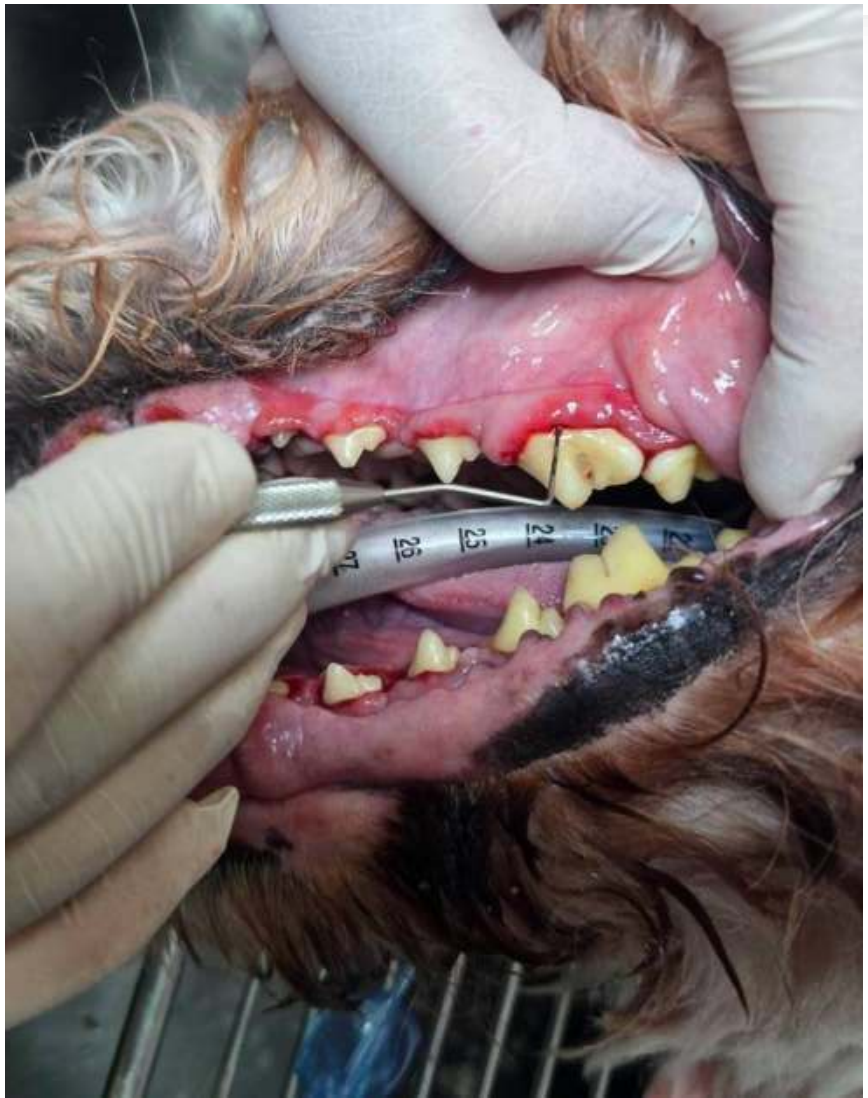


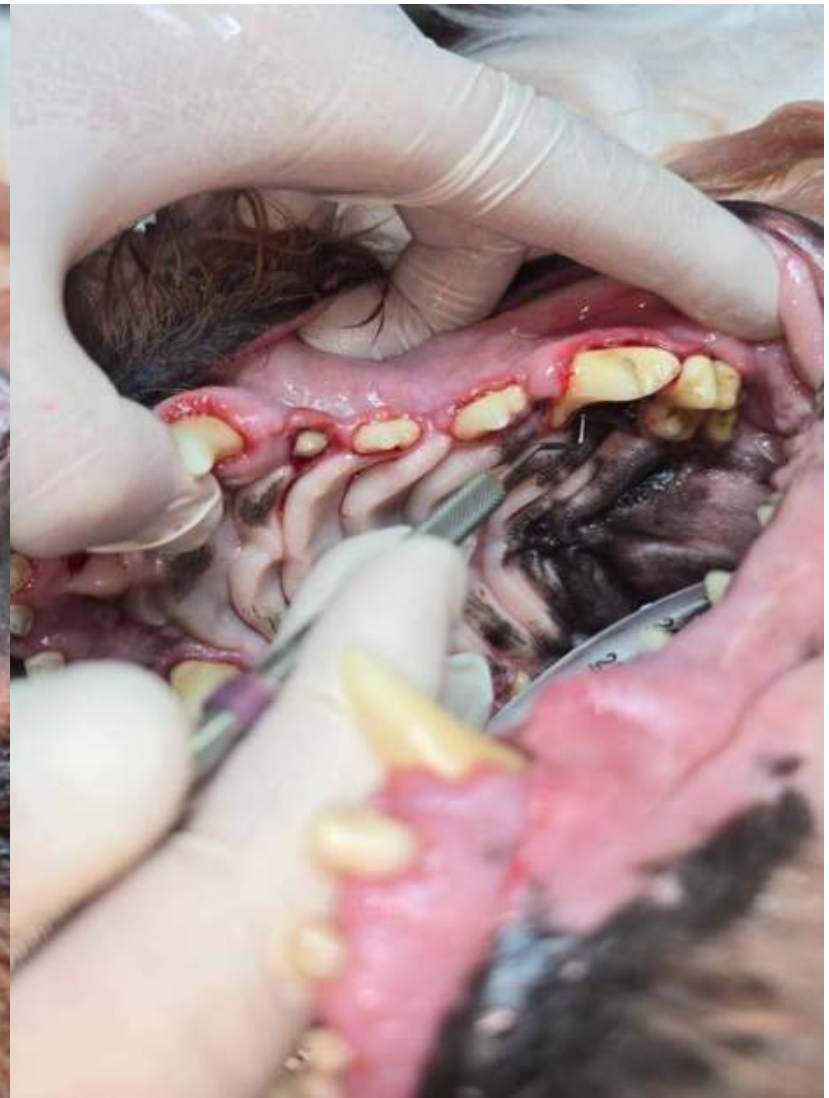


















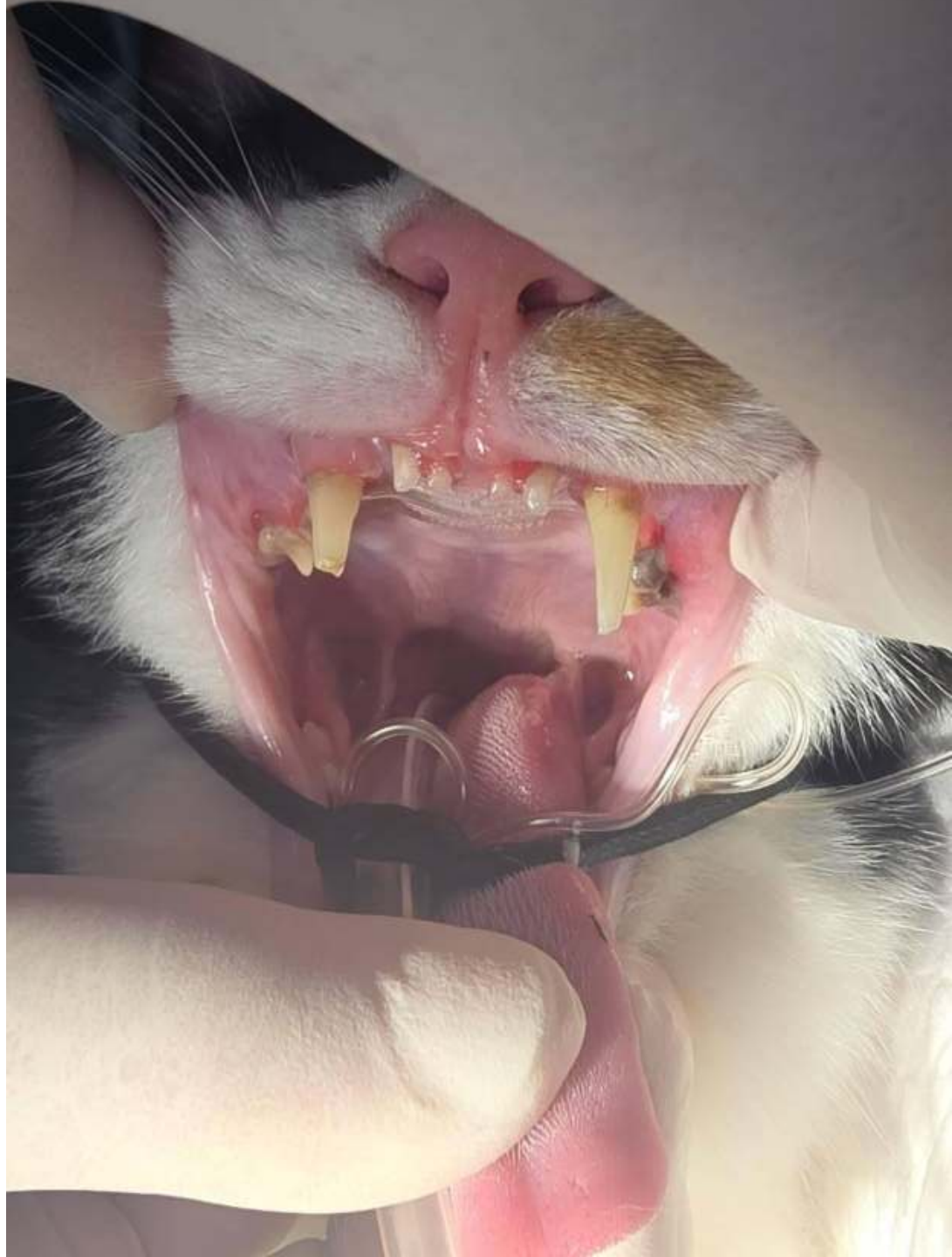


Treatment :extraction

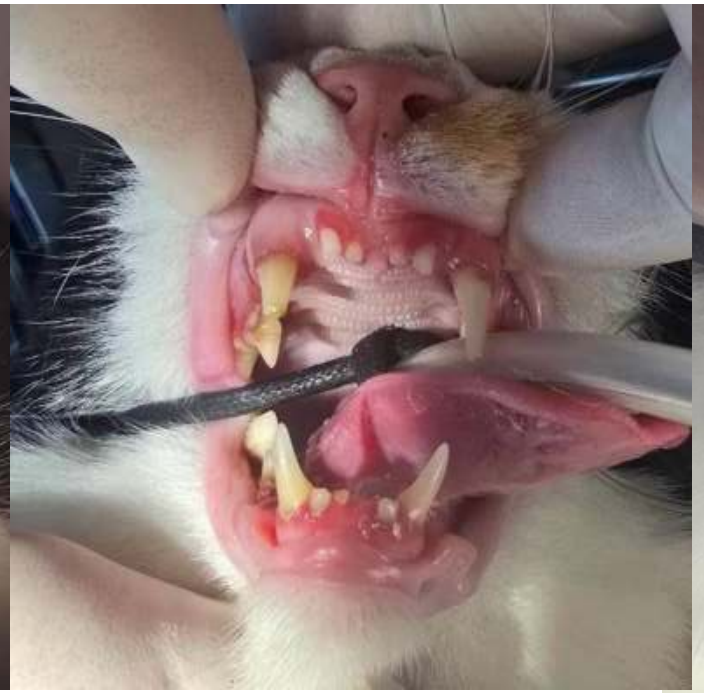


Case 2

- 可樂
- Cat
- Mixed
- Spayed female
- 3Y/O
- Maxillary canine teeth crown fracture


















HL





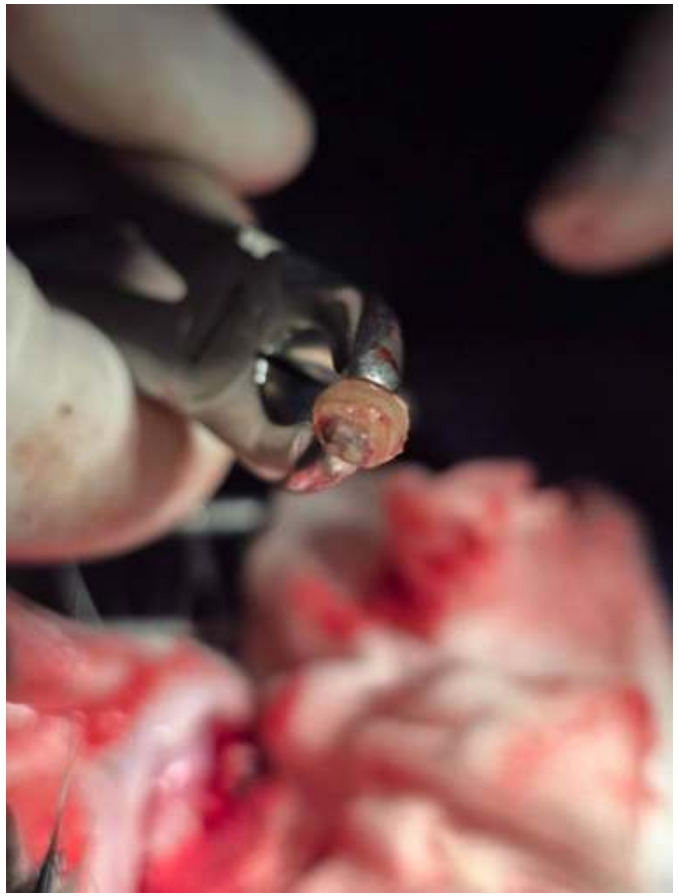



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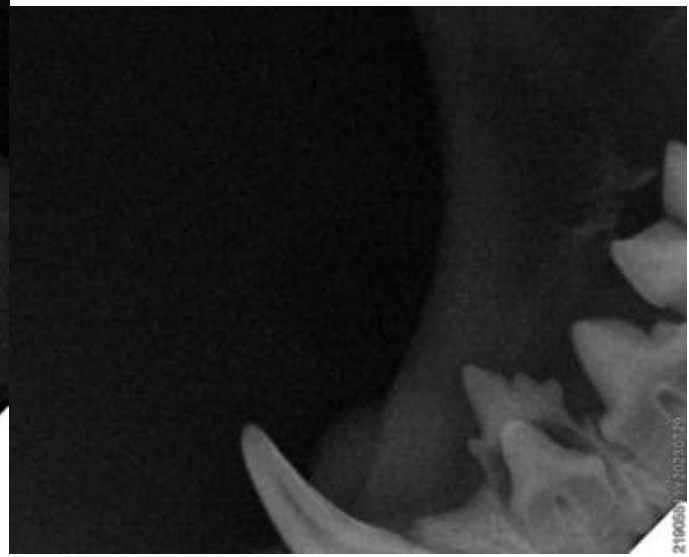
case3











謝謝您的聆聽!

祝 笑口常開

